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Risk Analysis of Applying Cryptocurrencies and Blockchain in Organizational Logistics Processes

Master's Thesis

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Contents

<i>Introduction</i>	1
<i>1. THE CONCEPT OF LOGISTICS</i>	3
<i>2. LOGISTICS SYSTEMS AND LOGISTICS PROCESSES</i>	5
<i>2.1. Subsystems of the logistics system</i>	9
2.1.1. Production Logistics	10
2.1.2. Procurement Logistics	12
2.1.3. Distribution Logistics	14
2.1.4. Warehousing Logistics	16
<i>3. OVERVIEW OF LOGISTICS PROCESSES</i>	19
<i>3.1. Inventory and Inventory Management</i>	19
3.1.1. Reasons for holding inventory in the warehouse	19
3.1.2. The Importance of Inventory Management	20
3.1.3. Inventory Management Models	21
<i>3.2. Warehouse Management</i>	22
3.2.1. The Importance of the Warehouse in the Logistics System	22
3.2.2. Storage Decisions	23
<i>3.3. Material Handling and Packaging</i>	24
3.3.1. Material Handling Objectives	24
3.3.2. Functions of packaging	25
<i>3.4. Transport</i>	26
3.4.1. Types of transport	27
<i>3.5. Order fulfillment</i>	30
<i>4. THE RELATIONSHIP BETWEEN LOGISTICS AND THE SUPPLY CHAIN</i>	32
<i>5. INTRODUCTION TO CRYPTOCURRENCIES AND BLOCKCHAIN TECHNOLOGY</i>	35
<i>5.1 Basic elements of the functioning of cryptocurrency systems</i>	36
5.1.1 "Blockchain"	36
5.1.2 "Consensus"	37
<i>5.2 Technological aspects of cryptocurrencies</i>	39
5.2.1 Hardware elements and requirements	39
5.2.2 Software elements	40
<i>5.3 Cryptocurrency exchange</i>	41
5.3.1 Cryptocurrency Price Formation	41
5.3.2 Places to trade cryptocurrencies	43
<i>5.4 Legal Aspects of Cryptocurrencies</i>	44
<i>5.5 Economic Aspect of Cryptocurrencies</i>	45
5.5.1 Advantages of cryptocurrencies	47
5.5.2 Disadvantages of cryptocurrencies	48
<i>6. APPLICATION OF CRYPTOCURRENCIES AND BLOCKCHAIN IN LOGISTICS</i>	51
<i>6.1 Ways to apply blockchain in the logistics sector</i>	52
<i>6.2 Advantages of Applying Blockchain in Logistics</i>	55

7. RISK MANAGEMENT in LOGISTICS and the Supply Chain	57
7.1 Identifying risks	58
7.2 Risk Classification	60
7.2.1 Ishikawa diagram.....	71
7.2.2 Application Procedure	72
7.3 Risk Analysis of Cryptocurrency's Impact on Logistics Processes – Determining Risks with with the greatest impact.....	78
7.3.1 FMEA method	78
7.3.2 Application Procedure	78
7.4 Defining and implementing measures to eliminate or reduce deviations/risks with a high level of risk	91
7.4.1 Measures to mitigate/eliminate the impact of the risk.....	94
7.5 Calculation of residual risk after implementing the planned measures	95
7.6 Defining the method for monitoring and verifying risks	97
Conclusion	92
References	94
LIST OF FIGURES	100
LIST OF TABLES.....	101

INTRODUCTION

In January 2009, as the world was recovering from a financial collapse of unprecedented proportions, an anonymous programmer/s introduced a new currency to the public - Bitcoin. This form of electronic money, although from an unknown source, soon gained popularity among the population as an alternative to traditional banking institutions and their system of operation, which they had begun to widely distrust in the post-crisis period.

Cryptocurrency essentially represents any form of currency that exists digitally or virtually and uses cryptography to secure transactions, control the creation of additional currency units, and verify currency transfers (Greenberg, 2011). This new type of virtual money is presented as a sustainable currency with broad applicability and a decentralized nature, which offers the freedom to issue new currency units and conduct transactions directly from the buyer to the seller without the need for a third party. One of the biggest problems that must be prevented in transactions is the "double-spending" problem, a manipulation technique that results in the same amount of money being spent from the same source twice. The established solution provided by the traditional banking system is the existence of a trusted third party, i.e., institutions that track and maintain records of account balances and the validity of transactions. However, the drawback of this method is that it involves authority over all user funds and their personal data. With the advent of cryptocurrencies, a way was found to avoid the aforementioned data sharing. This, combined with the shorter time required to complete payments at significantly lower costs and the possibility of getting rich with relatively small investments, led to rapid popularization and the gain of numerous supporters, especially among younger people, from the very first days of its use.

While some dismissed the new currencies as a trend that wouldn't catch on, others believed that the blockchain technology brought by cryptocurrencies was truly the future of money and commerce, destined to completely replace cash. Various examples have emerged of users who have profited unexpectedly by using the mysterious Blockchain technology, thus encouraging numerous other users to invest in it.

Today, there are more than 7,000 types of cryptocurrencies, with the number of users who own or invest in them exceeding 320 million worldwide. On average, around 280,000 crypto transactions are recorded daily, with a market capitalization that has reached \$2.65 trillion.⁽¹⁾ In 2014, companies such as Overstock accepted Bitcoin as a payment method for online purchases. Soon after, well-known companies such as Master Card, Pavilion Hotels, AKSA Insurance, Starbucks, Visa, PayPal, and the like followed suit.

The emergence of cryptocurrencies provided a new way of doing business based on a new, innovative technology. Their market brought many new customers and made international trade faster and easier. In this way, entrepreneurs around the world have been encouraged to pursue international markets instead of strictly adhering to national frameworks. This has enabled merchants to build relationships and foster trust with markets that were never before accessible, which is particularly significant for developing countries.

In addition to providing a new investment opportunity, Blockchain technology is slowly making its way into all modern financial flows. The ability to permanently record transactions in a "public ledger" has changed the way business processes function in many different industries, especially in logistics.

The supply chain is tied to the complex processes of creating and distributing goods. Depending on the product, the supply chain can include many stages, multiple geographic locations, several accounts and payments, a large number of individuals, companies, entities, and modes of transport, which can cause the procurement process to be delayed for several months. Due to the complexity and lack of transparency of traditional supply chains, it is of great interest to the stakeholders involved in the process

¹ https://ycharts.com/indicators/bitcoin_transactions_per_day#:~:text=Bitcoin%20Transactions%20Per%20Day%20is,2.04%25%20from%20one%20year%20ago.

logistics to introduce into their business and develop blockchain technology to improve supply chain logistics processes, making them more sustainable.

"Blockchain-driven innovations in the supply chain will have the potential to deliver enormous business value by increasing supply chain transparency, reducing risks, and improving efficiency and overall supply chain management"^[64]. From simplifying supply chain management and managing multiple cloud platforms to tracking various products from point A to point B, blockchain and cryptocurrencies go hand in hand to ensure the secure and efficient movement of goods through the supply chain. However, how does its existence and use relate to the transportation and supply chain markets?

Blockchain technology can be used for any data exchange, whether for contracts, shipment tracking, or financial exchanges (payments). Every action is recorded in a block, and the data is distributed across many nodes (computers), making the system transparent. Each block is linked to the preceding and following block, which makes the system more secure. Blockchain can increase the efficiency and transparency of the supply chain and positively impact all logistical processes, from storage to delivery and payment. In addition to the increased transparency and security achieved through Blockchain, it is also possible to speed up the physical flow of goods^[72]. Tracking goods through Blockchain can improve the decision-making process with the end result being a more satisfying service for the end-user. Blockchain technology has the potential to create new logistics services as well as new business models. It also offers companies the ability to customize the technology to their needs. As a relatively new technology, it is designed to achieve decentralization, real-time peer-to-peer operations, anonymity, transparency, immutability, and integrity in a widely applicable way. However, there are still vulnerabilities and challenges associated with this technology that should not be ignored^[73].

The future of the supply chain and logistics services inevitably includes the application of cryptocurrencies and blockchain technology. The question arises: what risks come with implementing these innovations in logistics processes, and how should organizations approach them to best leverage the new opportunities?

It is important to determine how to ensure that participants in the supply chain achieve the maximum positive results from implementing this innovation, at a time when it is impossible to prevent it. This topic forms the basis of this paper, in which the concept of cryptocurrencies and all the necessary elements for understanding how this system functions are presented and analyzed in detail, with an emphasis on their application in logistics processes, i.e., the supply chain, which are also defined in detail.

Throughout the paper, the Ishikawa method will be used to graphically represent all risks that could potentially lead to changes in logistics processes during the implementation of cryptocurrencies and blockchain. The FMEA method will be used to identify and analyze risks to determine which ones can have the greatest impact on the implementation process. This will allow for the design of adequate measures to successfully establish a new way of operating logistics processes while mitigating or eliminating the negative aspects of crypto technology.

1. THE CONCEPT OF LOGISTICS

The origin of the word "logistics" is multifaceted. It comes from the Greek word "logistiko," which translates to "skillful and experienced in assessing all the elements necessary for making optimal strategic and tactical decisions," and "logos," which represents the science of the elementary principles and forms of correct thinking and judgment, as well as the French word "loger," meaning to live, to lodge overnight, to settle down...²



Figure 1 – Different participants in the logistics chain ³

However, the concept of logistics itself did not have a proper name for a long time, and its content was often equated with the transportation and warehousing functions which, viewed from any point in the distribution process, have a two-way orientation. A particular interest in managing logistics activities has led to various interpretations of the concept of logistics itself.

As a result, confusion arose regarding the term, as numerous terms were used to denote the process of the physical movement of goods from suppliers, through manufacturers and intermediaries, to the end consumer.

Among the significant terms used to denote the different components of the process of the physical movement of goods are:

- physical distribution,
- physical distribution management,
- industrial logistics (industrial logistics),
- business logistics,
- marketing logistics,
- materials handling,
- strategic logistics management,
- supply chain management, etc.

In our professional literature, there are different understandings and interpretations of the term logistics. In principle, we can say that the concept of logistics has two meanings.

First, this term denotes a scientific discipline, and on the other hand, a specific function within a company that is intended to contribute to the achievement of business goals. More recently, the term integral logistics has been increasingly used.

A number of authors associate the concept of logistics with the time of Napoleon, when this term was used as a synonym for the organization of army supply and rear support for the front. In the technical sciences, logistics is defined as the discipline that studies the operation, functioning, and working conditions of technical systems. It provides integral support to the system, supplying consumables, fuel, and spare parts. In this way, the term enters the field of economics. From the above, it can be concluded that the implementation of logistics in various forms of activity has also led to its differentiation.

In this sense, in addition to military and technical logistics, business logistics has also developed, which manages the flow of raw materials from suppliers to manufacturers, as well as the movement of finished products from manufacturers to consumers.

Given the various interpretations of logistics over time and by different authors, for the purposes of this paper, the definition in accordance with the directives and recommendations of the Council of Europe is adopted, which reads: "*Logistics*

²Prof. Dr. Sanda Renko, Zagreb 2011, *Business Logistics*;

³<https://www.barkodskeneri.com/2019/08/26/gdje-uciti-o-logistici/>;

represents the management of the flows of goods and raw materials, the processes of producing finished products, and related information from the point of origin to the point of ultimate use in accordance with customer needs. In a broader sense, logistics includes the return and disposal of waste substances."²

- According to a functional perspective that emphasizes the various functions of logistics within a company, in accordance with the division of labor. This division represents the most comprehensive approach to the study of intralogistics, as it connects and includes all material flow activities in the reproduction process, from the receipt of materials through production consumption to finished products and their delivery to customers.

Certainly, individual parts and activities of some logistics areas overlap, for example, production logistics includes transportation logistics, distribution logistics includes picking logistics, and packaging logistics includes spare parts logistics, etc. Depending on the chosen company, its complexity, process structure, and the required level of detail, the boundaries of the analysis are defined.

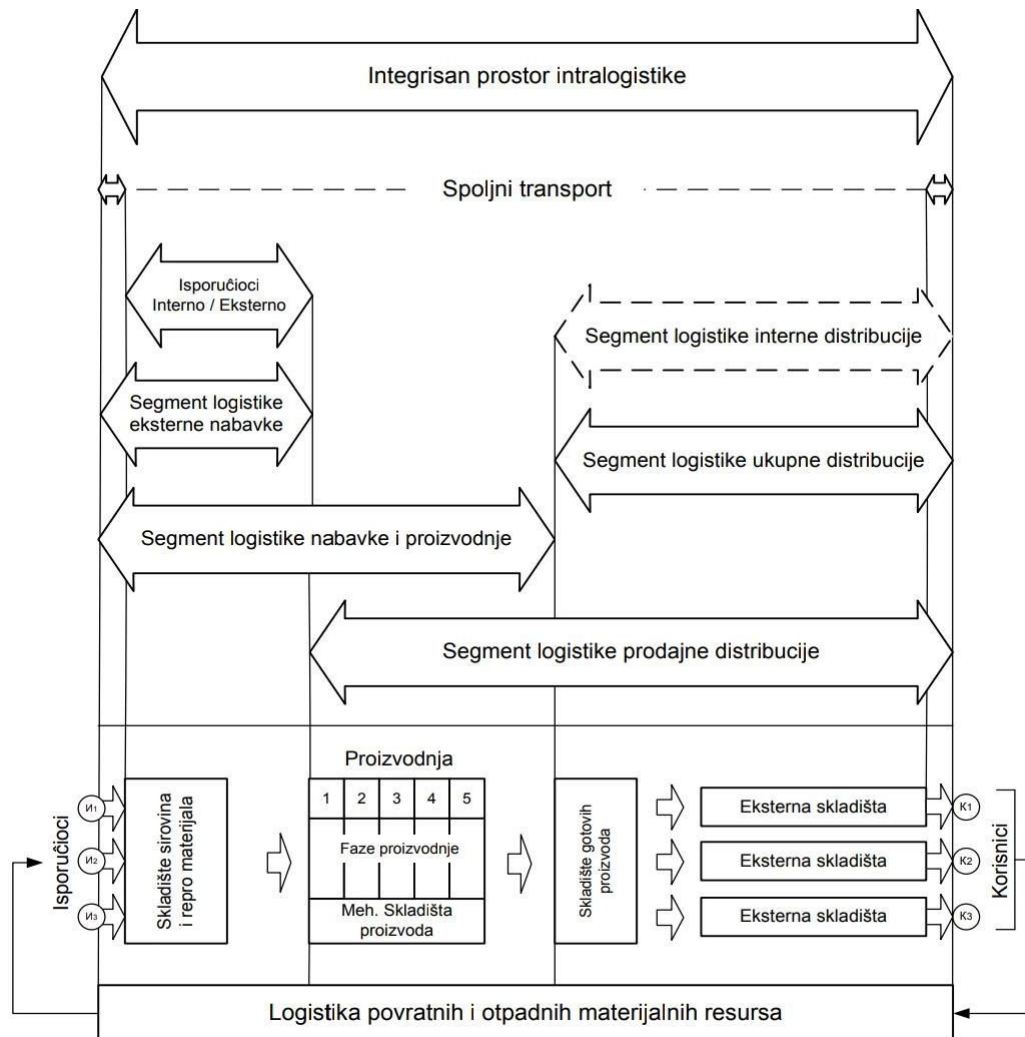


Figure 2 – Functional Decomposition of Logistics ¹⁸

- [1] **Purchasing logistics** - with material flows from the moment of arrival at the company to its receiving warehouses or directly to production sites within the plant
- [2] **Production Logistics** - with material flows from receiving warehouses to initial workstations where they are processed, fabricated, or consumed, transforming into a semi-finished or finished product, and with finished product flows from final workstations to the finished goods warehouse
- [3] **Distribution logistics** - with product flows from the finished goods warehouse to loading onto carriers' vehicles, end-user vehicles, or the company's own fleet of vehicles

- [4] **Logistics of return and waste material resources** - which includes activities that support the occasional movement of products or materials back to manufacturers or suppliers.⁴

2. LOGISTICS SYSTEMS AND LOGISTICS PROCESSES

To carry out logistical activities, various systems have been developed, which are referred to in the literature and in operational practice as *logistics systems*, *logistical systems*, or *logistical support systems*. These terms are equated, which is essentially correct, as the definition of the term depends on the system to which it belongs and the author who defines it.

There are different concepts for defining a logistics system, as well as numerous definitions, whose approaches are fundamentally based on general systems theory and cybernetics⁵.

In the literature, "a *logistics system is defined as an adaptive system with feedback loops, intended to perform certain logistics functions and operations, which usually consists of several subsystems and has strong ties to the environment.*" From a business perspective, "a *logistics system is defined as a system whose elements are flows over which logistics operations are carried out. In it, the elements are interconnected based on common goals and success criteria*"⁶

However, an increasingly common, more general methodological approach to defining a logistics system in business systems is used: "A *logistics system is a complex, organizational, and economically integrated system, which consists of elements that are interconnected by processes for managing flows, and whose functioning is determined by the specific goals of the business organization. The logistics system is based on the flow of processes, which are carried out by qualified personnel and utilize various technical means. Infrastructure facilities are also included in the logistics system*"⁽¹⁹⁾.

"Logistics systems are systems for the spatio-temporal transformation of goods, and the processes that take place in them are logistics processes"⁷.

It can also be said that a "logistics system is a set of elements of technical, technological, organizational, economic, and legal nature with the goal of optimizing the flows of materials, goods, information, energy, and people in a specific area in order to achieve the greatest economic effects"⁽⁸⁾.

An enterprise is a business system composed of several integrally connected subsystems: procurement, production, sales (delivery), and finance. This system is based on the flows of materials, energy, information, and value.

The logistics system is, in essence, a subsystem of this broader system and as such must be aligned with the other subsystems and subordinate to the goal of the broader system's operation (**Figure 3**). Thus, the goal of any logistics system is to support the core activity of the system to which it belongs. Therefore, a logistics system is:

- an integral part of the business system,
- dynamic and open,
- in constant harmony and interaction with other business subsystems.

It operates through various forms of technological processes and services. The system is hierarchically structured according to its competencies, and at the same time, it is conceptually spatially distributed, where each of its organizational units is self-contained and universal under given conditions. In any system, including a logistical one, it is necessary to identify its basic (constituent) parts, their interrelationships, the principles on which they operate, and the instruments used to regulate the relationships between the parts. On the other

⁴Prof. Dr. Branko Davidović, University of Technology Kragujevac 2012, *Intralogistics - Internal Transport*;

⁵ Dr. Petar Stanojević, Dr. Vasilije Mišković, Belgrade 2000., *Fundamentals of the Methodology for Designing Organizational-Technological Systems*;

⁶ Anikin B. A., Rodkina T. A., Russia 2008., *Logistics* featured in Baranowski S., Busko E., Polish Academy of Sciences – Logistics module 2014., *Directions for the improvement of logistics systems*;

⁷ Pfohl, H.Ch., Springer Verlag, Berlin 1985., *Logistiksysteme*;

⁸ Krpan Lj., Furjan M., Maršanić R., 2014., *Tehnički glasnik Vol. 8., The Potential of Reverse Logistics in Retail*;

Pages, the logistics system is a relatively independent entity, i.e., a subsystem of a larger economic system (which also contains other subsystems).

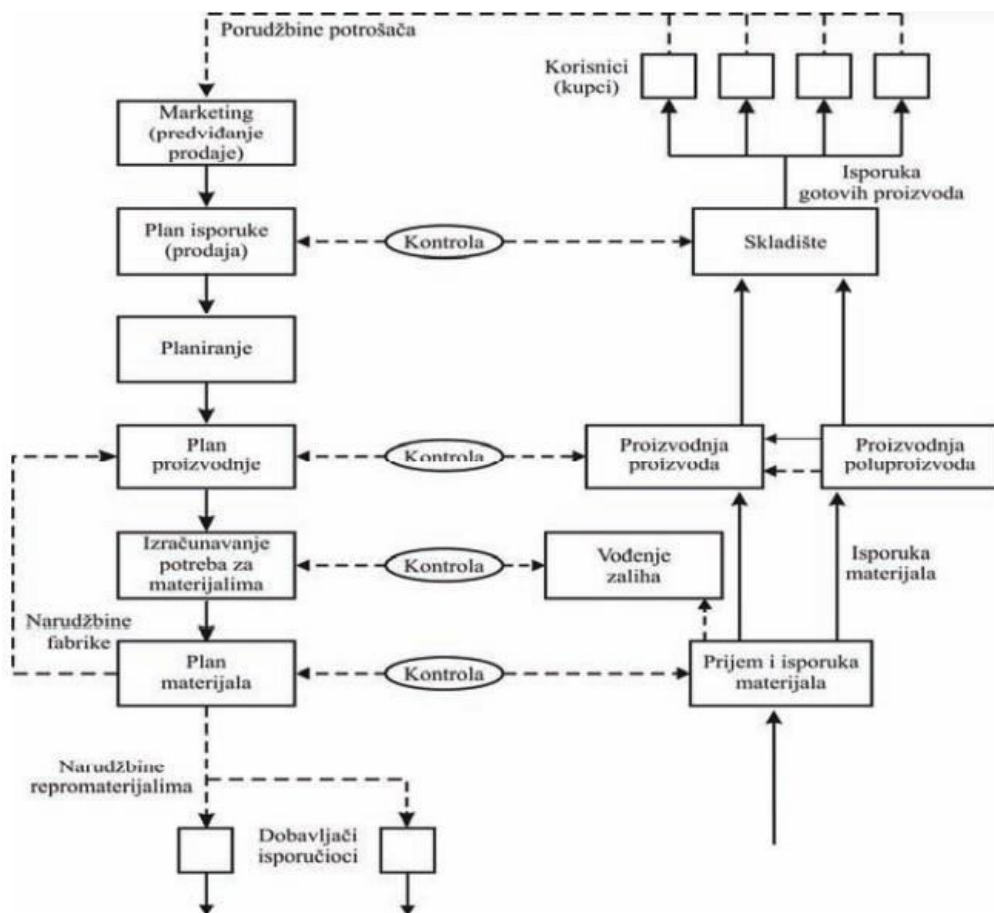


Figure 3 – Schematic representation of the company's logistics system⁹

The system's structure is composed of various system elements and the relationships between them, which can be represented as a network.¹⁰ Accordingly, *the structure of a logistics system*, or logistics network, is composed of certain elements. These are interconnected logistics locations organized into a network for the exchange of products and information.

A logistics location, as an elementary part of the logistics network (system), can be defined as a place where material and immaterial goods are created, upon a prior order or request, using materials and resources such as people, space, machines, and equipment. Multiple logistics sites located in one area can be called a logistics hub or a logistics center.

Multiple logistics centers that create similar effects or cover similar parts of the logistics chain can be distinguished and identified as a logistics organizational unit.

A business entity's logistics network is part of a global logistics network characterized by a multitude of different owners and users (participants). From a distribution standpoint, the network takes the form of a chain. In essence, the movement (transportation) and storage processes, whose understanding of interconnections is key to understanding the logistics system, are represented by this network. Objects move through the network and are held at its nodes, from which they are dispatched on routes to various destinations. In addition to goods, energy, information, and people also move through the network.

⁹Prof. Dr. Dušan Ragodić, Belgrade 2014, LOGISTICS – Fourth Supplemented Edition,

¹⁰Russell L. Ackoff, 1994., *Systems thinking and thinking systems*;

The network carries the logistical costs incurred from the execution of tasks at the logistics location and the mutual exchange ¹¹.

According to the above understanding of the network, the **basic structures of** logistics systems can be distinguished:

- **Single-stage logistics systems** - space and time are bridged by a direct flow of goods between the delivery point and the receiving point. At the delivery point, goods are prepared, and at the receiving point, they are used
- **Multi-stage logistics systems** - the flow between the delivery and receiving points is interrupted at least once, with the task of regrouping goods into smaller quantity units or concentrating them into larger delivery units. This is conditioned by the needs of the recipient at the receiving point.
- **Combined logistics systems** - in which elements of both single-stage and multi-stage systems appear simultaneously ¹²

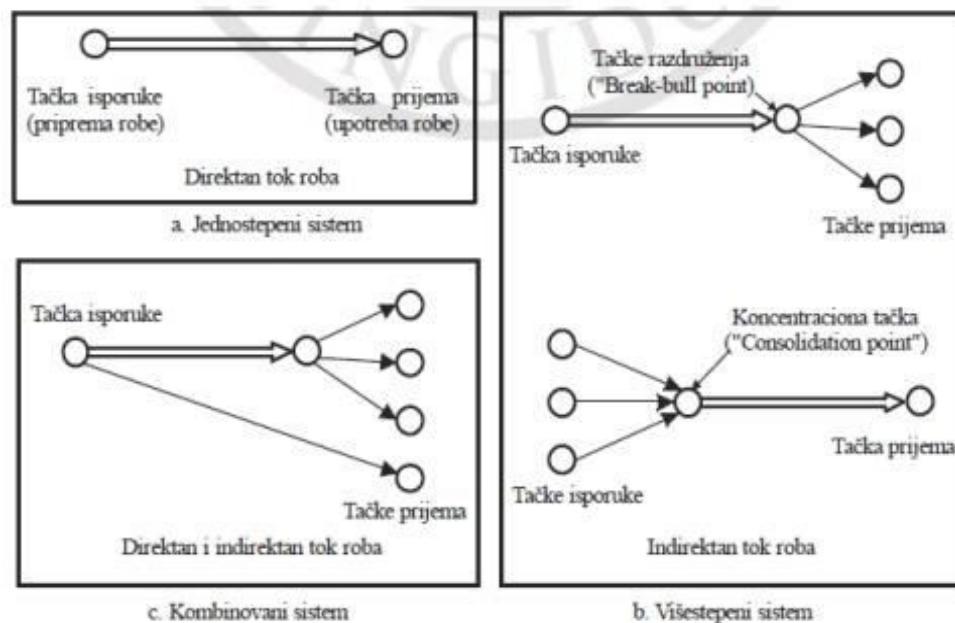


Figure 4 – The processes of movement (transportation) and retention (storage) in logistics systems of different structures ¹³

This system consists of the following components:

- **Inputs – incoming components** that are processed, distributed, and controlled to be ready for sale to intermediaries or end consumers. These are:
 - raw materials and other materials
 - tools and other means of production
 - energy
 - assets (capital)
 - information
 - personnel, and their knowledge and experience
- **Outputs – the output components** are realized through logistical activities. They include competitive advantages for the company that stem from a marketing orientation, manufacturing effectiveness and efficiency, time and space utility, and efficient movement to the consumer, increasing the company's assets. All other outputs encompass the mix of logistics services provided.

The subject of inbound logistics activities is:

¹¹ Gudehus T., Kotzab H., 2012., *Comprehensive logistics*;

¹² Ballou R.H., Ohio 1997., *Business Logistics Management*;

¹³ Dr. Zdenko Segetlija, Faculty of Economics, Osijek 2002., *Introduction to Business Logistics*;

- finished products
 - services
 - information
 - knowledge
 - means
 - energy
 - spare parts
 - scrap products and repair parts
 - consumable parts and materials and
 - sales equipment
- **The transformation process** – Spatiotemporal transformation is **a core function** of logistics systems. Fulfilling this includes the functions of changing the quantity and type of goods, as well as facilitating the transformation of goods.
It is performed:
 - by classical change
 - by relocation
 - storing until a specific time and
 - increasing

services¹⁴ These functions are performed in the processes:

- transportation, transshipment, and storage where the processes of the flow of goods are essential,
- packaging and signing where support processes for product flow are essential
- delivery and order processing where information flow processes are important¹⁵.

Logistics systems are inherently complex and dynamic systems characterized by multidimensionality and the presence of a large number of diverse elements, material and immaterial flows, which are subject to constant changes and modifications, the complexity of interactions between individual elements, and the complexity of the functions the system performs, complex organizational management, and the influence of a large number of environmental factors in which the system is located. They are also characterized by a multitude of specific constraints, elements, and processes that interact simultaneously to achieve a goal.

The fundamental properties of any logistics system are:

CRITERION:	PROPERTY:
Relationship to nature	OPEN
Time dependence	DYNAMIC TIME-DEPENDENT
Linearity	NONLINEAR
Structure	SPATIALLY BRANCHED
Type of relationship	MATERIAL
Purpose of the system	PLACE OR TIME SHIFT

Table 1 – Characteristics of a logistics system

¹⁴ Dr. Dušan B Regodić, 2019., Development and Integration of Logistics in Organizations;

¹⁵ Dr. Zdenko Segetlija, Dr. Maja Lamza-Maronić., Faculty of Economics, Osijek 2002., Distribution, Logistics, Informatics;

Elements of a lower-level system are subordinate to elements at a higher level. Such systems are, in essence "artificial" systems, are significantly influenced by management decisions and the feedback loops created. The coordinated functioning of all system parts in a way that achieves the greatest effects with the least energy input guarantees the effectiveness of logistics systems.¹⁶

2.1. Subsystems of the logistics system

By differentiating the logistics system, its subsystems, their interaction, the structure of technological production elements, processes, and activities, are examined, environmental influences, and critical logistical capabilities that will represent relevant constraints in the functioning of the enterprise, which is characterized by multifunctionality and multidirectionality in its transformation processes, from material procurement, through production material flows, to consumption. The goal is to identify the key elements that achieve spatial linkage, as well as the quantitative, temporal, and assortment equalization of production and consumption demands.

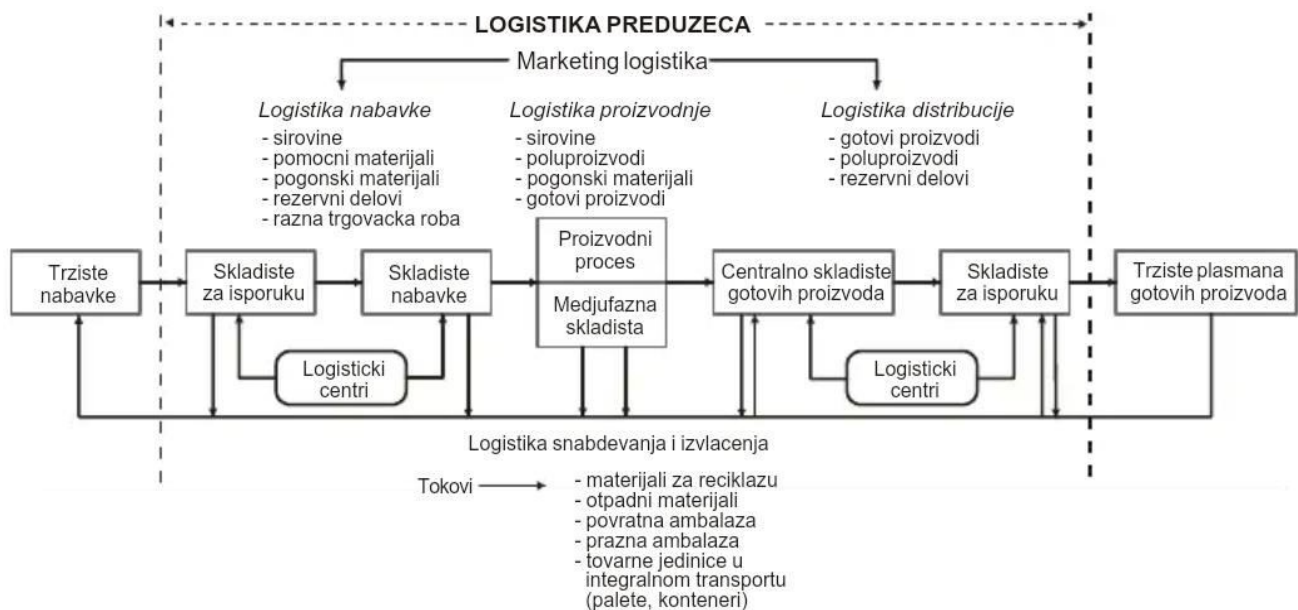


Figure 5 – Subsystems of the logistics system of an industrial enterprise¹⁷

From a functional perspective of logistics division, the subsystems that constitute it are as follows:

1. **Purchasing Logistics** - the spatial and temporal assurance of the delivery of goods to the site of their processing and manufacturing.
2. **Production Logistics** - flows of materials and information from the material warehouse to the production process and the finished goods warehouse
3. **Distribution logistics** - delivery of goods to the sales market, i.e., to trade units located between the sales market and the place of production
4. **Logistics of return and waste material resources** - return of goods to the supplier, return of returnable packaging, waste materials, as well as goods that can be reused¹⁸

According to some authors, **Warehouse Logistics** can be singled out. The aforementioned logistics subsystems will be individually explained in more detail in the following **Chapter 2**.

Logistics connects or networks business functions within business systems. In modern companies, logistics is integrated into business functions. procurement, production, distribution, and

¹⁶Dr. Petar Stanojević, Dr. Vasilije Mišković, Belgrade 2000, *Fundamentals of the Methodology for Designing Organizational-Technological Systems*;

¹⁷ Prof. Dr. Dušan Ragodić, Belgrade 2014., *LOGISTICS – Fourth Supplemented Edition*,

¹⁸ Assoc. Prof. Dr. Goran Avlijaš, 2019., *Integrated Supply Chain*;

warehousing (**Figure 6**). The tasks of each individual logistics subsystem overlap. They all involve the entirety of logistical tasks and procedures in the preparation and execution of each individual business function.

A systems approach to observing logistics systems makes it possible to identify the interdependencies of system elements that would otherwise be very difficult or impossible to notice in the usual way. Without applying this approach, the logistics system could not be viewed as a whole, i.e., it would be impossible to understand the relationships between the individual elements of that system.

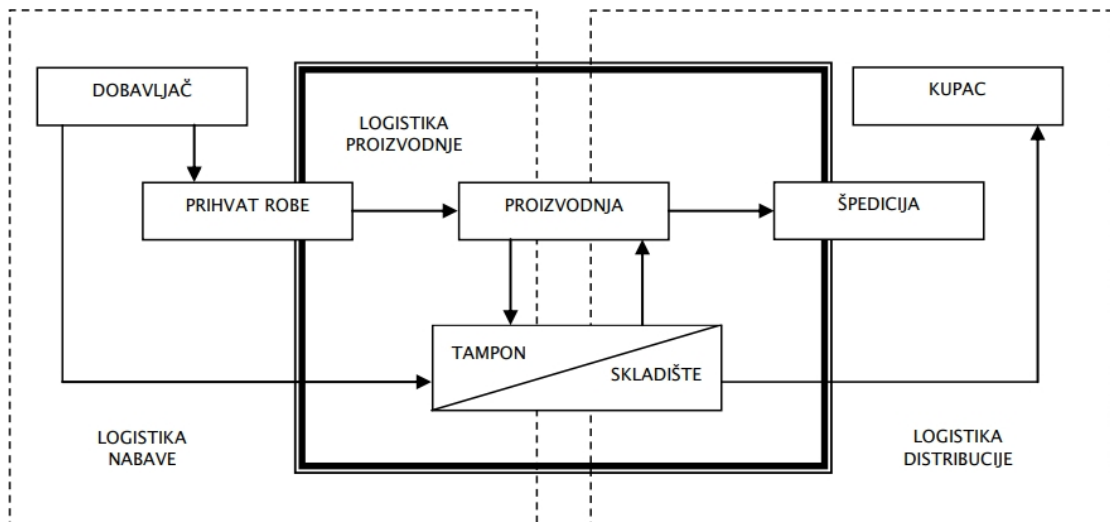


Figure 6 – Logistics within the business system ¹⁹

The logistics subsystems shown belong to an industrial enterprise, while for a trading company, production logistics is excluded, and for a service-based company, there is no distribution logistics either. Namely, in a functional sense, the logistics system is always a subsystem of the enterprise's business system, just as other subsystems are: sales, research and development, financing, and information. ²⁰

2.1.1. Production Logistics

Production logistics involves the planning, execution, control, and management of the flow of materials and information through the production process, and within and between production facilities. Simply put, production logistics provides the production process with everything necessary to smoothly produce quality products within the specified time frame and at the lowest possible cost.

It is responsible for activities such as procuring raw materials, distribution within the production process, and in some cases, delivery to customers. Therefore, logistics covers many areas within the production environment, such as: procurement, goods receipt, work-in-process, inventory control, finished goods storage, and distribution to customers.

This involves several techniques such as Just-In-Time (JIT) and Material Requirements Planning (MRP).

Production logistics includes:

- Consideration: to produce the products or to procure them from a supplier - The most important question in production logistics. This question is actually a matter of cost-effectiveness. To make the correct decision, the purchase price and the cost of producing the product in-house are considered.

¹⁹ Buntak K., Šuljagić N., Tehnički glasnik, Vol. 8 No. 4, 2014., *Economics of Logistical Functions in the Company*;

²⁰ Prof. Dr. Dušan Ragodić, Belgrade 2014, *LOGISTICS – Fourth Supplemented Edition*,

production. If the needs are smaller, then in most cases the decision is made in favor of procurement.

- Structuring Production According to Logistical Aspects
- Production Planning and Management
- Shaping the physical and information flow through production

According to (**Figure 7**), the process begins with procurement, through the warehouse, to production preparation. In the production process (PR), quality control (Q) is performed with the support of information technologies (IT). Therefore, logistics is composed of several distinct processes that contribute to the optimization of the production process.²⁹

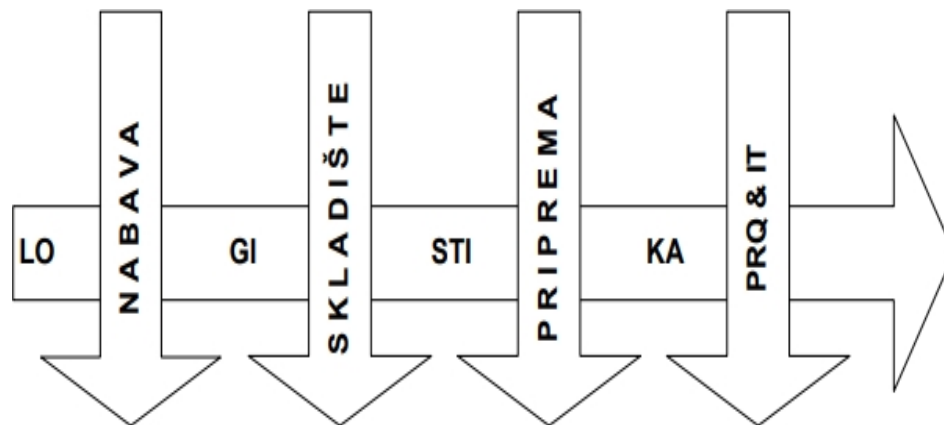


Figure 7 - The Place of Logistics in Process Manufacturing²¹

The main goal of production logistics is continuous production in accordance with contracted quantities while minimizing the inventory of work-in-process and finished goods.

Individual objectives:

- ❖ Operational planning and management in the shortest possible time
- ❖ Defect-free production, without rework or repairs
- ❖ Employees who can work excellently in multiple positions (multi-skilling) and flexibility regarding customer requirements
- ❖ Constant production readiness through the integration of internal material and product handling
- ❖ Minimal costs for internal transport, handling, and storage processes
- ❖ Minimizing production-related capital²²

Process production organization is a system with a large number of operations that are a source of high costs. This stems from a system that begins with procurement, continues with material control and preparation, and ends with the final allocation to individual operations according to a specific program. All of this is controlled to achieve the quality of not only the product but the entire process, all while keeping costs optimal. The consideration of production or process logistics begins with an analysis of the process production itself.

Depending on the quantity of the product and the frequency of repeating the same method of production, the following are distinguished (**Figure 8**):

²¹Lacković Z., Faculty of Economics, Osijek 2008, *Fundamental Elements of Business Logistics in Production*;

²²Dr. Zdenko Segetlija, Dr. Maja Lamza-Maronić., Faculty of Economics, Osijek 2002., *Distribution, Logistics, Informatics*;

- **Mass production** – in which logistics is tasked with supplying production sites with input materials over long time periods. For production as such to be able to function a high degree of mechanization must exist.
- **Batch production** – where the biggest problem is the batch size. Logistics is primarily responsible for the resulting inventory by ensuring the availability of batches of specific types of materials and (semi)finished products.
- **Individual production** - is almost always made to order, which requires logistics to be flexible and able to supply production workstations with different materials at all times.
- **Process y** - a highly mechanized and automated system that involves large capital investments. By its nature, it is very efficient as it operates 24 hours a day, all year round except in maintenance time and thus results in a high production volume²³



Figure 8 – Comparison of characteristics of different types of production systems²⁴

Quality management affects the overall efficiency of the enterprise and is an integral part of the production logistics system. Profit size directly depends on the revenue from the sale of products or services and on production costs, so the study of revenue and costs, as key factors of business success, is a very significant part of the economics of production logistics.²³

2.1.2. Procurement Logistics

Purchasing is a function that, in a narrow sense, is responsible for the continuous supply of materials, equipment, services, and energy necessary to achieve the goals of the business system, as well as their timely delivery to the appropriate place, at the appropriate price. In a broader sense, its subject matter includes all of the company's inputs (raw materials, materials, services, personnel, information, etc.) that it procures from the market. The procurement problem is reduced to supplying production with material resources, on the one hand, and linking the material flows in the preparatory phase with the material flows in the technological and final phase of the reproduction cycle, on the other hand.

²³ Buntak K., Šuljagić N., Technical Gazette, Vol. 9, 2015., *Production Logistics Economics*;

²⁴ Fraidoon M., English 1998., *Engineering Management*;

Individual objectives:

- ❖ Delivery of materials and necessary parts in accordance with the contract for the ordered products
- ❖ Delivery of materials or parts directly from the supplier (Just In Time) and from its own warehouse at the lowest preparation costs
- ❖ Use of ready-made components for production (reduces production time)
- ❖ Minimizing internal and external transportation related to procurement
- ❖ Minimizing the inventory of raw materials and parts required for production
- ❖ Agreeing on short delivery times with suppliers and creating the ability to respond quickly to customer requests²⁵
- ❖ The "zero defects" principle in the preparation of raw materials and parts required for production through strict quality control upon receipt of goods²⁶

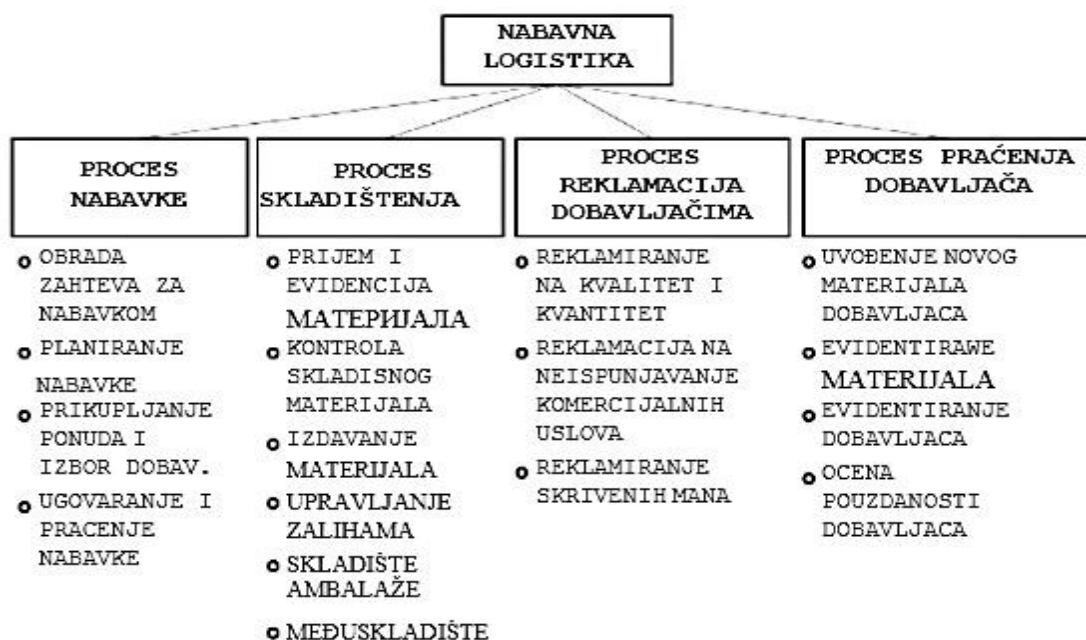


Figure 9 – Structure of Procurement Logistics Activities

The planning, organizing, and execution of the activities listed in **Figure 9**, using appropriate economic methods to reduce procurement costs and decrease average working capital, constitutes the content of procurement logistics. Understood in this way, procurement logistics intertwines with, and partially overlaps, some of the classic functions within a company, primarily with the financial and human resources functions.²⁷

Tasks of procurement logistics:

- 1) Securing the purchase of material resources at the lowest prices and with the highest quality
- 2) Ensuring the timely delivery of materials for production needs
- 3) Adhering to the terms of purchase for raw materials and components;
- 4) Determining optimal inventory levels;
- 5) Aligning inventory levels with production needs;

²⁵Prof. Dr. Sanda Renko, Zagreb 2011, *Business Logistics*;

²⁶ Buntak K., Šuljagić N., Tehnički glasnik, Vol. 8 No. 4, 2014., *Economics of Logistical Functions in a Company*;

²⁷Dr. Zdenko Segetlija, Dr. Maja Lamza-Maronić., Faculty of Economics, Osijek 2002., *Distribution, Logistics, Informatics.*;

- 6) Quality control of raw materials and components;
- 7) Ensuring the storage of material resources (finished products) at minimal cost.²⁸

Purchasing logistics engages existing procurement capacities, creating the necessary flows of materials and (semi-)finished products to make the purchased goods physically available to the company. This external material preparation refers to raw materials, auxiliary and operating materials, parts, and merchandise, and is carried out in 3 ways, as follows:

- *Individual Purchase on Demand* - the advantage for logistics is that small quantities are stored, so capital tie-up and inventory costs are low. Problems that arise can include long delivery times and underutilization of production equipment.
- *Procurement with inventory holding* - separates internal from external material preparation, and the company is less sensitive to changes and uncertainties in supplier delivery.
- *Make-to-order procurement* - possible with reliable suppliers and their close cooperation with customers. Inventory is held only in the form of low safety stocks, which significantly reduces the cost of capital tied up in inventory ma.²⁹

2.1.3. Distribution Logistics

One of the many definitions of distribution is that it represents the phase that follows the production of goods from the moment of sale to the delivery to consumers, facilitating the selection, purchase, and use of goods.³⁰

The primary role of distribution and product sales logistics is to link production with the consumption of final products, positioning itself between the two. (**Figure 10**) As such, it represents the final point, which is of great importance because it is through this point that sales and end-users are reached.

Distribution is a complex system that must align set objectives and balance a series of spatial, temporal, quantitative, and structural imbalances.

Its task is to maximize the alignment of supply and demand for products and services in the market (between production, trade, and the end consumer).³¹

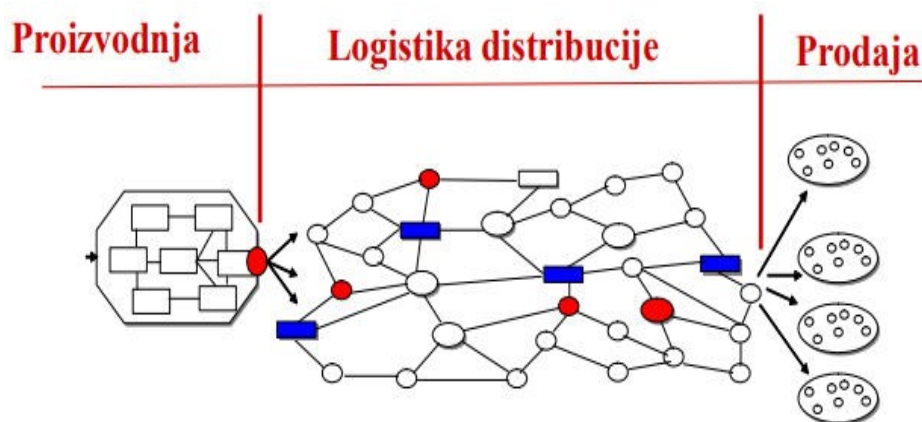


Figure 10 – The place of distribution logistics in the supply chain y³²

²⁸Prof. Dr. Dušan Ragodić, Belgrade 2014, LOGISTICS – Fourth Supplemented Edition,

²⁹Dr. Zdenko Segetlija, Faculty of Economics, Osijek 2002, Introduction to Business Logistics;

³⁰ Savremena administracija, Belgrade 1975., Economic Lexicon;

³¹Buntak K., Šuljagić N., Technical Gazette, Vol. 8 No. 4, 2014, Economics of Logistical Functions in the Company;

³²Prof. Dr. Dr. sc. Kata Ivić, Faculty of Economics in Osijek 2020, *Distribution and Trade Logistics*;

The main goal of distribution logistics: continuous sales readiness and maximum quality of the logistics service, while simultaneously minimizing the costs of finished goods inventory, packaging, transportation, and handling.

Individual objectives:

- Delivery in accordance with customer preferences (in the appropriate quantity, quality, and price, at the right time)
- Quality storage of finished products at minimal cost
- Minimizing finished goods inventory
- Minimizing packaging costs and the use of recyclable materials
- Minimizing external transportation costs
- High level of service
- Minimizing the costs of commercial activities related to sales³³

Depending on the relationship between the manufacturer and the distributor, or the client (seller), three characteristic types of distribution can be distinguished:

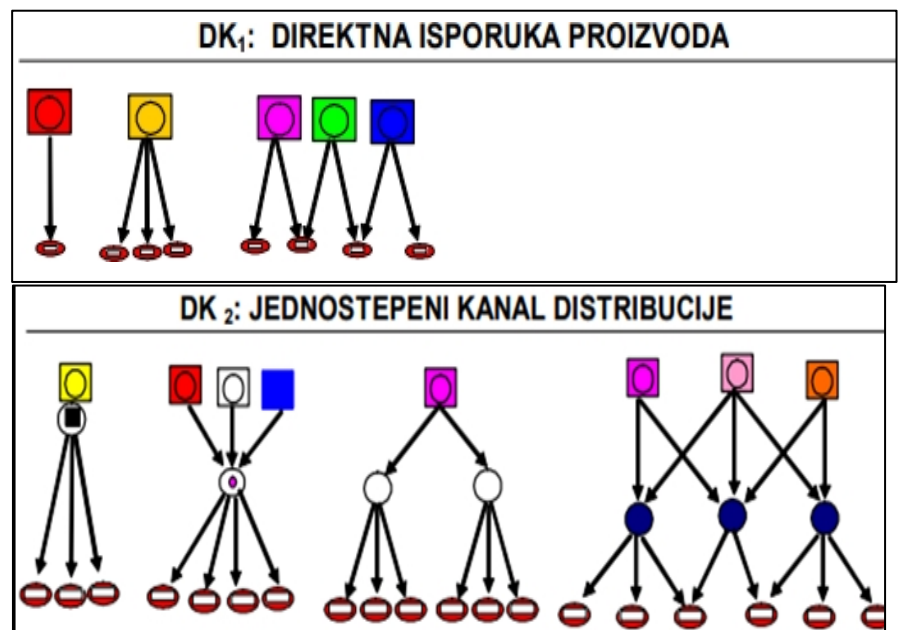
- **Open distribution** - means that the manufacturer sells its products to all retailers that meet the sales requirements. Likewise, retailers stock different brands of the same product.
- **Exclusive distribution** - involves a special agreement between the manufacturer and a retailer, whereby the manufacturer grants the retailer the exclusive right to distribute the product in a specific geographic territory. Likewise, the retailer agrees to distribute only this brand of the product.
- **Selective distribution** - means that the number of distributors is limited, respecting territorial exclusivity, and is dependent on the market's demands and needs, the brand image, and the required quality of delivery service³²

The sale, marketing, and distribution of the product is carried out through specific distribution channels and chains.

A **distribution channel** is an organized set of entities whose attributes can be spatial, temporal, quantitative, capacity-related, cost-related, qualitative, technological, organizational, and informational.

The structural elements of the distribution channel are:

- manufacturing, trading, or service companies,
- transportation, freight forwarding,



³³Prof. Dr. Sanda Renko, Zagreb 2011, *Business Logistics*;

warehousing, transshipment companies,

- distribution centers,
- goods terminals,
- freight transport centers,
- transportation infrastructure, facilities, and equipment.

The distribution network, through which specific distribution chains operate, is composed of the aforementioned elements in various combinations, hierarchical relationships, and technological, organizational, economic, informational, and cooperative ties.

The distribution chain includes a series of logistics processes and activities: ordering, packing, unitization, loading, transportation, unloading, warehousing, picking, inspection, and the preparation of accompanying documentation.

In practice, various distribution channels and chains exist. In terms of hierarchical structure and degree of distribution, distribution channels can be classified into five categories³⁴. (Figure 11)

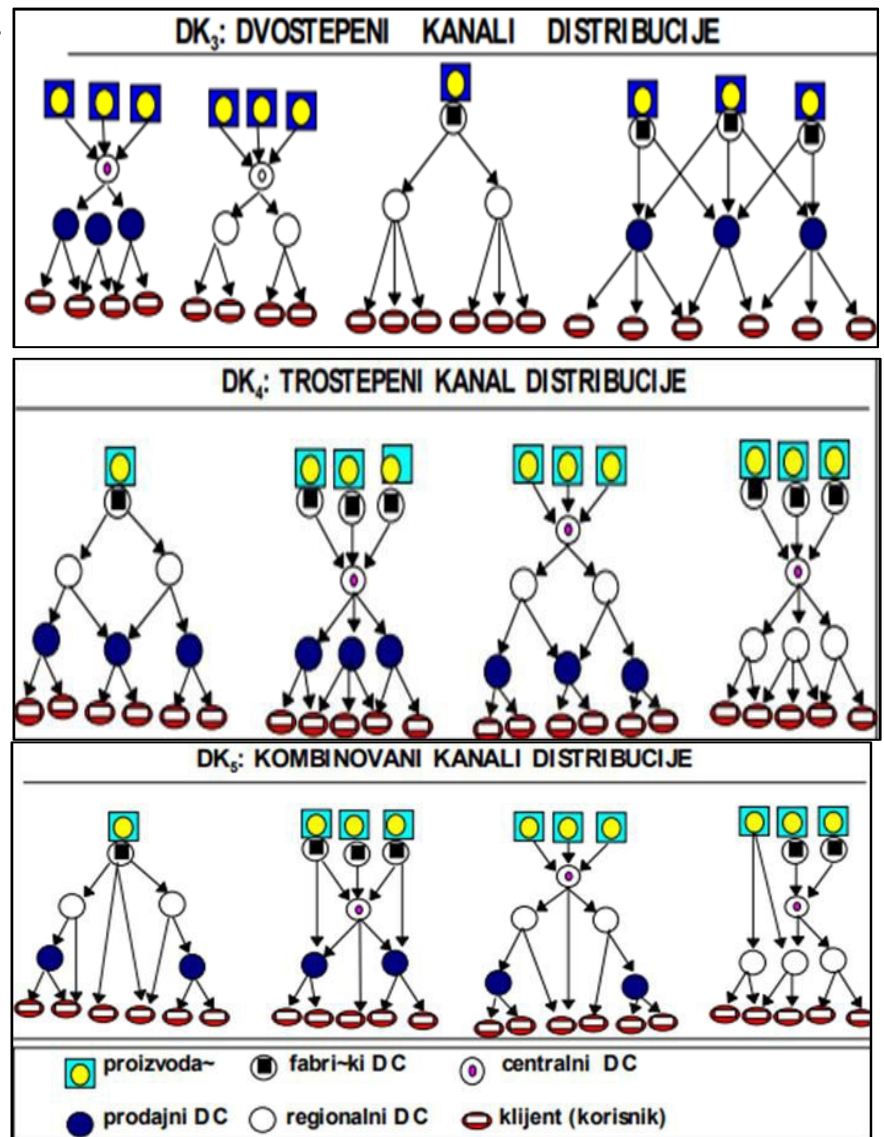


Figure 11 – Classification of distribution channels based on hierarchical structure and degree of distribution³⁴

2.1.4. Warehousing Logistics

The primary task of a warehouse is to economically reconcile differently sized cargo flows. The warehousing function includes the tasks of storing and safeguarding materials, parts, and other physical resources directly related to the procurement function, and the storage of finished products or goods directly related to the sales function.

The motives for holding inventory can be represented through the following functions:

- **Levelling** - a significant function in the event of a discrepancy between the demand for material and its supply in terms of quantity and/or its time distribution.
- **Safety** - significant in the event of unpredictable risks during production, fluctuations in sales market demand, and delays in procurement market deliveries.
- **Classification** - significant for shaping the product assortment in retail or for determining product types in industrial enterprises
- **Speculative function** - is significant when an increase in prices is expected on the procurement market or the sales market

³⁴Prof. Dr. Dr. sc. Kata Ivić, Faculty of Economics in Osijek 2020, *Distribution and Trade Logistics*;

- **Value-added processing function** - significant in cases where stored goods undergo a change in quality³⁷

Warehouse logistics is a subsystem within the logistics system whose task is to design and optimize systems for all types of storage, picking, and transportation of goods, from the entry of goods into the warehouse through all reprocessing, i.e., holding, to the exit of goods from the warehouse.

System optimization includes both hardware and software logistics tools. Warehouses with a favorable location are often more expensive, so it is necessary to determine the most optimal location solution.

Warehouse logistics includes the following activities:

- Finding a suitable warehouse
- Ensuring warehouse functionality
- Standardization optimization
- Decision-making regarding the quantity of products in storage
- Achieving optimal storage and picking
- Finding the most economical means of transport, and³⁵

Main objective: to maintain product inventory in order to ensure the synchronization of processes that precede storage and those that follow it, while reducing costs, improving service quality, and ensuring the conditions for the successful execution of these processes.³⁶

Warehouse logistics is related to and supports (**Figure 12**):

- **Logistics of procurement** - by providing for and storing procured materials in incoming warehouses
- Warehouse **logistics** is **related** to and supports procurement logistics (Figure 12): By providing and storing procured materials in incoming warehouses xml-ph-0000@deepl.internal By providing and storing products (36)
- **Distribution logistics** - by providing and storing the products³⁶

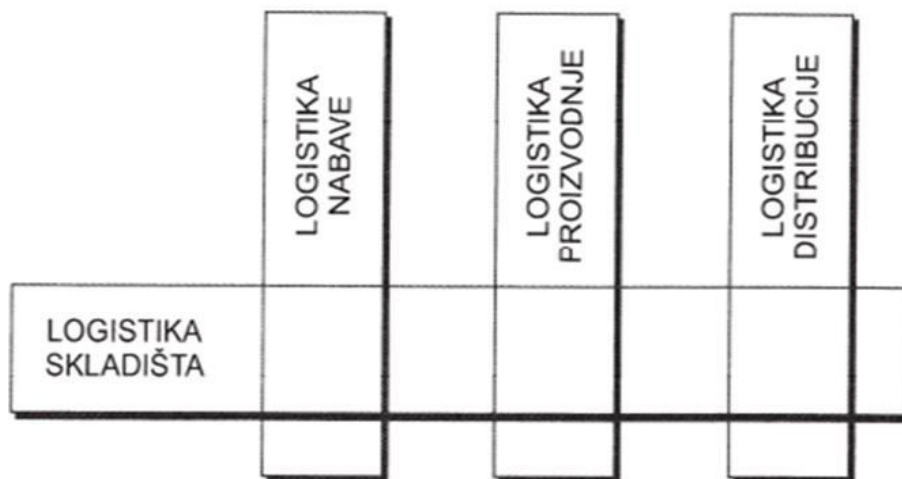


Figure 12 – The connection of warehouse logistics with procurement, production, and distribution logistics³⁵

Figure 13 shows the forms of connecting individual processes in the logistics system by engaging the warehousing subsystem. Warehousing processes do not typically connect directly with other processes, but rather through the transport subsystem, whether it is internal or external transport.

³⁵ Buntak K., Šuljagić N., Tehnički glasnik, Vol. 8 No. 4, 2014., *Economics of Logistical Functions in the Company*,

³⁶ Prof. Dr. Dušan Ragodić, Belgrade 2014, *LOGISTICS – Fourth Supplemented Edition*,

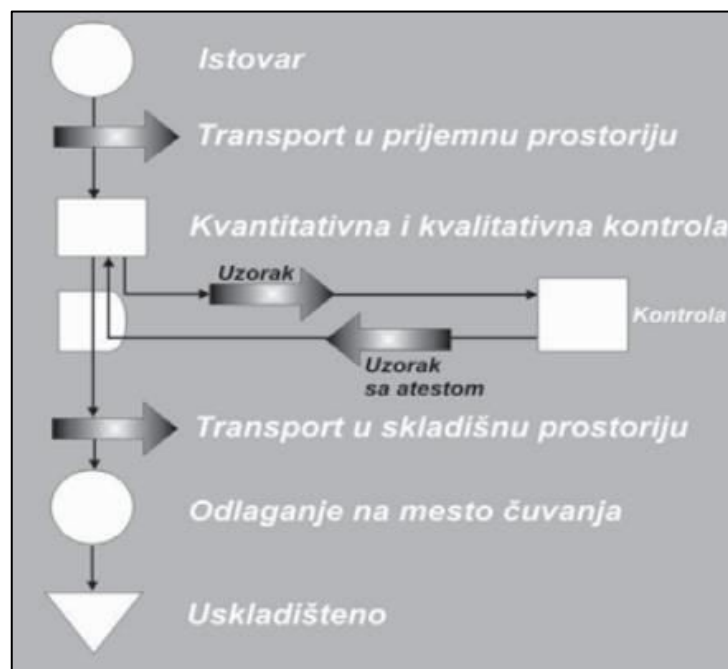


Figure 13 – Warehouse Receiving Operation³⁷

According to the above scheme, the basic processes that take place in the warehouse are:

- Receipt of procured quantities
- Incoming inspection
- Storage (housing and sorting)
- Storage and maintenance during the holding phase
- Moving and relocation *f*
- Preparation for delivery
- Delivery
- Record of work through all the listed phases³⁷

Warehouses are divided by function in the logistics system into:

- ❖ **Issuing warehouses** - are most often located next to manufacturing plants or retail businesses, as they are used to store finished products intended for distribution and sale to consumers.
- ❖ **Transshipment warehouses** – they receive goods for a short period to transship them from one mode of transport to another. The speed of transshipment is more important than the warehouse's capacity.
- ❖ **Distribution warehouses** – Goods are collected from various suppliers in large quantities and forwarded to one or more manufacturing or trading companies in smaller quantities. Likewise, goods can be received in a warehouse, for example, from small-scale buyers in small quantities for consolidation into large shipments that are then sent to distribution centers or processing plants. These warehouses play a key role in the logistics system.

³⁷Prof. Dr. Dušan Ragodić, Belgrade 2014, LOGISTICS – Fourth Supplemented Edition,

3. OVERVIEW OF LOGISTICS PROCESSES

The basic logistics processes of a logistics system, whose interrelationship is shown in **Figure 14**, include:

1. **Inventory Management**
2. **Warehousing**
3. **Transloading, packing, and dispatch**
4. **Transportation**
5. **Order Fulfilment³⁸**

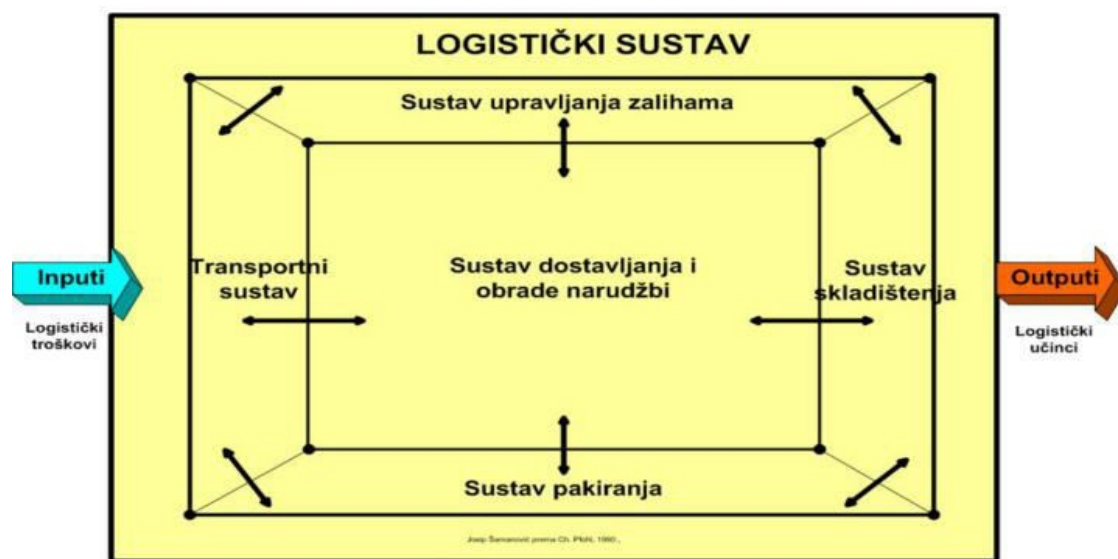


Figure 14 – Functional Interrelationships of the Logistics System³⁹

3.1. Inventory and Inventory Management

3.1.1. Reasons for holding inventory in the warehouse

Inventories are buffers between the input and output flows of material goods. These buffers arise as soon as there is a discrepancy between the temporal and quantitative structure of the input and output flows. If the input flows in all parts of the company were completely synchronized with the corresponding output flows, inventory would be unnecessary. The so-called "just-in-time" production is based on the complete synchronization of these flows⁴⁰.

Inventories should be held for the following reasons:

1. Many products have a seasonal production pattern and a constant demand throughout the year (e.g., fruits and vegetables).
2. Some products have a seasonal demand pattern and a steady production flow throughout the year. Producers justify their regular year-round production due to high demand during the season
3. Sometimes goods are stored as a precaution against expected price increases and as a protection against later shortages

³⁸Prof. Dr. Dušan Ragodić, Belgrade 2014, LOGISTICS – Fourth Supplemented Edition,

³⁹Dr. Zdenko Segetlija, Faculty of Economics, Osijek 2002., *Introduction to Business Logistics*;

⁴⁰ Sotirović Velimir, Egić Branislav, Technical Faculty Zrenjanin 2005., *Business Logistics Management*;

4. In some cases, storage occurs as a necessary consequence of procuring in quantities larger than those needed to meet immediate needs, due to the desire to obtain a quantity discount on purchases and lower transportation costs.
5. Many products require special storage equipment during transport to the market. Fresh fruit, vegetables, eggs, and butter, for example, require refrigerated trucks to protect them properly.
6. Other goods require heated trucks to avoid exposure to extreme cold during the winter months
7. From the time of production until final consumption, some goods require "conditioning." This conditioning, which occurs with goods such as meat, cheese, tobacco that must be "cured," and some alcoholic beverages, improves quality and makes the products much more suitable for consumption.
8. Finally, many goods are held in stock for purely speculative reasons—so that their producers can achieve much more favorable prices

3.1.2. The Importance of Inventory Management

Inventories are a complex economic category that appears in various forms of engaged resources. The relationships through which they are involved with other categories in economic processes are relationships of multiple interdependence.

The relevant factors for determining storage and inventory policy must not be viewed solely from the perspective of an enterprise, as an element of the production system, but much more broadly. This concerns the joint storage and inventory policy of a number of entities, which are closely connected, or rather integrated, into a reproductive whole (**Figure 15**). Therefore, as such, the policy of storage and inventory encompasses all relevant decisions and programs of measures, means, and deadlines that relate not only to the economy as a whole but also to social activities.

For many reasons, both economic and technological, a full alignment of production and consumption cannot be achieved, either in time or in space. What is produced is not, as a rule, consumed at that moment or in that place. Therefore, larger or smaller inventories are necessary to ensure the continuity of the production process. Poor inventory control by large or small, manufacturing or commercial enterprises, results in a loss of a significant portion of income due to the high cost of inventory carrying charges.

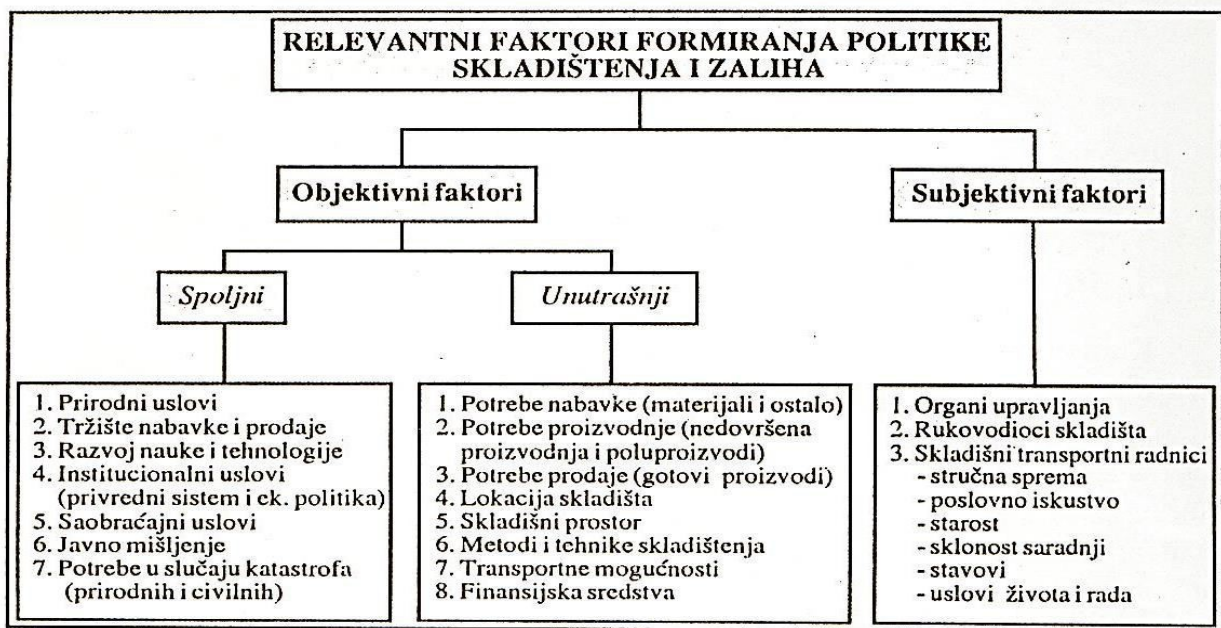


Figure 15 - Classification of relevant factors of storage and inventory policy

To secure them against uncertainty, supply chain disruptions, increased demand that exceeds expectations, or in the event of speculation, companies often accumulate inventory. Logistics faces a very difficult and important task related to inventory optimization, namely finding an inventory management model that will, under the given circumstances, ensure sufficient quantities of goods and satisfy consumer demand at the lowest possible total cost.

Inventory generates extremely large losses and costs, which account for 20% to 30% of total logistics costs. Research shows that the standard cost of these expenses is around 25% of the total inventory value, where capital costs amount to about 15%, warehouse space costs 2-3%, taxes 2-6%, insurance costs 1-3%, and inventory counting and administration costs around 2%. In certain industries, the costs of holding inventory risk, obsolescence, theft, and spoilage account for a significant share—around 6%, 6-12%, and 3-6% respectively—so the total cost of inventory can be considerably higher. The percentages stated clearly show that any reduction in inventory provides significant savings and benefits [66].

3.1.3. Inventory Management Models

Optimal inventory is the quantity of products in stock that will incur the lowest total inventory holding costs for the business, without jeopardizing the continuity of business operations. The optimal inventory size can be calculated using:

- **Deterministic models** - characterized by the constancy and predictability of conditions and circumstances
- **Stochastic models** - characterized by variability and unpredictability of conditions and circumstances

Deterministic inventory management strategies are:

- **Economic Order Quantity (EOQ)** – Used to determine the optimal order quantity for a specific inventory item, given its estimated usage, procurement cost, and holding cost. The exact quantities can be determined using mathematical methods.

$$EOQ = \frac{\sqrt{2DS}}{H}$$

where:

EOQ – Economic Order Quantity D –

Total demand

S – procurement costs

H – holding costs

The order quantity must be optimized so that the total cost, composed of procurement and holding costs, is minimal, and a stockout never occurs.

- **Continuous Reordering System** - After each inventory drop, it is replenished to a predetermined level, with the order quantity being fixed. The objective is to determine the specific point at which, upon its reaching, an order for a fixed quantity of goods is placed. The reorder point is calculated by adding the safety stock to the average incoming delivery lead time.

- **Periodic monitoring system** – This is characterized by replenishing inventory at precisely defined times or time intervals (e.g., daily, weekly, monthly, etc.). Inventory levels are monitored at fixed intervals, and an order is placed for the quantity needed to reach the maximum inventory level. Maximum inventory represents the quantity that should cover demand and ensure an adequate buffer until the next procurement (delivery) of goods. Periodic ordering is appropriate when the seller or manufacturer has routine orders from customers that are evenly distributed over time.

Stochastic inventory control systems are:

- **MRP I (Material Requirements Planning)** - Its basic function is to guarantee the availability of required material. MRP is used to plan the procurement or required quantities

raw materials in time for internal use, assembly, production, or distribution. This process involves monitoring inventory levels and automatically generating a purchase order for raw materials, or for the production of a specific work-in-process item or part of it. By operating in this way, MRP attempts to achieve a balance

- **MRP II (Manufacturing Resource Planning)** - This system is more developed than MRP I, a superset that considers the situation several steps above MRP I. MRP I takes into account raw materials and product components; however, the MRP II model also considers production capacity, worker capacity, and financial indicators. By using MRP II, a company integrates its financial and logistical plans, reduces inventory costs, decreases production downtime, and increases planning flexibility. Integrating the MRP II and JIT models attempts to develop models with an even faster response time.
- **DRP (Distribution Requirements Planning)** – These models forecast for each product in inventory and are based on: demand forecasting for each individual product, the current inventory level of each product, target safety stock, recommended reorder quantity, and lead time. Since it makes demand forecasts, the moving average method is most commonly used. This method is based on the average consumption or sales over several time intervals, and this average is then used for demand forecasting. DRP models allow for improved delivery service, reduced transportation costs, and enhanced operations in distribution centers. DRP models are often used in combination with MRP models, resulting in the efficient integration of the entire system, lower costs, and so on.
- **JIT System (Just-In-Time)** – This model was developed to meet consumer needs with minimal delay. The JIT model is based on holding minimal inventory levels and producing only when demand arises. Coordination and alignment of all entities within the system are essential. Inventory should be available to the company when it is needed, no sooner, no later. The proper use of the JIT production model leads to increased quality, efficiency, productivity, and most importantly, a reduction in costs. The model works by producing exactly the amount of product that is needed, or in other words, as much as the market demands. There is no creation of inventory or excess production. The lead time from ordering a product to its final delivery is shortened, thereby optimizing inventory throughout the entire supply system. By shortening production time, safety, maximum, and other inventories are reduced, which increases business revenue.⁴¹

The traditional EOQ inventory management model is increasingly being supplemented by or replaced with modern management models: Just-In-Time (JIT), Material Requirements Planning (MRP), and inventory planning and control based on distribution market conditions (DRP). Modern inventory management models, especially the DRP model, are critical to the profitability of commercial enterprises.

3.2. Warehouse Management

3.2.1. The Importance of the Warehouse in the Logistics System

The warehousing function is of great importance, but its significance is often overlooked, mainly because warehousing (holding goods until they are needed) is considered a passive function. The value added to goods while they are in storage is not immediately apparent, nor is warehousing easily understood as an economic function. However, warehousing is the "heart" of the logistics and physical distribution system. Almost every delivery of goods originates from or is destined for a warehouse. Every product that is sold passes through or is at least stored in one warehouse, and often in several. Warehouses are facilities for storing and handling goods, and warehousing is the set of activities related to storing and handling goods in a specific space. Until recently, the only term used to denote the concept of warehousing was "inventory holding." For this reason, warehousing was often defined as "the *holding of goods before their use*." However, today, inventory holding is the largest part of warehousing activities, but not the only one.

⁴¹Goran Gorički, 2017, University of Zagreb, Varaždin, *Inventory Management Models*;

From a macroeconomic perspective, warehousing performs a very important function. It creates time utility for raw materials, industrial goods, and finished products. The assumption is that market-oriented storage enables the customer to be served in a shorter time period. More importantly, storage increases the utility of goods by extending their availability to potential consumers. In other words, by using the company's warehouses, goods can be made available when and where consumers demand them. This function becomes increasingly important as companies use customer service as a competitive tool in the marketplace.

The basic requirement for storage depends on the quantity and variety of goods that need to be housed in the warehouse. Accordingly, warehouses exist because companies hold certain products in inventory.

3.2.2. Warehousing Decisions

Warehouse management involves making decisions about:

1. **Ownership**
2. **Number**
3. **Size**
4. **Warehouse Location**

1. **Ownership Decision** - To determine ownership of the facility, an organization has two basic alternatives: to use *private* or *public* warehouses. The choice of one of these alternatives, or a combination of them, is the first and main storage decision. Many companies combine public and private warehouses due to changes in regional market conditions, as well as for other reasons (e.g., seasonal impact). Certain operations require storing goods in private warehouses, while for others, the best solution is public facilities where companies rent space or sign short-term contracts as needed. If they opt for private storage, companies also make decisions about *determining the necessary space* and *the interior layout* of the chosen warehouse. Also, an important question is whether the company will use a *centralized* or *decentralized* approach to supplying its warehouses (this decision primarily concerns how many warehouses are needed to supply the company).
2. **Decision on the number of warehouses** - The company must coordinate the decision *on the number* of warehouses with the decision on transportation alternatives. For example, air freight allows for the rapid coverage of the national market from one or two strategic warehouses. Although the costs of air freight are relatively high, a company can offset them with savings in storage and inventory costs. The number of links or transportation alternatives makes the decision on the number of warehouses a real challenge, especially when considering that this decision is linked to the public-private warehouse alternative. Closely related to the decision on the number of warehouses, two other decisions are whether to centralize or decentralize: *the decision on size* and *the decision on location*.
3. **Size decisions** - If a company uses both private and public warehouses, the issue of size is important. A company can usually expand or contract space according to its needs at different times if it uses public warehouses.
4. **Location decisions** - The company must provide the desired level of service to consumers at the lowest possible distribution cost. By analyzing the desired warehouse functions, a firm can determine general locations, such as placing high-service facilities near the market, blending raw materials closer to production, or a combination of other factors. The choice of a specific location will depend on factors such as transportation, the market, and local characteristics. These decisions, once implemented, can be changed, but at high costs (especially in a private warehouse). Therefore, careful consideration of all factors is essential.

In addition to the warehouse decisions mentioned above, companies must decide how to arrange the warehouse interior. In other words, the company must decide on the space for moving between goods in the warehouse, on

shelving, equipment, and all other physical dimensions of the warehouse interior. The second decision is how to most efficiently house inventory in the warehouse, i.e., the method of product layout.

Other storage decisions relate to *the items (products)* that the company should keep in stock and how much inventory it should have in different warehouses. A company with multiple locations must decide whether to store all products from the product line in every warehouse, whether each warehouse will be specialized in storing only certain products from the product line, or whether to combine both approaches.

Storage decisions are important and require significant attention. The primary focus of warehouse management is to increase efficiency and productivity. Properly planning space utilization in advance by determining inventory levels and considering distribution operations is more important than later constructing additional facilities. Furthermore, warehousing decisions are closely related to other areas of the logistics system.⁴²

3.3. Material Handling and Packing

Material handling is very important for warehouse operations (moving goods to and from the warehouse) and for moving goods to different locations within the warehouse itself. The very term "material handling" is somewhat difficult to define. Some consider material handling to refer only to the equipment used to move goods within the warehouse, such as forklifts or conveyors. Others literally understand it as the "manual" manipulation of goods. In fact, developed mechanical equipment, manual labor, or a combination of both can perform material handling operations.

Simply put, material (goods) handling is *"the efficient movement of goods, which typically takes place within a facility (factory, warehouse) and between the facility and transportation agencies."* In a modern logistics system, specially designed equipment performs the movements, and for this reason, material handling has often been equated with equipment. However, manual movement is also material handling.⁴³

Most often, the logistics manager's responsibility for material handling is manifested within and around the warehouse or in the factory's storage departments. The logistics manager for material handling may also collaborate with other individuals, such as the production manager, regarding the procurement and maintenance of equipment.

3.3.1. Materials Handling Objectives

One of the primary goals of material handling is to increase the usable capacity of a storage facility. A warehouse has a fixed internal length, width, and height, and a specific capacity. Space utilization minimizes the costs of operating the warehouse. Many warehouses do not maximize the available space because they do not store goods as high, i.e., as vertically as possible. Horizontal space (area) in a warehouse is the easiest to fill, so it is necessary for warehouse managers to pay significant attention to the vertical dimension (volume) as a separate challenge.

The goal of material handling is also to reduce the number of operations when handling products. A company typically first moves products into the warehouse and places them in storage, then moves them to the order picking area, and finally moves and prepares them for delivery to the customer. Thus, this process involves several unavoidable product moves. However, in some warehouses, the company may move products multiple times within each zone. It must avoid this additional product handling if it wants to operate efficiently. Therefore, the design of any product handling system, as well as its related activities, should minimize the number of product moves to, within, and outside the warehouse. In some cases, additional moves are unavoidable due to overcrowding, so firms must temporarily store goods and move them later. However, an efficient material handling system should minimize the number of moves

⁴² Ljiljana Stošić Mihajlović, 2020, *Business Logistics – Packaging*;

⁴³ Prof. Dr. Dušan Ragodić, Belgrade 2014, *LOGISTICS – Fourth Supplemented Edition*,

goods and to enable their rapid and efficient flow through the warehouse. The solution for minimizing the number of product transfers is control.

Material handling improves efficiency by enabling the logistics system to respond more quickly and effectively to the demands of factories and customers. It plays a key role in delivering products to customers on time and in the right quantities. Material handling is a crucial support for outbound logistics in the efficient movement of goods in the warehouse, locating inventory, accurately filling orders, and quickly preparing orders for delivery to customers. Service is a major focus for logistics managers. The manager must ensure that the material handling system can always respond quickly and efficiently to the needs of customers and production. Companies must integrate material handling needs not only with the needs of their departments but also with the needs of their customers.

Effective material handling can contribute to cost reduction by increasing productivity (a greater and faster flow of goods). Additionally, more efficient use of space and the relocation of less-moving goods reduce costs.⁴⁴

3.3.2. Functions of packaging

Packaging goods has several functions:

1. **Production**
2. **Marketing**
3. **Usage**
4. **Logistical**

1. Production function - Thanks to the proper packaging of raw materials and materials, i.e., the packaging of workpieces, production can be organized without a special organizational unit whose task would consist of the quantitative preparation of workpieces, in accordance with the needs of the technological process. Proper packaging of the workpiece, coordinated with the quantities entering the technological process, connects the preparation phase with the production phase in a simple and rational way. On the other hand, packing finished products in quantities that match the pace of product completion blurs the line between the technological and final merchandising phases, thereby contributing to faster and more rational material flows in this part of the reproduction cycle. The packaging of goods is often integrated into the technological process itself, so its characteristics affect not only the length of production time but also its economy.

2. Marketing function - Unlike the production function of packaging, which is equally important for packaging the workpiece and for packaging finished products, the marketing function is significant only for the packaging of finished products. Packaging, or product packaging, as a form of communication with the market and the consumer, is an extremely important instrument of a company's sales policy. Through its visual characteristics, the product stands out from other products of competing companies, and its aesthetic qualities influence an increase in the motivation to purchase.

3. Usage function - The usage function of packaging refers to the ability of the packaging, after the product it protects has been unpacked, to also satisfy other personal or general needs of the consumer. Thus, for example, various boxes or bags can be used by consumers to pack other products, cans and plastic packaging can be used for various purposes in households, and sacks and barrels can be reused on farms, etc. In recent years, there has been an increasingly strong general need to adapt the characteristics of industrial, or commercial, packaging to the requirement of preserving the human environment, and to the requirement of reducing and recycling industrial waste.

⁴⁴ Sotirović Velimir, Egić Branislav, Technical Faculty Zrenjanin 2005, *Business Logistics Management*;

4. Logistics Function - Packaging is very important when it comes to logistics activities. The packaging logistics functions are reflected in the following:

- Protection of the product from spillage, damage, and environmental influences
- Facilitating storage - warehouse activities
- Enabling transportation and handling of goods⁴⁵

The size, shape, and type of packaging will affect the handling of materials and products, as well as the company's ability to use pallets or different types of material handling equipment. Therefore, it must take these packaging characteristics into account to make the most efficient use of its warehouse and transportation vehicles. Many companies choose packaging that is too wide or tall to efficiently utilize their vehicles and warehouses, so linking packaging activities with storage and transportation is very important.

Also, packaging is important for protecting products from damage, not only in storage but also during transport. Damaged products can reduce future sales, so packaging is important because it must prevent products from reaching consumers damaged. On the other hand, poor packaging can affect the costs of handling goods.

The goal is to achieve the optimal level of protection that will meet the desired service standard for consumers at minimal packaging cost. Due to the unpredictability of hazards in distribution, only the most critical and very expensive items should be packed with 100% protection. For most other items, a built-in damage rate is desirable, indicating that the packaging requires special handling or other instructions. This rate varies widely and can be as high as 15% for cheaper and easily replaceable items.⁴⁶

3.4. Transport

The transportation function deals with the movement of goods from one economic process to another, i.e., the creation of spatial utility. Raw materials and auxiliary materials are brought to the processing site (the production facility), then goods are moved within the production process itself, and finally, finished and semi-finished products are transported to the market by various means of transportation. Transportation also ensures the workforce for all social and economic activities and the movement of people from one location to another for any reason. Due to this immediate need to move the objects of labor, means of labor, labor force, and production goods, transportation stands out as a separate activity of material production, often called *the transportation industry*.

The development of the transportation industry has accelerated the development of industry in general, as well as its connection to other economic activities on one hand, and to the world market on the other.

When transporting goods from the producer to the consumer, a number of options are used to employ different modes of transport. All these options are based on certain common elements, such as:

- use of different types of vehicles, depending on the type and quantity of goods being transported, transshipment at external transport points between the service user and the carrier, as well as between different transportation equipment, which is carried out on a piece-by-piece basis,
- the use of different technical handling equipment adapted to specific types of goods and various types of vehicles, and the necessary packaging to protect goods during transport.

The common elements of the transportation process, as well as the heterogeneity of its organization, suggest that such transport cannot be treated as a continuously integrated goods process. This, on the one hand, leads to an increase in transportation costs, and on the other hand, also affects other elements of logistics.

Table 2 shows the costs of domestic and international transport as a percentage of the product price. It can be seen that the share of transport is quite high in the structure of the retail price. But not only that,

⁴⁵ Ljiljana Stošić Mihajlović, 2020, *Business Logistics – Packaging*;

⁴⁶ Sotirović Velimir, Egić Branislav, Technical Faculty Zrenjanin 2005, *Business Logistics Management*;

Transportation also directly affects the costs of other logistics activities. For example, inventory holding costs can be reduced through careful planning, i.e., by making deliveries in smaller quantities, which can significantly impact the total costs of the entire physical distribution system.

Sektori visokih troškova	Kamen, glina, proizvodi od stakla	27%
	Naftni proizvodi	24%
	Građevinsko drvo i proizvodi od drveta	18%
	Hemijski proizvodi	14%
	Hrana i srodni proizvodi	13%
	Nameštaj i instalacije	12%
Sektori srednjih troškova	Papir i srodni proizvodi	11%
	Industrija primarnih proizvoda i sirovina metala	9%
	Proizvod tekstilnih fabrika	8%
	Proizvodi prerađenih metala	8%
	Raznovrsna proizvodnja	8%
	Transportna oprema	8%
	Gumeni i plastični proizvodi	7%
Sektori niskih troškova	Proizvodi od duvana	5%
	Mehanički isključujući električne	4%
	Instrumenti	5%
	Odeća i drugi tekstil	4%
	Štampanje i publikovanje	4%
	Električna i elektronska oprema	4%
	Koža i proizvodi od kože	3%

Table 2 – Domestic and International Transportation Costs

Regular transportation can also reduce inventory in transit. Obviously, inventory savings can be compared and offset against different levels of transportation costs. A proper decision regarding transportation can help company managers determine safety and other types of inventory, thereby avoiding large variations in demand.

Obviously, modern economic development, the developed social and territorial division of labor, as well as changes in population agglomeration and the rise in the standard of living, have created entirely new demands for transportation services. These changes have been reflected in new technology, or rather, in the necessary adaptation to new demands.

It is observed that there are two basic categories of transport channels in the process of distributing final products. In the *procurement* phase, only direct channels exist, as every manufacturing company supplies itself with raw materials and semi-finished goods via direct transport. This is also dictated by the nature of the raw materials themselves, which generally do not tolerate transshipment and storage during transit.

In the distribution of final consumer products, between the producer and the final consumers, intermediaries can be wholesalers, department stores, and retailers. In this case, broken transportation channels are present. They differ, first, in the volume of goods, and second, in the dynamics of transportation. Transporting goods from the producer to large consumers (on the first part of the broken transport channel) to warehouses is characterized by large masses of goods in individual shipments and longer time intervals between shipments. This requires larger capacities for transport vehicles and carriers of the transport function. On the second part of the broken transport channels, from wholesalers to retail stores, department stores, or large consumers, the transport process is characterized by: smaller loads, more frequent transport, and more flexible adaptation of transport to consumption fluctuations. The logical choice of mode of transport is trucking, using one's own vehicles. For these reasons, the mode of transport must be chosen to bring products to the production facility or to deliver them to commercial enterprises and consumers.

Manufacturers and retailers must constantly examine the productivity and efficiency of their transportation system. By maximizing transportation efficiency, transportation costs can be reduced by 10% to 30%. Savings are achieved by reducing the number of carriers, negotiating favorable contracts, and eliminating redundant domestic and international transportation services.

3.4.1. Types of Transport

The transport of goods should proceed as smoothly as possible, delivery deadlines must be met, and the risk of damage or loss of goods must be minimized. Therefore, it is essential to choose the appropriate mode of transport that will ensure the safest, most efficient, and most economical delivery.

First, it is important to consider the type, size, and quantity of the goods being shipped. For example, transporting food, fruit, or flowers requires choosing the fastest and shortest mode of transport. Likewise, large items, as well as large quantities of goods, cannot be transported by air.

Another important element is the transportation costs. They will depend on the choice of transportation method, the value of the goods, transportation risks, the destination, and the laws of the destination country, as well as a number of other factors.⁴⁷

There are different types of transport, namely:

1. Rail transport

Rail transport is used to transport heavy, bulky goods that have a relatively low value per unit of weight (coal, grain, etc.) mainly to quite distant locations. It is a low-cost mode of transportation, but it is not economical for small quantities of goods or for shorter distances.

- Advantages of rail transport:
 - Low prices
 - Easy transport of all types of goods of various dimensions
 - High precision in transport times
 - Ability to transport large quantities of goods at once
- Disadvantages of rail transport:
 - Railroads often do not cover all necessary destinations, especially in remote regions
 - Additional costs for transporting goods to and from the railway
 - A mechanical failure can cause much greater delays, and thus higher transportation costs

2. Road Transport

Road transport is the most widespread and flexible way of transporting goods. It is provided by cars (with or without trailers), vans, trailers, trucks, and refrigerated trucks. Road transport carriers are focused on transporting light goods over shorter distances. Compared to rail, road transport offers better door-to-door delivery options and greater flexibility in transporting goods to various geographical destinations.

- Advantages of road transport:
 - Saves time and money
 - Detailed road network and door-to-door delivery capability
 - Flexibility to change location, speed, and direction of movement
 - There are a number of specialized vehicles for transporting all types of goods
- Disadvantages of road transport:
 - Not suitable for long distances
 - Smaller cargo space capacities
 - Road and weather conditions affect transportation

⁴⁷ Djordje Sumic, 2020, Article: *Types of transportation*;

3. Water transport

It is divided into maritime, river, and lake transport. Water transport is used to transport bulky and low-value goods. It "competes" with rail transport for the transport of bulky goods such as iron, grain, cement, coal, oil, etc. These products are transported by barges, bulk carriers, and tankers for oil and liquefied natural gas. As with rail transport, containers are also used here to speed up the handling of goods and provide protection from moisture and theft.

- Advantages of water transport:
 - The cheapest transport to the most remote destinations
 - Ability to deliver large quantities of goods at low prices
 - Cargo containers can be used for further transport by road or rail
 - No restrictions on the type and size of goods
- Disadvantages of water transport:
 - Delivery is much slower than other modes of transport
 - Bad weather can cause additional delays in the delivery of goods
 - It is difficult to track the cargo
 - Often, additional overland transport of goods is required
 - In addition to mandatory insurance, additional cargo insurance is often required.

4. Pipeline transport

Pipeline transport is mainly used for natural gas, chemicals, liquid fertilizer, and products that can be transported in liquid form. Unlike other modes of transport, it is characterized by constant movement and no stock accumulation, a one-way flow, and the transport of only a single product.

- Advantages of pipeline transport:
 - Reliability
 - Low maintenance and operating costs
 - Ability to operate continuously (24 hours a day, 7 days a week)
- Disadvantages of pipeline transport:
 - High fixed capital investment
 - Low throughput speed through the pipeline
 - Limited range of goods that can be transported through them

5. Air transport

Air transport is the mode of transport that has shown the fastest expansion. It is used for transporting high-value goods, perishable goods, and for urgent deliveries over long distances. It is used more for transporting passengers and less for transporting goods.

- Advantages of air transport:
 - It saves transportation time and is ideal for long distances
 - Provides a high level of security for sensitive goods
 - It plays a significant role in the transport of money, valuables, and special products
- Disadvantages of air transport:
 - High transportation costs and high fees
 - Flights are subject to delays and cancellations, depending on weather conditions,

- This mode of transport is not suitable for all types of goods
- Additional costs for transporting goods to and from the airport⁴⁸

Combined transport is a mix of road, rail, air, and water transport according to shipping needs and conditions. This type of transport is quite complex and requires intricate planning.

3.5. Order fulfillment

The order is the foundation of the information flow in logistics and serves as the input for the logistics system. The purchase order is, therefore, a source of information and the initiator of all other timely activities related to sales. Order fulfillment encompasses all activities within the company regarding the movement of information and products, from the receipt of the order, its processing, through the procurement and preparation of necessary resources, the production of the product, as well as the preparation and shipment of the product to the customer, all the way to invoicing.

Order information includes:

- Order number and date
- Customer address and number
- Salesperson and sales territory
- Item code and number
- Quantity of the item and gross price
- Terms of purchase, discounts, etc.
- Mode of transport, shipping cost calculation
- Shipping title, delivery date

When executing the order, the sales contract is concluded according to the following steps (**Figure 16**):

- a) The buyer requests a quote from the seller
- b) The seller provides the buyer with an offer
- c) The buyer sends the order to the seller
- d) The seller sends the buyer an order confirmation
- e) The seller sends the invoice to the buyer



Figure 16 – Order fulfillment scheme – conclusion of the sales contract

⁴⁸Sotirović Velimir, Egić Branislav, Technical Faculty Zrenjanin 2005, *Business Logistics Management*;

Order fulfillment functions refer to the realization of information flows before, during, and after the product flow, based on which the product flow should be planned, managed, and controlled. The information flow before the product flow involves the timely notification of all participants involved in the product flow to carry out activity preparation (planning). During the product flow, the information flow includes information for the operational execution of transportation, transshipment, and storage (proper handling of hazardous materials), and the control of the product flow up to the point of receipt. The information flow after the product flow includes information on the shipping invoice, critical points in transportation, complaints, and necessary information for other departments (market research). The information path for an order during its fulfillment is shown in **Figure 17**.

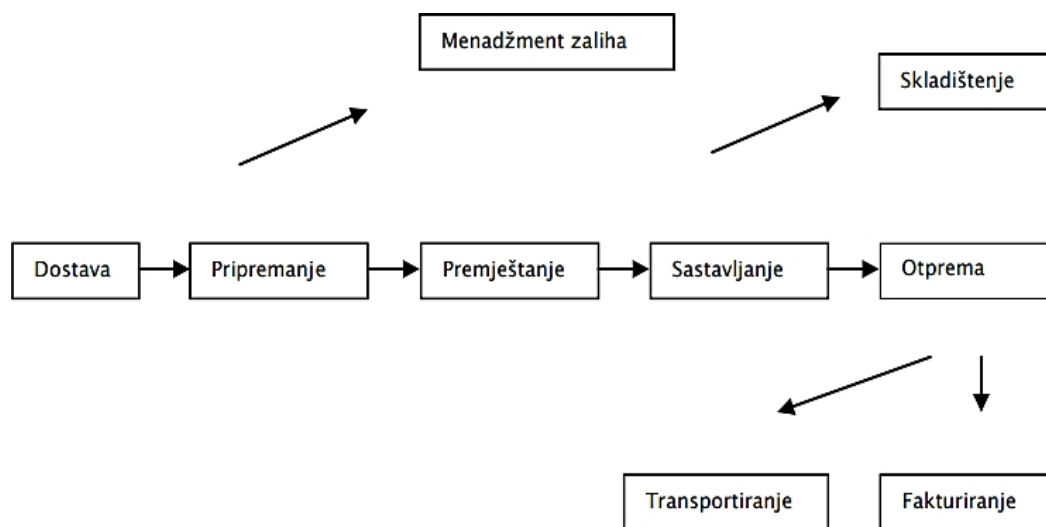


Figure 17 – Order fulfillment tasks

The order fulfillment tasks are therefore:

- Order Delivery - can be performed by mail, telegram, telephone, fax, email, or e-order to the supplier's headquarters, sales outlet, or supplier's representative
- Preparing - involves adapting the order to internal requirements (supplementing, verifying price, customer creditworthiness) and planning in the logistics system (whether the product is in stock). The information flow goes to inventory management.
- Movement (transfer) - occurs after preparation. It can be manual, mechanical, or electronic. It includes order confirmations and processing documents (shipping notice and all shipping documents)
- Picking (commissioning) - involves assembling the ordered goods and preparing them for shipment, as well as sending information for warehouse operations (for warehouse accounting).
- Shipping - involves completing shipping documents, providing information on cargo, transportation, and timing, selecting optimal modes of transport and shipping routes, and maintaining close coordination with the carrier.
- Inventory Management
- Transport
- Invoicing - Can be done after shipment or in parallel with the order processing stages (if there are sufficient quantities in stock). Parallel invoicing can delay the delivery of goods and incurs postage costs.
- Warehousing⁴⁹

⁴⁹ Prof. Dr. Dušan Ragodić, Belgrade, 2010, *LOGISTICS – First Edition*;

Order processing methods can be:

- Manual document writing - the process of writing a document with copies
- Computer processing - is related to the electronic data processing (EDP) system. It can include activities related to decision-making, verification, recording, and delivery of information. The advantages of computer processing are the ability to process a large number of operations at high speed and to use information (statistics).
- Electronic Data Interchange and Order Fulfillment - a modern form of business that involves the intensive application of information and internet technology. It involves the exchange of information and the automation of business transactions (automatic ordering and payment). Modern e-business is based on computer networks (the ability to exchange data and information between computers). Local, national, and international networks are distinguished.⁵⁰

4. THE RELATIONSHIP BETWEEN LOGISTICS AND THE SUPPLY CHAIN

A widespread definition of the supply chain is: "A supply chain represents a network of organizations that are involved, in upstream and downstream links, in various processes and activities that produce value in the form of products or services aimed at the end-user or consumer."⁵¹ (Figure 18).

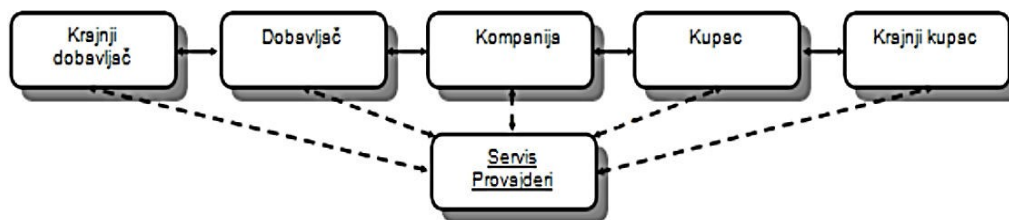


Figure 18 – A simple illustration of the elements of the supply chain ⁵²

Supply chains are interconnected associations of individual businesses. The concept of a network indicates the introduction of coordination in processes and relationships. An upstream link means moving "against the flow" and refers to the relationship between a company and its suppliers, as well as a subcontractor's relationship with its suppliers. Downstream ties, or "with the flow," refer to the business-to-consumer relationship. Combined upstream-downstream ties can occur, as is the case with companies that have returnable containers, pallets, or products for internal exchange.

The purpose of supply chains can be broken down into: end-customer satisfaction, chain efficiency, and supplier chain flexibility. Successful supply chain management must result in improvements in the business's touchpoints.

The primary focus of supply chain management is on: increasing profit, better resource utilization, and cost reduction.

Primarily, the goal of any supply chain is the satisfaction of end consumers: the end customer is the only one who injects the "existing" money that drives the chain of activities for all partners. The money is distributed among the chain members proportionally to their added value.

⁵⁰ Ljiljana Stošić Mihajlović, 2014., *Business Logistics*;

⁵¹ Rogers D., Tibben-Lembke R., Reverse Logistics Executive Council, Pittsburgh 2005., *Going Backwards: Reverse Logistics Trends and Practices*;

⁵² Dr. Dušan B Regodić, 2019, *Development and Integration of Logistics in Organizations*;

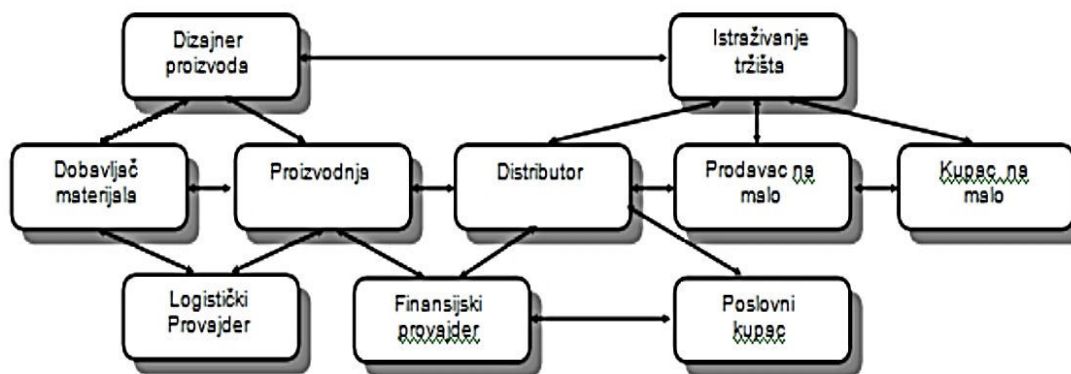


Figure 19 – Supply Chain ³¹

Logistics deals with the movement of goods from the perspective of a single company, meaning the movement of materials and goods that a business receives and manages internally, as well as when it delivers those goods to a customer. The supply chain, on the other hand, is a network of businesses that sequentially handle the production or distribution of goods or services. In short, logistics is generally a single company's concern, while the supply chain is a concern of multiple companies. Although logistics can be coordinated across part or even the entire supply chain, each segment is under the authority of a single entity until it hands over the material or product to another entity in the supply chain.⁵³

Logistics is responsible for the flow of materials, i.e., their movement and storage, through the supply chain, and is temporally linked to resource positioning or the strategic management of the entire supply chain. A supply chain is a series of events in the flow of materials, information, and services aimed at satisfying the customer.⁵⁴

Supply management, manufacturing operations management, and integrated logistics management are focused on:

- a) on the coordination of procurement of materials from external sources,
- b) on eliminating bottlenecks and reducing their impact on consumers and
- c) on the continuous flow of products and services to, through, and from the firm

The supply chain management process begins with intensive price negotiations to conclude long-term contracts with first- and second-tier suppliers. Supply chain managers often differentiate between "approved suppliers," "preferred suppliers," or "supplier alliances." By doing business with With "accepted" and "preferred" suppliers, the supply chain manager can analyze the quantities and prices of competitors' products and use the resulting information to manage their own company. This approach may require the firm to interact only with first-tier suppliers. However, doing business with a supplier alliance requires more complex interactions for the firm with both first- and second-tier suppliers.

By fully implementing the supply chain management model described, the fragmentation and suboptimization of product and service flows are minimized. When a firm builds a unified system of goods and information flow with other chain members, it actually increases the efficiency of managing the entire chain. However, the full implementation of the supply chain management concept (Figure 20) also requires:

- developing strong business relationships between supply chain members
- development of high-quality products and services
- the sharing of information between supply chain members
- reducing order cycle time
- minimizing inventory levels throughout the supply chain
- reduction in the number of suppliers and carriers

⁵³ Abby Jenkins, Product marketing manager, 2020., *Logistics for Business Defined: Importance, Role & Benefits*;

⁵⁴ Prof. Dr. Dušan Ragodić, Belgrade, 2010, *LOGISTICS – First Edition*;

- increasing adherence to the concept of supply chain management.

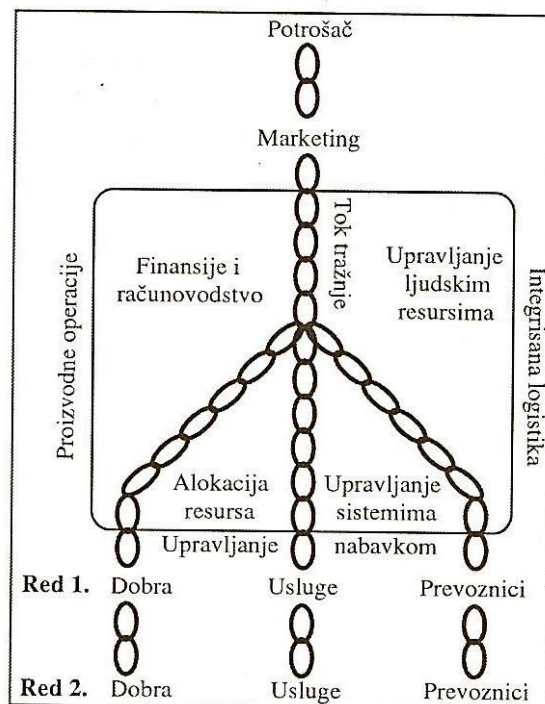


Figure 20 – Supply Chain Management Model

If applied correctly, the supply chain management concept enables the faster movement of products and services to end consumers. In this way, the value of the entire system for the consumer is increased and a strategic competitive advantage for the chain is created.⁵⁵

⁵⁵Sotirović Velimir, Egić Branislav, Technical Faculty Zrenjanin 2005, *Business Logistics Management*;

5. INTRODUCTION TO CRYPTOCURRENCIES AND BLOCKCHAIN TECHNOLOGY

In recent years, we have witnessed the continuous and ever-faster development of what are called cryptocurrencies or, as the name suggests, "hidden" digital currencies. Cryptocurrencies are created so that their value is not tied to gold or any other form of tangible assets and is not under the jurisdiction of government institutions and regulatory bodies. The way their value is formed is based on supply and demand in the crypto market, where there are various trading platforms, the most well-known of which today is „Binance" [11].

Cryptocurrency is a digital currency, usually decentralized, designed for use on the internet that does not rely on banks and other intermediaries to verify transactions. Since most cryptocurrencies are not regulated by national governments, they are considered *alternative currencies*, i.e., financial means of exchange that exist outside the boundaries of state monetary policy.

In essence, the term "cryptocurrency" was created by combining the word "*crypto*" and "*currency*." "Crypto" comes from the science of cryptography, which enables a secure process for executing digital transactions, while the term "currency" refers to a medium of payment and gives monetary value to the digital record. The cryptocurrency system uses cryptographic protocols or highly complex coding systems that encrypt sensitive data to secure the medium of exchange.

The creator of this technology, Satoshi Nakamoto, intended to empower users with their finances by eliminating the need for a third party, such as banks, to facilitate transactions. The goal was for cryptocurrency transactions to operate on a "peer-to-peer" basis, meaning direct communication between the buyer and seller where their personal data and identities are encrypted and not transmitted. In this way, users could control their own funds without other entities being informed of the purpose for which the funds are being used. This leads to the main difference between cryptocurrency transactions and those made with credit or debit cards, as cryptocurrencies allow the user to send exactly what they want to the recipient without any additional information. The use of cryptocurrencies does not require providing any names or other information, only a digital wallet address (European Commission, 2020).

After the initial launch of Bitcoin in 2009, officially the first cryptocurrency, this payment system gained the potential to become an alternative to government-issued money in just a few years. This would reshape the financial world as we know it and call the existence of traditional financial infrastructure into question. The number of people trading these digital currencies to achieve above-average profits, as well as the number of countries legalizing them, is constantly growing. It is noticeable every day that an increasing number of companies worldwide are offering the option to use cryptocurrencies as a means of payment. Furthermore, Bitcoin *futures contracts* have even been listed on two U.S. exchanges (CME and CBOE). Today, there are about 7,800 cryptocurrencies, which are traded on over 200 virtual exchanges worldwide.

Unlike physical, tangible money that is transferred and exchanged in real time in the form of coins and banknotes, cryptocurrency payments exist exclusively as digital entries in an online database that describe specific transactions which are verified by network nodes. When transferring cryptocurrency funds, transactions are recorded in a decentralized, public ledger of transactions called a "blockchain"⁽¹²⁾. The blockchain essentially represents a database that is regularly updated with information about all transactions that have occurred since the last update. Each set of recent transactions is combined into a "block" using a complex mathematical process. To produce new blocks, cryptocurrencies rely on individuals to provide computing power. Essentially, the cryptocurrency system rewards people who supply this system with computing power. These individuals who trade computing resources for currency are called "miners."

Since cryptocurrencies are virtual money, they can only be accessed digitally. Consequently, it is necessary to have the software and hardware technological infrastructure to use cryptocurrencies. Hardware

⁵⁶A cash settlement in USD based on the Bitcoin Reference Rate (BRR), which serves as a reference rate once a day for the price of bitcoin in US dollars [28].

infrastructure refers to a computer that has internet access and meets the appropriate technical specifications. To receive and store cryptocurrencies or to send them, specific software in the form of a "*digital wallet*" is required. (see **5.2.2 Software Elements**). To enable transactions between different digital wallets, a decentralized Blockchain system is used, and the "production" The production of cryptocurrency takes place through a process called "mining" (see **1.1 Basic Elements of the Cryptocurrency System's Operation**), which involves using the power of computer systems to solve complicated mathematical problems that generate new coins. Users can also buy currencies from brokers and then store and spend them using digital wallets.

All of the previously mentioned different concepts related to the functioning of the cryptocurrency market will be explained in more detail below.

5.1 Basic Elements of How Cryptocurrency Systems Work

5.1.1 "Blockchain"

The term "Blockchain" comes from the words "block" and "chain" because it represents a chain of interconnected data where one piece of data follows another. It is a decentralized and public digital ledger that is used to record transactions of digital currencies across numerous computers.

Simply put, a blockchain is a publicly accessible database where all transactions between users have been recorded. Cryptography, or encryption, ensures that these transactions cannot be deleted or altered, but instead remain permanently stored as an electronic record in digital format.

The key difference between a typical database and a blockchain is in the way the data is structured. A database typically organizes its data into tables, while a blockchain collects information in groups, known as "blocks," which contain sets of information. Blocks have a specific storage capacity, and when they are filled, they are closed and linked chronologically to the previously filled block, forming a data chain. All new information added to the database after a freshly formed block is collected into a new block, which will also be added to the chain as soon as it is filled. This data structure creates an immutable timeline of data, with each new block becoming part of it upon completion, assigned a timestamp of its exact addition to the chain.

Each record block consists of four data items:

- A reference to the previous record block
- A hash of the transactions contained in that block
- Timestamp of that record block's creation
- Proof of creation of the respective block of protected records⁵⁷

This type of database is actually a distributed system, that is, a system consisting of multiple entities that process transactions. The entities of this distributed system are computers that are networked and jointly confirm transactions. The data in these systems remains unaltered, transparent, and secure.

[15]

What blockchain essentially enables is for the data held in that database to be distributed among multiple network nodes, or computers, in different locations, with an accurate and transparent order of events. The data in the database is protected from any subsequent changes unless the entire chain is synchronized and the changes are made unanimously. This further implies that any change to a particular block would be impossible without modifying all previous blocks, i.e. without the participation of the entire network, so that no single entity or group has direct control over the data. This ensures that each node, or user participating in the blockchain, is also a custodian of all information about a given transaction. In this way, if one user modifies a transaction record, all other nodes would easily determine which of them has the incorrect information. The aforementioned system ensures data security

⁵⁷Pavić, 2020.

and trust that they will be protected from any malfeasance that would harm users without the need for a trusted third party.

One of the main characteristics of cryptocurrency payment systems is the elimination of a third party, most often banks, when it comes to the use of merchants for other activities. In a situation where every bank customer decided to withdraw their balance from their bank account at the same time, the banks would not have enough money to pay each customer their money. Most of that money is not physical cash in the form of banknotes, but only a digital record of a particular customer's money. Furthermore, in today's world, a significant percentage of the world's money exists only as a record of who owes what amount to whom, and banks and similar institutions are the central entities that maintain these records. In classic banking transactions, where, for example, Person A transfers money from their bank account to Person B's account, the bank is the entity that must inevitably verify that Person A has enough money in their account for the transaction to be successfully completed. Additionally, the central entity, i.e., the bank that maintains the records of transactions, can change the current balance of bank accounts, make amendments to the list of debtors and their debts, alter transaction history, and decide who can and cannot perform a transaction, thus having unlimited control over the bank accounts. Blockchain is a technology that allows for decentralization and the execution of transactions without a central entity, but rather by using decentralized computers that confirm the transactions.

There are several types of blockchains, and they are:

- *Public blockchain* - Allows anyone to have insight into transactions
- *Private blockchain* - Can only be used internally, for example, within a specific organization, so it may have certain limitations for participants
- *Consortium blockchain* - Controlled by multiple different organizations, each of which has authorized access, so anyone can create a node on such a network⁵⁸

5.1.2 "Consensus"

Since, as already explained, the cryptocurrency system is decentralized and distributed, the information entered on the blockchain must be immutable, i.e., it must not be changed afterward. To ensure their immutability, all added transactions must be cryptographically protected. However, to ensure this characteristic is met, a consensus among the network's participants is required.

Each block of the cryptocurrency chain operates using a "consensus" mechanism and is of utmost importance to the entire blockchain network. This mechanism includes protocols, algorithms, or other computer systems that allow the cryptocurrency trading system to exist and be implemented correctly. This system is based on "agreements" which determine the validity of transactions and the management of the blockchain.

There are different types of consensus mechanisms with various advantages and disadvantages. This mechanism is used by cryptocurrencies, or more specifically, by blockchain users, to verify the authenticity of transactions and maintain the security of the underlying blockchain. In essence, when speaking of consensus in the context of cryptocurrencies, the term refers to the method by which blockchain participants "agree" on what should be included in a particular chain. This is also why these systems are referred to as "systems of proof" [17].

The significance of this mechanism stems from the fact that it forms the basis for ensuring that all new legitimate transactions are recorded on the blockchain and that all miners agree on the next block of transactions. Additionally, the mechanism distributes information about each new block to all other miners, allowing any user on the network to download a copy of the blockchain to their computer as a node, which is an exact match of the main chain. At the same time, consensus ensures that each copy of that blockchain contains all valid transactions while ensuring agreement with past transactions. It also affects how transactions are verified, how much energy is used, network fees, transaction speeds, and other details related to the currency and network applications. Computers that confirm new transactions on most blockchains are called "*miners*."

⁵⁸Živkovic, 2018.

The two most prominent consensus mechanisms currently used by crypto networks are **Proof-of-Work** and **Proof-of-Stake**. Besides the ones mentioned, there are other forms of consensus mechanisms such as "Proof-of-Authority," "Proof-of-Capacity," "Proof-of-Activity," "Proof-of-Burn," etc.

5.1.2.1 "Proof-of-Work" algorithm (PoW)

The Proof-of-Work algorithm is used to achieve consensus among computers in the network and to verify transactions. It is a very energy-intensive consensus mechanism because it involves "**mining process**," but on the other hand, it provides a high degree of trust.

Participants in a network based on this mechanism can choose to be "**miners**." This system works by having all the computers in the network—the miners—compete against each other in solving "compression functions" to verify the next block of transactions^[17]. It is a function whose solution is very demanding and complex, but verifying its solution is a very simple process. The purpose of solving these mathematical problems is to simulate the device, in this case the corresponding computer, consuming electricity. In return, the node is rewarded with a certain amount of cryptocurrency, including the transaction fees. This process of creating new cryptocurrency units is called "**mining**." When a miner "guesses" the solution, a new block is created on the network's blockchain, where other miners then confirm the correctness of the given solution. The "winning" miner earns a mining "reward," which is paid by the users who initiate transactions on the network. For example, when person A sends Bitcoins to person B, person A pays a certain fee in Bitcoins for this transaction. The miners who confirm the transaction are then rewarded with that fee. This fee, or "**block reward**," is in the form of a certain amount of newly created cryptocurrency, which is used as an incentive for participants to continue the mining process and help maintain the chain. The mining software transfers all earned rewards to the crypto wallet address specified by the user. The reward amount is variable, and currently, it is possible to receive 6.25 Bitcoins for a single mined block on the Bitcoin blockchain⁵⁹.

The concept of distributed computers mining cryptocurrency enables the decentralization and security of the blockchain, but it comes at a cost. Since solving the hashing function requires a lot of processing power, the average computer is less likely to find a solution. For this reason, special, extremely expensive computers with advanced graphics cards are typically used for mining. The more computing power a node has, the more likely it is to solve the mathematical problem first and, in turn, receive a reward faster. This is also the reason why in PoW-based blockchain systems, significant pooling occurs (in a "**mining pool**"). It is a collective group of cryptocurrency miners who combine and pool their computing resources over a network to increase the probability of finding a block or otherwise successfully mining for cryptocurrencies^[24]. This method of pooling is typically used by miners of large cryptocurrencies like Bitcoin, for which the process of solving mathematical calculations requires special computer equipment that is difficult to acquire as an individual.

Besides the cost of the computer, the most expensive factor is undoubtedly electricity, as the special computers dedicated to mining consume significant amounts of this resource. The Bitcoin network currently uses more than 73 TWh of energy annually^[6]. Since the mining process itself is not cheap, the mining process is one of the elements that gives cryptocurrencies a certain value as well as a degree of security to users. This is achieved by the fact that the only way for miners, i.e., the participants who maintain the network, to cheat the system is with 51% of the total computing power of the entire network. Even then, it is impossible to change already-entered transactions; it is only possible to halt subsequent ones. Considering the amount of capital that must be invested, there is no economic incentive to go against the system.

In addition to the high cost of equipment and the amount of electricity required for the mining process, there are other drawbacks to the algorithm that are important to consider. The larger the network, the more energy is required to confirm transactions, and the more time it takes for transactions to be confirmed, which in turn means fewer transactions per second. These disadvantages are also the reason this method is questioned, as well as whether there is an economic sense for cryptocurrencies based on it to be used for exchange.

⁵⁹ <https://www.binance.com/en/price/bitcoin>

Overcoming these inherent flaws or problems of this method directly depends on the increasing computing power of devices that people own worldwide. Despite the aforementioned shortcomings, this mechanism is the most widespread, and the most popular cryptocurrencies today are based on it: Bitcoin, Litecoin, Bitcoin Cash, Ethereum, Game Credits, and others.

5.1.2.2 "Proof-of-Stake" algorithm (PoS)

The Proof-of-Stake algorithm operates on the principle of the largest monetary stake, based on which its owners validate new blocks. In this way, users with the largest stake in the network are "rewarded," and transactions are faster and cheaper^[17]. PoS is a built-in consensus mechanism that uses a blockchain designed to reduce network congestion and environmental pollution compared to the PoW algorithm. It cannot be earned in any way, but it helps secure the network and allows users to receive rewards by using a cryptocurrency client that participates in PoS validation or by becoming validators themselves^[21].

The use of the previously explained "Proof of Work" algorithm involves a mining process, which means that transaction confirmation is relatively slow and requires owning extremely expensive computer equipment. Besides this drawback, the given algorithm has a negative impact on the environment because its operation requires consuming a significant amount of electricity. "A cryptocurrency's dependence on large amounts of electricity is unsustainable, and can only survive in an ultra-stable world" (Škvorec, 2018.).

Given the above, an increasing number of cryptocurrencies today are beginning to adopt and implement a new way of verifying transactions—the "Proof of Stake" concept, which does not involve the classic mining process at all and does not require solving cryptographic-mathematical puzzles, and is therefore faster and more efficient. Thus, the need for high electricity consumption and the use of expensive hardware is automatically eliminated. Users who were miners in the "Proof of Work" algorithm have the role of "**validators**" in this algorithm. Validators confirm (eng. Validate) transactions and, as with the PoW algorithm, receive a reward for this from the user initiating the transaction. To be considered trustworthy, validators must lock up a certain amount of their funds, i.e., make a stake (eng. Stake) in the form of a specific amount of cryptocurrency^[22]. If a validator confirms a false transaction, their stake is forfeited, so given the risk of potentially losing their stake, it is not in a validator's interest to engage in any malfeasance. In this way, the integrity of the entire concept is maintained.

PoS can also take into account other determining criteria besides the criterion of the wealthiest nodes, including the length of time a node has staked its funds, as well as pure randomization. The block reward in PoS refers to the network fee that the blockchain awards to the person who submits a valid block, similar to the PoW mechanism. In PoS, block selection is based on coin ownership, so exchanges offer staking services, which allow users to stake cryptocurrencies on their behalf in exchange for more consistent rewards. Multiple interested parties can join a staking pool to combine their computing resources and maximize their chances of being rewarded. In other words, they pool their staking power during the verification and validation of new blocks to have the best possible chance of receiving a block reward.

5.2 Technological Aspects of Cryptocurrencies

5.2.1 Hardware Elements and Requirements

The previously explained mining process uses vast amounts of processing power to secure and verify transactions, as well as to simultaneously create new coins. Given this, users must meet various requirements regarding the necessary hardware and its performance if they want to engage in a mining process that would be economically viable.

When Bitcoin mining began in January 2009, no special equipment was required to perform the process. Mining was done by computer processors with the specifications of an average home computer (**CPU mining**). At that time, no one saw an interest in optimizing the mining process and making it more efficient. In mid-2010, it began to slowly develop

The price of Bitcoin has made mining slowly profitable, and users are looking for ways to earn a larger amount of coins through this process. The solution was found in graphics cards (**GPU mining**), as it was discovered that with the right software, graphics cards can mine cryptocurrencies significantly more efficiently than processors. Since then, users have been in a race to find and develop the most powerful graphics cards with better performance to gain the greatest possible advantage over other miners. However, although a large number of cryptocurrencies are still mined this way today, GPU mining has several drawbacks—graphics cards consume a significant amount of electricity, are very noisy, and generate a huge amount of heat that needs to be dissipated, so the search for an alternative was intense.

FPGA boards appeared on the market in 2013 in response to the need to improve the mining process. Although they were not very powerful, they consumed significantly less electricity, but they never managed to completely replace graphics cards. Partly because they were relatively expensive, and partly due to the emergence of **ASIC machines**, which represent a far better option than FPGA and GPU mining. Graphics cards were not purpose-built for cryptocurrency mining, so although it is possible to do so, their performance is limited. On the other hand, FPGA boards are composed of components that can be connected to form a single machine exclusively configured for the mining process—a "**rig**." ASIC devices have taken this idea a step further, as they are designed and manufactured solely for a single specific function: in this case, mining.^[30]

5.2.2 Software components

5.2.2.1 Mining Software

As explained, the "mining" process represents the PoW consensus, i.e., the creation of new cryptocurrency units by solving mathematical problems that simulate work, for which the computer consumes electricity. However, in addition to the hardware components, the corresponding "mining" software is also required to enable the computer hardware to be used for managing the blockchain technology and solving complex equations in order to earn a reward—new coins.

Most Bitcoin mining software is free and works on all major operating systems. In the early days of cryptocurrency, it was possible to use the standard processors found in ordinary home computers for mining; however, today these programs require increasingly powerful computer hardware as more users join the process.

Some of the most well-known crypto mining programs today are: "CGMiner," "Awesome Miner," "Easy Miner," "Kryptex Miner," ECOS, etc.^[38]

5.2.2.2 Digital Wallets and Digital Wallet Technology

A prerequisite for owning and trading cryptocurrencies is for the user to open a unique "wallet," or digital wallet, which is installed on a computer or mobile device or created online.

Crypto wallets do not store cryptocurrencies in the same way a physical wallet stores cash in someone's pocket. Instead, they store "public" and "private keys"—cryptographic methods that make up the wallet and are necessary for purchasing cryptocurrencies, providing digital signatures that authorize each transaction. In essence, the person who owns the cryptocurrency actually only owns a digital record of the cryptocurrency on this software. Wallets also provide an easy-to-use interface for managing crypto asset balances for buying or selling and support cryptocurrency transfers over the blockchain.

When cryptocurrency transactions are made, they do not represent the "sending" of crypto tokens, i.e., a specific number of cryptocurrency units, from a person's mobile phone or computer to another person's mobile phone/computer. When sending tokens, the user's private key signs the transaction and broadcasts it to the blockchain network, i.e., the public ledger. The network includes that transaction and reflects the updated state of the sender's and receiver's addresses.

Since the entire process of receiving, sending, and storing cryptocurrencies is conducted via the internet, it is necessary to ensure that a third party cannot access or alter the transaction or the amount of funds on

digital wallets. To achieve this, cryptocurrencies use a mathematical approach to protect information using "cryptography." The cryptographic algorithm used to create digital wallets, i.e., to generate private and public keys, is called the "Elliptic Curve Digital Signature Algorithm" (Sharma Kumar, 2018).

Sending or receiving cryptocurrency can be likened to sending messages that represent transactions. In this process, they need to be signed or, to put it simply, verified with a "secret or ...private key." This key is a fundamental cryptographic tool whose task is to enable the sender and receiver to communicate securely over a communication channel such as the web, in a way that a third party that may potentially have access to the communication channel cannot understand their messages (Dujella, n.d.). The rules for encryption and decryption are a pre-arranged "secret or private key" or a set of specific codes through which the recipient verifies their address in order to execute a transaction^[25]. Only individuals who have the secret key will be able to decrypt the encrypted messages and learn their content.

Two people who want to exchange messages can pre-arrange the encryption rules they will follow. For example, each letter of the message being sent can be replaced with a number (the letter P with the number 1, the letter A with the number 2, the letter S with the number 3, etc.). If they agree on this rule in advance, and one person sends another the message "123," the other person will know that 1 represents P, 3 represents A, and 2 represents S, and will thus be able to decrypt the message. If the message is sent over the internet, a third party monitoring the network will only see the encrypted message "123" and will not be able to interpret it correctly, meaning they will not know the message actually says "PAS". This is an example of **symmetric message encryption**, which is the simplest yet very secure method of encryption.

In addition to the "private key" to carry out a transaction, a "public key" is also required. This is the recipient's address, which can be sent to any sender from whom a payment to a digital wallet is expected, or it can be published publicly. The sender can be any user making a purchase with cryptocurrency, an exchange through which cryptocurrency is bought, and so on.

On the other hand, there is also an **asymmetric encryption** mechanism. In this case, the same goal needs to be achieved, but the method will differ from the previous example. It is applied in situations where a cryptocurrency transaction needs to be made online, where the sender and receiver do not know each other and it is impossible to pre-arrange an agreement on the encryption rules, i.e., the secret key.

With asymmetric encryption, the sender's private key is used only to encrypt the message, while the recipient has their own private key that can only be used to decrypt the transaction. The essence is that there is not one key that the parties have pre-arranged, but rather each person has their own unique private key with which they can decrypt the message, or in this case, access the funds in a digital wallet. With this method of encryption, the two parties, the sender and the recipient conducting the transaction, do not need to share a private key.

Asymmetric encryption is performed using an algorithm that employs a *one-way function* to make it impossible to decrypt the message using only the public key. A one-way function is any function that is computationally impossible to calculate in a reasonable amount of time (Code School, n.d.).

5.3 Cryptocurrency exchange

Besides mining, individuals who want to own crypto coins can acquire them by purchasing from other users—either miners or users who possess a certain amount of coins they simply want to sell—as well as through an exchange. This leads to the field of cryptocurrency trading.

5.3.1 Formation of the cryptocurrency price

Today, all currencies in use in the world are fiat currencies. This is money that is not backed by any specific commodity like gold or silver, but its value is given by the guarantee of the government institution that issues it, usually the Central Bank⁶⁰.

⁶⁰ <https://www.centarzlata.com/fiat-novac/>

By the same principle, the value of cryptocurrencies is also not based on any single commodity; however, there are significant differences compared to fiat currencies.

If we compare the value of cryptocurrencies and fiat money, the biggest difference is that fiat currencies are backed by Central Banks and adopted as legal tender. Their value essentially stems from the fact that the Central Bank and the state have declared that they have value and stand behind them. In the case of a transaction, the sender and receiver base their trust in the value of fiat currencies on that fact.

Cryptocurrencies, on the other hand, are not controlled by a Central Bank or authorities, and most regions do not accept them as legal tender. Trading in these currencies actually has significant similarities to trading in stocks, and it often happens that the cryptocurrency market moves in the same direction as the stock market during times of crisis. Unlike stocks, banks do not favor clients who trade cryptocurrencies because they represent competition to the banking sector, as transaction fees do not go to banks but to miners. Additionally, the price of a stock is significantly dependent on the performance of the company that issued it, whereas this is not the case with cryptocurrencies, and several factors influence their price changes, such as:

- The market's supply and demand for it
- The cost of creating currency units through the mining process
- The number of competing cryptocurrencies
- Regulations governing the sale and use of cryptocurrency
- Media and news ^[26]

The supply of cryptocurrencies in the market plays a vital role in determining their prices. It is highly likely that cryptocurrencies with a scarce supply will have high prices, while those available in abundance will have low prices.

The places where it is possible to track the supply and demand of cryptocurrencies are cryptocurrency exchanges, where a large number of users place their orders, which are like buy or sell orders for cryptocurrencies (Orders). This information can be interpreted as the buy and sell rates at a given moment. If someone were to decide to buy, for example, 50 Etheriums, the price would rise because the order would be filled and the subsequent orders would move. The same applies in reverse; the price would fall if someone were to decide to sell a large amount of Ethereum.

In addition to tracking the value relationship between different cryptocurrencies like Ethereum and Bitcoin, it is also possible to track the dollar-to-Bitcoin or Euro-to-Bitcoin ratio. The mid-rate, or current price, is the price at which the last order was executed, and the cryptocurrency's value is tracked at that price. One of the most accurate methods for tracking cryptocurrency values is the "*CoinMarketCap*" web application. This application works by taking the average rate for each cryptocurrency from all available cryptocurrency exchanges where it is traded and then calculating the average rate. Since cryptocurrencies are continuously being bought and sold, the exchange rate is constantly changing. For this reason, the web application refreshes several times a minute, and often the exchange rate for the same cryptocurrency is different with each refresh. Cryptocurrencies are characterized by frequent and rapid price changes, which is why it is possible to make large gains or large losses by trading cryptocurrencies.

As with other products, production costs play a fundamental role in determining the price of cryptocurrencies. For example, in the case of Bitcoin, the production cost is approximately the sum of the direct fixed costs of the infrastructure and electricity required for mining the cryptocurrency, and the indirect costs associated with its algorithm's difficulty level. Different levels of a cryptocurrency's algorithm difficulty can speed up or slow down the production rate, or the mining process, and affect the total supply, which in turn can affect the price in the case of a PoW mechanism. The higher the coin's mining difficulty, the harder it is to increase the supply and cause upward price pressure when demand is high.

When we talk about legal regulation as a factor influencing cryptocurrency prices, it is important to emphasize that cryptocurrencies largely remain an unregulated area. Nevertheless, developments in this area can indeed drive changes in cryptocurrency prices. For example, in the US, decisions about cryptocurrencies made by the Securities and Exchange Commission (SEC) can affect cryptocurrency prices. In October 2021, the price of Bitcoin rose to \$69,000 a few weeks after the SEC approved the first American ETF

tied to Bitcoin: ProShares Bitcoin Strategy ETF (BITO)⁶¹. Another situation that significantly lowered the price of Bitcoin was the ban on trading and transactions in Bitcoin by the Chinese government in September 2021, which affected the supply and demand of cryptocurrencies. Chinese miners were forced to pack up their machines and move them to countries friendly to cryptocurrency trading and mining. This was reflected in the price of Bitcoin, which dropped to around \$51,000 in early September and then to about \$41,000 by the end of the month.

Although Bitcoin is the world's most well-known cryptocurrency, there are numerous other cryptocurrencies to invest in that are awaiting user investment. In 2017, Bitcoin dominated the global cryptocurrency market, accounting for 80% of the market capitalization; however, by 2022, that share had fallen to less than 50%. The main reason for this is increased user awareness of the existence of alternative coins and the opportunities to invest in them.

In addition, the price of cryptocurrencies is significantly influenced by the media or public sentiment. If a particular currency receives negative publicity, the price of that coin will generally fall. Conversely, if the same coin were to receive significant support, positive user reviews, and good media coverage, its price would almost certainly increase. This means that cryptocurrency prices are heavily influenced by human emotions and hype.

5.3.2 Places to trade cryptocurrencies

A newly created cryptocurrency wallet will always be empty, and if you want to own a specific cryptocurrency, this is possible either by mining or by purchasing it. If you don't want to mine and decide to buy, it can be done in several ways. The most ideal and simplest method is to find a person who is willing to sell cryptocurrency, meet with them, and complete the purchase by giving them money and having them send the cryptocurrency to our digital wallet. What greatly simplifies the search for someone willing to sell cryptocurrency are cryptocurrency trading venues, most often in the form of an online web platform. These venues are divided into:

5.3.2.1 Cryptocurrency Exchanges

Cryptocurrency exchanges are private companies that allow clients to buy or sell cryptocurrencies for cash. An example of one such exchange is "Bitcoin store," which has a website where you can buy 68 different cryptocurrencies by depositing funds into the exchange's account or by paying with a card. There is also an option to sell, which first requires sending the cryptocurrency to the exchange's digital wallet, after which the exchange deposits the proceeds from the sale into the client's bank account. Such companies take a commission of about 1 to 5% on each transaction (Bitcoin store, 04/05/2023.). Besides online exchanges, there are also physical exchange locations, such as branches or ATMs, for buying and selling cryptocurrencies for cash. Due to the significantly higher fees for using these types of exchanges, most people still opt to use online exchanges or purchase through cryptocurrency exchanges.

5.3.2.2 Cryptocurrency Exchanges

Cryptocurrency exchanges differ from exchanges in that cryptocurrencies are bought and sold from other users, not from the exchange itself. The exchange simply acts as an intermediary, providing a web platform for trading, and charges only a small (often around 0.1%, Binance, n.d.) fee on each buy and sell, which is significantly lower than the fees for using cryptocurrency exchanges. A sale or purchase is executed by placing a sell or buy order, where the user chooses the price at which they want to sell or buy the cryptocurrency and waits for someone to execute that order, i.e., to buy the cryptocurrency from the user or sell it to the user.

In addition to the ability to buy or sell cryptocurrencies, exchanges also offer the opportunity to predict cryptocurrency prices and bet on their rise or fall. For example, if a user predicts that

⁶¹ <https://www.proshares.com/press-releases/proshares-to-launch-the-first-u.s.-bitcoin-linked-etf-on-october-19>

a specific cryptocurrency will fall 5% during the observed period, it is possible to bet on that and actually profit when that cryptocurrency falls.

Exchanges that offer betting also very often provide the option of **leveraged trading**, which is attractive to users because it allows them to achieve significantly greater buying and selling power with a smaller amount of money.

To open a \$10,000 position in Bitcoin, a \$1,000 base is required. If the price of Bitcoin rises by 20%, you will earn a net profit of \$2,000 (minus fees), which is much more than the \$200 that would be possible to earn if you were trading with the \$1,000 of your own capital without leverage.

However, if the price of BTC falls by 20%, the position would lose \$2,000. Since the initial capital is only \$1,000, the 20% drop would trigger a liquidation (the account balance falls to zero), which can be avoided by adding more funds to one's wallet to increase the margin. This method of making money on the exchange is attractive, however, it carries a high risk and often leads to losses for inexperienced users^[28].

5.4 Legal Aspects of Cryptocurrencies

Each country has its own specific rules for dealing with domestic and foreign currencies. Foreign currencies are generally regulated by special laws. Citizens can conduct transactions in any foreign currency according to rules that are often linked to other laws, for example, investment restrictions, sector-specific investments, and corporate laws.

The legal status of cryptocurrencies varies from country to country and is still undefined or in a state of change in many of them. This means that cryptocurrencies are not explicitly declared illegal, but that does not mean they are legalized. A large number of countries, such as Croatia, tax profits earned from trading cryptocurrencies but have not officially adopted them as legal tender (RRIF, 2018). While most countries do not legally prohibit the use of cryptocurrencies, their status as money is not firmly established and remains undefined.

In countries such as Morocco, Bolivia, Ecuador, Kyrgyzstan, and Bangladesh, the use of Bitcoin is illegal and punishable by law, while it is legal in Brazil, Chile, Colombia, and Argentina.

Taking this into account, the emergence of cryptocurrencies on the market has raised legal questions related to the trading of this form of money and its largely unregulated status.

Public interest in cryptocurrencies, such as Bitcoin and Ethereum, is constantly growing as more industries explore how to leverage the new technology. As public interest increases, new uses for cryptocurrencies are developing and emerging almost daily. With their emergence, potential legal obligations also arise.

The response of central banks in various countries to these issues has been lukewarm. While some central banks strongly support cryptocurrencies and look forward to their development to align with a modernized financial system and speed up payments, other states are cautious due to extreme levels of volatility and potential security risks. The need for control over capital and the taxation of assets acquired through crypto trading has increased concerns in this area.

Some of the legal issues related to the crypto market are:

- *Contractual issues*

One of the most striking features of blockchain technology and cryptocurrencies is their self-executing "smart contracts." *Smart contracts* are a set of promises, usually listed in a digital format, that serve as the basis on which parties in a transaction fulfill their specific promises. A smart contract automatically pays the other party when it fulfills its contractual obligations. Due to the unique nature and inherent complexity of smart contracts, it is difficult to determine whether they fit within the legal framework of traditional contract law. For example, the law of certain U.S. states provides for the limited legal enforceability of smart contracts. However, since the legal validity of smart contracts is unclear, they are likely to lead to protracted litigation.

- *Jurisdictional issues*

The main idea behind the blockchain technology that supports cryptocurrencies is that it does not include a way to precisely determine the actual location of the public ledger. Consequently, transactions conducted on the blockchain offer greater privacy than transactions carried out on traditional platforms. But this advantage presents a complex challenge for the relevant regulatory institutions. First, because the nodes of a crypto transaction are located in different jurisdictions, they can be subject to conflicts between different legal frameworks (for example, one country has a regulated status for cryptocurrencies while another does not, or it simply differs significantly). Second, it is difficult to determine the "country of residence" for cryptocurrency software due to the lack of a physical location on the ledger. Third, the transnational nature of the blockchain makes it extremely difficult to determine applicable laws and select the correct jurisdiction for enforcement. The absence of regulation means that cryptocurrencies can be freely used across state borders and are not subject to the same controls that governments impose on fiat currencies. For any state, enforcing the law among blockchain users, transactions, or projects involving cryptocurrencies is currently a difficult task due to the technology's cross-border nature.

- *Data theft, financial fraud, and the financing of criminal activities*

Data theft and financial fraud are additional urgent legal issues related to cryptocurrencies. Also, the promise of blockchain anonymity and its apparent freedom from regulation may lead many users involved in illegal activities such as drug and human trafficking, money laundering, terrorist financing, and more.

In 2017, a researcher from Cornell University identified a serious security flaw in the Ethereum blockchain that put \$250 million at risk of theft. Similarly, the crypto wallet manufacturer "Ledger" recently exposed one million email addresses, compromising data security. Additionally, personal data, such as full names, mailing addresses, and phone numbers, were stolen.

9,500 customers. It is unclear whether existing data laws can prevent data theft and financial fraud stemming from cryptocurrency trading.

In light of the above, the European Central Bank issued a warning to point out the possible risks to which individuals may be exposed when handling or trading cryptocurrencies (European Central Bank, 2015, p. 4). First, if a platform that holds or trades one's virtual currencies fails or ceases operations, there are no specific regulatory protection mechanisms to secure owners against losses. The warning highlights the fact that users often lose significant amounts of money held on these platforms because they are often not subject to any legal regulations. There are many recorded cases of various platforms or exchanges ceasing operations due to the bankruptcy of the exchange owner and sometimes due to hacking attacks. In the event that money disappears from a particular trading platform or it goes bankrupt, there is no specific legal protection in the form of a deposit insurance agency that would provide insurance for cryptocurrency owners. Furthermore, losing a key or password can mean the permanent loss of virtual money, as there are no agencies that keep records of passwords or issue new ones^[20].

5.5 The Economic Aspect of Cryptocurrencies

Today, just 14 years after their creation, the connection between cryptocurrencies and the economy is visible in many sectors. For example, education is an area where the potential use of cryptocurrency technology is rapidly increasing. Recently, some universities in Cyprus, Switzerland, the USA, and Germany have accepted cryptocurrencies as a form of tuition payment on their websites, while other online educational institutions accept payment via Bitcoin. Some transportation companies have also approved the use of crypto-currencies as a means of payment, and the ability to purchase airline tickets, hotel reservations, car rentals, and cruises with these currencies is accelerating the system's expansion.

The housing and real estate sector is another area where cryptocurrencies are being increasingly applied. With the growing number of companies accepting cryptocurrency payments, it is believed that investments in this area will increase.

Retail is also one of the important sectors showing interest in cryptocurrency technology. The first example in this regard began with the website *Overstock.com*, which provides furniture manufacturing and sales services, accepting Bitcoin payments. Following this, many retail portals, including major chains such as *Crate and Barrel*, *Nordstrom*, and *Whole Foods*, have added the option to purchase with cryptocurrencies.

Another area not to be overlooked is video games, where cryptocurrencies have created new opportunities for publishers and developers. Some online games have already started trading in cryptocurrencies, and one of the best examples of this practice is "Project Big ORB," a game that allows players to convert their in-game currency into other assets, including cryptocurrencies, and then exchange it for real-world money.

➤ **The Impact of Cryptocurrencies on the Economy Through Blockchain**

Blockchain, as the underlying technology behind the functioning of cryptocurrency mechanisms, is increasingly being adopted outside of that framework. Many experts predict that the use of this technology in other markets could potentially unlock billions of dollars in opportunities for those markets. So far, Blockchain technology has been shown to impact various business practices across multiple industries:

- Blockchain has improved cross-border transactions for financial institutions
- Transaction exchange applications have used the technology to benefit deals with private investors
- Leasing deals and car sales can apply Blockchain to simplify car leasing
- Cloud computing can use Blockchain to execute smart contracts and resist hacking attacks
- Government and public institutions to reduce paperwork and fraud while increasing accountability in record-keeping procedures
- Companies like Kodak intend to launch their own cryptocurrency to ensure that photographers are paid properly and decently for the services they provide

➤ **The Impact of Cryptocurrencies on the Labor Market**

The rise of cryptocurrencies has brought with it a whole new industry dedicated to monitoring and overseeing cryptocurrency exchanges taking place around the world. A number of early adopters of the technology have managed to get rich in a relatively short time, while a large percentage of other users have developed companies that rely on cryptocurrency trading as a significant source of income. The number of jobs in the Blockchain industry grew from just over 1,000 in 2016 to over 4,000 in 2017. Software engineers are the most directly and frequently sought professionals for the cryptocurrency industry. And while this job market has fluctuated over the past few years, interest in careers in this field has not waned. With the potential legalization of cryptocurrency trading, more global investment and the creation of a large number of new jobs in the field could be expected.

➤ **The Economic Impact of Cryptocurrencies on Unstable Domestic Currencies**

Nowadays, we are increasingly seeing a decline in public trust in Central Banks. In countries where the domestic currency is constantly fluctuating or hyperinflation is present, this leads to great dissatisfaction and a decline in the standard of living, causing citizens to often turn to cryptocurrencies to circumvent the situation. The complete decentralization of cryptocurrencies represents a significant advantage, especially for residents of poorer countries and those with unstable currencies, as it allows them to freely trade across borders with citizens of wealthier nations, thereby creating a level of economic equality.

➤ **The economic impact of cryptocurrencies through low transaction costs**

Transaction costs for cryptocurrencies are minimal for most users. Since cryptocurrencies and the Blockchain are decentralized, meaning they function without intermediaries, they do not require investment in physical infrastructure like

which are bank branches, nor paying their direct operating costs, such as utility bills, rent, and employee salaries. In the traditional system, these costs are partially offset by the high fees paid by banking customers, making the possibility of fast and cheap transactions very appealing. Furthermore, this can have a positive effect on the efficiency of exchange and increase the volume of transactions on a global level.

➤ **The Economic Impact of Cryptocurrencies through Transparency**

Blockchain technology and cryptocurrency transactions are automated, digitized, and tracked on a ledger that people, companies, or governments can never manipulate. This not only gives people power and freedom, but also reduces the risk of fraud and

a "ledger" that people, companies, or governments can never manipulate. This not only gives power and freedom to people, but it also reduces the risk of fraud and corruption since it is nearly impossible to cheat a system that cannot be altered. This fact about cryptocurrencies is particularly useful for underdeveloped and countries where citizens suffer the consequences of deeply corrupt governments. The cryptocurrency mechanism allows these people to invest and transact with the global economy, which can positively impact their countries' economies and improve their quality of life.

➤ **The Economic Impact of Cryptocurrencies on Entrepreneurs**

With its decentralized format, cryptocurrency is a global economy where all users can trade and exchange their crypto capital regardless of their citizenship or place of residence. This is especially important for entrepreneurs who are no longer limited to just national markets, but can turn to international ones where funds can be exchanged without issues with foreign exchange rates and international law.

In an increasingly digitalized world, the societal need for cross-border communication now manifests itself in financial needs, and traditional financial institutions are unable to keep up with it as cryptocurrencies can. There are no geographical barriers to cryptocurrency trading, which ensures easy and fast trading for corporations. Therefore, entrepreneurs are increasingly using the opportunities for investing, saving, and sending money across borders that the crypto market offers, thereby reshaping global business practices.

In today's world, as large companies increasingly dominate global markets, it is especially important to examine the impact of cryptocurrencies on small businesses. Cryptocurrencies can provide several very significant advantages to small businesses. The low transaction fees that come with cryptocurrencies are one of the biggest reasons why many businesses have decided to accept these digital currencies as a means of payment. While traditional forms of money, especially credit and debit cards, can cost businesses high processing fees, cryptocurrencies eliminate these costs. In addition to low transaction costs, crypto transactions can be completed almost instantly. While debit and credit card payments can take several days to process, crypto transactions are fast and efficient largely due to the absence of a third party. All of this contributes to an increase in trading volume.

Another major advantage of accepting cryptocurrencies is that it can open the doors of small businesses to a wider audience. Because cryptocurrencies are a universal, international currency, they can be used by anyone, allowing small businesses to trade with global customers. Additionally, since cryptocurrency is popular among younger people, accepting it can help a small business appeal to a younger audience. Essentially, using cryptocurrencies in business allows a company to reach a wider range of customers and demonstrate its ability to innovate and evolve in line with global trends.

The use of cryptocurrencies as money in the payment system is still quite limited compared to traditional systems. However, the discovery and growth of awareness about cryptocurrencies have only occurred recently. Some observers claim that the potential benefits of cryptocurrencies will be realized in the coming years or decades, leading to their widespread adoption. Skeptics, however, emphasize the obstacles to widespread cryptocurrency adoption and doubt they can overcome the flaws that separate them from official legalization and legal regulation.

Below are some of the potential benefits that cryptocurrencies can provide to the public and the economic system as a whole that have not already been addressed, as well as the drawbacks that need to be minimized or eliminated.

5.5.1 Advantages of Cryptocurrencies

- *High risk and potential for high rewards*

Today, there are more than 10,000 cryptocurrencies on the market, and each has its own unique characteristics. But all cryptocurrencies share a few common characteristics—a tendency to experience sudden spikes (and drops) in value. Prices are driven primarily by the supply of coins provided by miners and the demand for them from buyers, with this dynamic potentially resulting in high returns. The price of Ethereum, for example, roughly doubled from July 2021 to December 2021, which was an excellent situation for investors who got in at the right time.

- *The blockchain technology that underlies cryptocurrencies is very secure in itself.*

Some of the main advantages of cryptocurrencies are not related to the currencies themselves, but to the infrastructure that supports them – the Blockchain. The way this system is designed, as has already been mentioned, involves tracking and recording every transaction without the possibility of modification, with all information stored decentralized across multiple computers. The system constantly compares all data to ensure its legitimacy. This means that no individual who potentially wants to hack the system can access the entire chain at once, and all data is forever secure.

- *A cryptosystem as a fairer and more transparent financial system compared to traditional banks*

Following the recession at the beginning of the 21st century, the public's trust in the banking system significantly declined. Today, we are witnessing many people hesitate to keep their money in bank accounts and opt for other types of investments, such as real estate, gold, or cryptocurrencies. The financial system revolves around third parties, intermediaries who process transactions, which means that every user must trust one or more intermediaries when conducting a transaction. As a state-recognized system, citizens rely on trust in government authorities to ensure that this system of intermediaries, i.e., the banking system, is secure and the right choice for managing their capital. However, many want to avoid this system and are turning to crypto transactions without any intermediaries.

- *The crypto market operates 24 hours a day*

Another advantage that cryptocurrencies have over banks is that crypto markets are always open. Since coins are mined and transactions are recorded non-stop, there is no need to wait for the NYSE, Nasdaq, or any other exchange system to open for the day if you want to buy, sell, or trade cryptocurrencies. This has had such an impact that regular stock exchanges are considering the possibility of trading stocks outside of regular banking hours. For investors who are on the go 24 hours a day, 7 days a week, crypto could be the best way to generate returns outside of regular business hours.

- *Cryptocurrencies could help investors beat inflation*

Cryptocurrencies are not tied to any single currency or economy, so their price reflects global demand rather than national inflation. On the other hand, the risk of cryptocurrency inflation is very low because the number of coins that can be "mined" and put into circulation is generally limited, so the available supply cannot get out of control and inflation does not occur. Some coins (like Bitcoin) have a total cap, others (like Ethereum) have an annual cap, but in any case, this approach keeps inflation at bay.

5.5.2 Disadvantages of cryptocurrencies

- *Understanding cryptocurrency requires time and effort*

If you don't follow digital trends, the concept of cryptocurrency trading and Blockchain can be a very abstract field that is difficult to trust, let alone invest your money in. Trying to invest in something you don't actually understand is always a risk, which is why it's necessary to set aside time and effort and be willing to dedicate yourself to researching and learning about this system in order to truly understand the pros and cons of investing in cryptocurrencies.

- *Cryptocurrencies can be an extremely volatile investment*

While the price of a cryptocurrency can skyrocket and bring significant profits to investors, it can also experience frightening drops just as quickly. Unlike government-backed money, the value of virtual currencies is entirely driven by supply and demand. This can create uncontrolled fluctuations that bring significant gains for investors but also possible large losses. Therefore, if someone wants to achieve stable returns, this type of investment is not the best option. The cryptocurrency market is largely based on speculation, and its relatively small size makes it more vulnerable to price fluctuations, which in turn can cause disruptions in the coin's value. – This is one of the main disadvantages of cryptocurrencies, which is why the system is often seen as relatively unserious and untrustworthy. Investments in cryptocurrencies are subject to far less regulatory protection than traditional financial products like stocks, bonds, and mutual funds.

- *Cryptocurrencies have not yet proven to be a long-term investment*

Although cryptocurrencies have become well-known and are still gaining in popularity, it's worth remembering that they have existed for just over a decade. The concept only really emerged with the publication of the Bitcoin white paper in 2008. Exchanges, in contrast, can look back on centuries of history, and gold has been considered a store of value for millennia. It is still too early to speak of cryptocurrencies as a long-term investment because no one actually knows what will happen to them in the future, which is a significant risk for investors.

- *Crypto has serious scalability problems*

The transaction processing for these types of digital currencies operates at a high speed, but only up to a certain point. When that point is passed, major problems with processing capacity arise, leading to a slowdown in the process. The scalability problem refers to the network's limited ability to handle large amounts of transaction data on its platform in a short period of time. This is related to the fact that the records (known as blocks) in the blockchain are limited by the time required to create a block and its size.⁶² This can represent a negative experience for transaction participants, with potential financial losses that it can cause.

- *Crypto users are potentially vulnerable to security risks*

Using cryptocurrencies for trading carries a very low risk when it comes to the blockchain system itself, but this does not mean that security issues and hacking attacks cannot occur. A cryptocurrency owner can lose the private key that grants them access to their wallet, and with it, all their funds. Additionally, there is the possibility of hacking and phishing, as well as other malicious means, to obtain key information. Phishing is a type of cryptocurrency scam that involves manipulating victims into revealing and sending their private keys or personal information. The attacker typically pretends to be a legitimate entity or person to gain the victim's trust.⁶³

Cryptocurrency trading websites are located on the internet and are therefore dependent on internet security rather than cryptographic security. Digital wallets located on cryptocurrency trading websites are called "hot" wallets. The adjective "hot" comes from the fact that their private keys are online, making them a suitable target for criminal activity. New vulnerabilities in websites and internet platforms are discovered almost daily, and sometimes their discovery is exploited for a hacking attack. The largest hacking attack on a cryptocurrency exchange occurred in 2018 when as much as \$534 million was stolen from the Japanese exchange "Coincheck" (FortKnoxster, 2019).

- *Crypto mining is responsible for significant environmental pollution*

⁶²Antonopoulos, Andreas M. (April 2014). Mastering Bitcoin. Unlocking Digital Crypto-Currencies. O'Reilly Media

⁶³ <https://cointelegraph.com/learn/what-is-a-phishing-attack-in-crypto-and-how-to-prevent-it#:~:text=What%20is%20phishing%20in%20cryptocurrency,the%20trust%20of%20the%20victim.>

According to studies published in "Joule" and the American Chemical Society in 2019, the annual energy consumption for Bitcoin mining alone results in annual carbon emissions ranging from 17 to 22.9 MtCO₂, which is comparable to the emission levels of countries such as Jordan and Sri Lanka⁶⁴. In September 2022, a report in the journal "Scientific Reports" revealed that from 2016 to 2021, every U.S. dollar of Bitcoin mined also caused \$0.35 worth of climate damage. This is comparable to the gasoline industry, which is responsible for 41 cents of pollution per dollar. According to study co-author, economist Andrew Goodkind, compared to gold mining, Bitcoin's share of climate damage is an order of magnitude larger.

The annual amount of e-waste generated from bitcoin mining is estimated to be over 30,000 tons as of May 2021, comparable to the IT equipment waste produced by the Netherlands. One bitcoin generates 272g of e-waste per transaction. Due to the constant increase in the amount of computing power required to mine Bitcoin, mining devices are estimated to have an average lifespan of 1.29 years before they become unprofitable to use and need to be replaced. Other estimates assume that a Bitcoin transaction generates about 380g of e-waste, which is equivalent to 2.35 iPhones. Unlike most computer hardware, the application-specific integrated circuits for mining, since they are produced specifically for this operation, have no alternative use other than mining Bitcoin.⁶⁵

⁶⁴ The carbon footprint of Bitcoin; [https://www.cell.com/joule/fulltext/S2542-4351\(19\)30255-7?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2542435119302557%3Fshowall%3Dtrue](https://www.cell.com/joule/fulltext/S2542-4351(19)30255-7?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2542435119302557%3Fshowall%3Dtrue)

⁶⁵ Bitcoin mining producing tonnes of waste; <https://www.bbc.com/news/technology-58572385>

6. APPLICATION OF CRYPTOCURRENCIES AND BLOCKCHAIN IN LOGISTICS

In a globalized world, the logistics sector of the industry is growing at an unstoppable pace and becoming increasingly diverse and complex. It is projected that the expected growth will reach \$12 billion by 2027⁶⁶.

However, the less positive side of this part of the industry is the losses due to unnecessary intermediary costs, cyber-attacks, or errors in marking and transporting cargo. These problems translate into about \$50 billion in lost profits annually^[58]. Blockchain technology can completely change the way logistics processes work. The use of this technology has a clear impact on the profitability of companies in this sector, in addition to addressing trust issues, speeding up processes, reducing risks, and increasing transparency.

With globalization, logistics faces new challenges arising from a global network that must be planned, managed, optimized, and controlled. This network consists of a multitude of different companies, which are currently controlled individually with different central systems^[50]. Controlling the supply chain (SC) could increase its transparency and generate and maintain competitive advantages over other SCs. Supply chain management (SCM) could greatly benefit from Bitcoin's decentralized structure, high transparency, and security against manipulation^[52].

Blockchain in logistics comes in the form of a decentralized public ledger that documents all changes to records in real time. With this data, companies can create a strategy to implement faster transport routes, eliminate unnecessary steps in the delivery process, reduce errors, and generally save time^[45].

The question arises whether it is worth introducing Blockchain technology into the logistics and supply chain sector? Blockchain technology (BT) has the potential to reduce specific risks in the supply chain and enable the use of digital currency by solving the double-spending problem (the problem where the same token is spent multiple times) and eliminating the need for third parties (banks). In addition to process efficiency and cost reduction, Blockchain enables the exploration of new services and solutions that may not yet have been explored^[48].

Shipping documents, for example, can now be converted into smart contracts based on the BT model to mitigate transaction risk and uncertainty in decentralized environments^[48]. The new system, combined with BT, can reduce storage risk and achieve the same level of transaction security. Moreover, smart contracts based on Blockchain technology enable decentralized autonomous organizations (DAOs)⁶⁷ to engage in collaboration between business and employment relationships. Blockchain technology has recently received significant attention for its application in the logistics sector, which includes the exchange of digital documents and the tracking of goods.

Researchers identify the potential for Blockchain technology in logistics because shipments are constantly made worldwide, with multiple parties involved in the process. Participating parties use different channels for information flows and communications. Each company uses its own operational data and there is little interest in sharing information with partners to protect privacy. This leads to a process that lacks transparency and provides participants with asynchronous information⁶⁸. Another problem is the reliance on a trusted third party when transactions are conducted between two parties⁶⁹. These problems relate to the concepts of privacy, transparency, and trust. Blockchain technology has the potential to solve existing problems, but there are also obstacles, such as the issue of existing information system infrastructure in the logistics sector, which is not designed for new technologies like Blockchain, resulting in the need for organizational and technical adjustments.

Blockchain can help the goods and transport industry by redesigning the delivery process. With its approach to tracking items, production levels can be significantly increased and the efficiency of the chain improved.

⁶⁶<https://serbian.cri.cn/2022/11/29/ARTIpZMhEpVjkhwt8mLySY0n221129.shtml>

⁶⁷Decentralized Autonomous Organizations (DAOs) are a form of governance system run by their community. They can be independent of third parties and state corporations, and they also differ significantly from traditional organizations in their business and organizational model.

⁶⁸ Nakamoto, 2008.

⁶⁹ Svan, 2015.

supply chains by enabling faster and more cost-effective delivery. It also improves product traceability and allows for easier communication between partners. Blockchain technology, due to its decentralized nature, simplifies procedures that previously required several intermediaries.

Blockchain also helps companies track the trade of items. Once the goods are exchanged, smart contracts are concluded, and algorithm signatures that are impossible to forge are generated. Blockchain uses hash algorithms, which serve as individual fingerprints. Each hash fingerprint is unique to each supply chain, and every change is tracked. A hashing algorithm is a one-way program, so no one else can decipher it.

...scrambles the data and makes it unreadable⁷⁰. Hashing algorithms are one-way programs, so no one else can decipher and decode the text. Additionally, Blockchain technology records transactions in the supply chain with authors and timestamps, which are visible to everyone who has access to the supply chain. This data is also shared with consumers to maintain transparency.

Blockchain is more than a simple data storage system. It provides sophisticated methods for analyzing the data that is sent. It can help users develop projections and predictions based on past data as well as help users identify supply chain delays. This data analytics has proven to be essential for companies looking to reduce supply chain costs as they grow their operations, and it can also help increase consumer satisfaction. Retailers can use the Blockchain database to review where items are in production and delivery to better plan their delivery timeline. Finally, customers can also get access to certain data that is useful to them on the blockchain.

Blockchain keeps a detailed record of every transaction and communication between all participants in the supply chain, making it easy to track every step of the process. Inaccurate or falsified information can be easily detected thanks to the built-in security measures. Smart contracts are executed as soon as the buyer meets all the essential conditions, and payment is made as soon as the smart contract is executed, thereby reducing the risk of fraud and increasing the accuracy and timeliness of invoicing.

"Smart contracts" are simple programs that are stored on the Blockchain along with information about coins, tokens, and wallets. They are key to the development of decentralized applications based on cryptocurrencies. These are applications that ensure each party to the transaction fulfills their part. The simplest example of a smart contract is a transaction between a consumer and a business, where a sale is made. The smart contract executes the customer's payment and the business's shipment or transfer of ownership^{[[60]]}.

When applied correctly, Blockchain helps organizations involved in logistics and supply chain management increase efficiency, reduce costs, and increase revenue by developing new goods and services. In the future, Blockchain is expected to trigger an avalanche of changes in logistics. Through mutual trust among participants, it will increase the efficiency of the transaction verification process and can play a crucial role in the delivery of products and services.

6.1 Ways to apply blockchain in the logistics sector

Blockchain technology applied to the logistics and supply chain sector has numerous applications:

❖ *Inventory tracking*

The supply chain involves numerous stakeholders throughout the process, which is why Blockchain technology is very useful for building an efficient system that allows companies to track their products at any level. Such systems are applied by major global companies in their operations, such as Walmart, Nestle, or Unilever.

❖ *Improved Delivery*

Freight forwarding companies have already recognized the advantages of Blockchain, especially in international shipping, for tracking each item while simultaneously simplifying the current logistics process. One example is the giant Maersk, which uses it to control the movement of cargo across international borders.

⁷⁰[https://www.okta.com/identity-101/hashing-](https://www.okta.com/identity-101/hashing-algorithms/#:~:text=A%20hashing%20algorithm%20is%20a,and%20decoded%20by%20anyone%20else.)

[algorithms/#:~:text=A%20hashing%20algorithm%20is%20a,and%20decoded%20by%20anyone%20else.](https://www.okta.com/identity-101/hashing-algorithms/#:~:text=A%20hashing%20algorithm%20is%20a,and%20decoded%20by%20anyone%20else.)

This new system is expected to save freight companies billions of dollars, which will help reduce errors, improve delivery times, and detect fraud.

❖ Secure settlement and payments

Blockchain simplifies cross-border payments, while ensuring their security and transparency. An example is a company like "Visa" which has launched its own payment service, "B2B Connect," based on this technology and smart contracts, which helps manage billing and payments.

❖ Authentication

Blockchain technology also has advantages for end consumers, not just for companies. Customers can verify the authenticity of product provenance in a secure way, which is of great importance for luxury goods such as diamonds. This builds greater trust and credibility and avoids counterfeiting or illegal trade.

❖ Secure tracking of shipments/receipts of goods

One of the major problems in the logistics industry is disputes over lost cargo or delays. FedEx has launched an initiative to address this problem, which involves a blockchain-based ledger that collects information from the parties involved in shipping and receiving. This eliminates the need for a third party and reduces opportunities for fraud^[45].

❖ Transparency

Transparency is an issue that is often discussed in the logistics industry and is highlighted as a problem that is being sought to be solved. Thanks to Blockchain technology, this no longer has to be the case. Through its implementation, every stakeholder can access and track deviations or variations in the supply chain. They can also verify ownership of various assets, which is perhaps the most important benefit for logistics activities.

❖ Reduced transaction costs

The logistics industry suffers from high processing and administrative costs. To receive payment for an average invoice, a company has to wait an average of 42 days, while more than \$140 billion is tied up in transportation-related disputes.

Solutions that support Blockchain also help to solve this problem. Not only will payments be made directly and automatically, but this will also lead to a higher level of profitability. It is estimated that the logistics industry will save more than \$500 billion annually once Blockchain networks reach a higher level of adoption^[58].

One example of Blockchain's practical application in logistics is comparing how the food industry's supply chain operates without and after the implementation of Blockchain. A very similar example is also found in the service sector when it comes to the delivery of postal shipments. **Figure 21** shows the supply chain without Blockchain technology in the food industry, although this configuration can be completely "mirrored" onto the supply chain in international postal traffic. The image shows how the product moves from the manufacturer and from imports to retail and food establishments (in MPS: for example, how a shipment moves from the postal operator's warehouse to the recipient). **Figure 21** shows that activities in the food industry supply chain begin with the import of raw materials and semi-finished products for production. With this type of configuration, for example, suppliers cannot be directly linked to retail, from which they would obtain information about customer demand in order to make forecasts for future supplies. Supply mainly depends on wholesalers and distribution centers, as information about customer demand is obtained from the wholesale sector (wholesale goods are sent to retail outlets, i.e., to customers). If this way of operating the supply chain is applied to the service provision sector, such as MPS, the name and to some extent the function (activities) of individual entities in the chain are potentially changed. The recipient of the postal shipment cannot directly obtain information related to the postal operator's service provision directly

from the sender. Information about delivery time and costs can be obtained from the postal operator, who is the intermediary in this configuration (having functions similar to wholesale in a manufacturing SC).

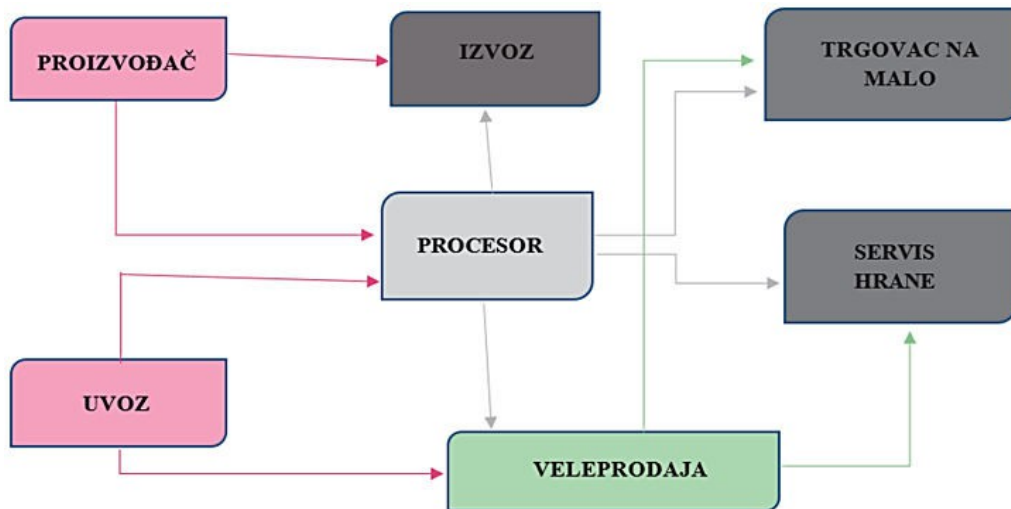


Figure 21 - Supply Chain without Blockchain Technology^[52]

The recipient of the postal shipment cannot directly obtain information related to the postal operator's services from the sender. Information about delivery time and costs can be obtained from the postal operator, which is an intermediary in this configuration (it has functions similar to a wholesaler in a manufacturing supply chain). Supply chains in international postal traffic are particularly sensitive, threatened, and vulnerable due to the fact that the majority of user requests are handled "from the armchair" (from home). Therefore, the delivery of goods (including food products) is increasingly being handled by courier services. Increasing the use of Blockchain technology is significant for efficiency and security in the execution of business activities in an exposed and vulnerable supply chain, as well as for resource provision, humanization of work, etc. ^[56]

In comparison with the analysis of the functioning of the traditional supply chain, **Figure 22 - Supply Chain with Blockchain Technology^[52]** shows how the same configuration would function if the processes were carried out using Blockchain technology. The supply chain structure is fully decentralized and all transactions (money, information flows) are stored on the Blockchain. Each member of the chain can create its own transactions using Blockchain technology, however, the members of the chain can "read"/have access only to those blocks with which they have a direct connection.

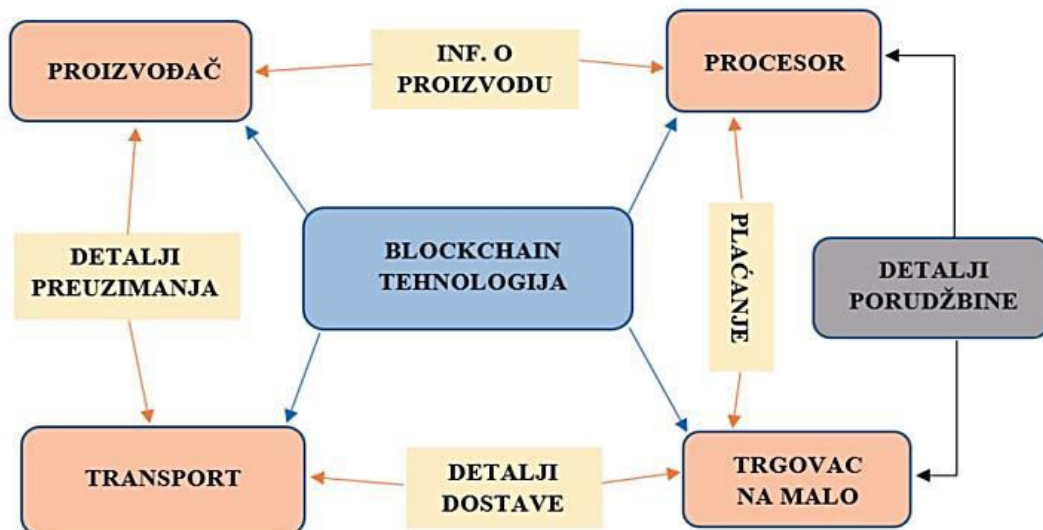


Figure 22 - Supply Chain with Blockchain Technology^[52]

Within the chain in international postal traffic, this significantly facilitates and accelerates the execution of all activities from the sender to the recipient of the shipment. It is not new for information about a postal operator's costs to be available to the sender (e.g., in the case of online shopping, the cost of shipping is increasingly available in addition to the product price, but at least one participant in the chain had to contact the postal operator about these costs—in this case, the sender). With the application of Blockchain technology, the sender and recipient communicate directly without the intermediary of the postal operator, with the information being available to all members of the chain. Monetary transactions (payment) take place directly between the respective entities in the chain. The money for the delivery service is sent directly to the postal operator from, for example, the shipper (when the delivery cost is included in the product's price). For large companies, "smart contracts" are of particular importance, where, in this case, the contracting parties would be the shipper and the postal operator.

Applications of Blockchain in logistics are being seen with major global companies. Walmart, the American retail giant, introduced a pilot project for the distributed ledger technology to track the origin of pork in China and its production in the U.S. This is one of the first significant test projects for applying Blockchain technology outside the financial sector. This technology can enable more efficient handling of errors and missed deadlines, thereby making the supply chain more efficient. In fact, this sector in particular is a primary target for Blockchain-based applications. According to the latest data, 42% of companies in retail and manufacturing plan to spend \$5 million on this type of technology⁷¹.

Everledger is a startup that aims to reduce risks and fraud in banking and open markets. This company wants to make the diamond supply chain more transparent and help eliminate the use of forced labor across Africa⁷². The platform is based on a global, digital public ledger that allows for the tracking of goods throughout the entire supply chain. Records on the Blockchain consist of defined characteristics, history, and ownership, and all parties in the supply chain can access them at any time⁷³.

Provenance is a London-based company that aims to make the supply chain more transparent by establishing a blockchain that makes companies more honest in their operations, including the environmental impact, the product's place of origin, and who produced it. The company emphasizes that it is "very concerned about how data is presented to consumers and that they are very focused on the availability of that information and how it is presented, on the product or in the store" ⁽⁷⁴⁾. Furthermore, "Provenance" will use Blockchains to eliminate labor exploitation or other bad practices, because the information obtained in the Blockchain ledger can be diverse and used in different ways.

6.2 Benefits of Applying Blockchain in Logistics

Blockchain technology is revolutionizing the current business landscape - Besides cryptocurrencies, their applications in any sector are countless, and logistics is no exception. Some of the most important advantages are:

- 1) *Transparency and Traceability* - With Blockchain technology, the falsification of documentation is drastically reduced, allowing every part of the supply chain to track the status of transport in real time, efficiently and reliably. All information is available to every participant in the chain and can only be changed with the appropriate "permissions." Blockchain technology is a platform, or database, that shortens waiting times for all participants in the chain by providing transparent, accurate, reliable, and up-to-date information. For example, during a shipment delivery, the recipient has information about the type and quantity of the shipment they are expecting, all cost elements, the payment method, its real-time location, and more. One of the essential positive characteristics of the business is the process automation, which ensures shorter timeframes and lower costs for all participants in the chain.

⁷¹Ramamuti, 2016.

⁷²www.everledger.io

⁷³ Dixon, 2016.

⁷⁴ Aloson, 2016.

- 2) Verification of provenance and quality assurance - During transport, product damage or spoilage is quite common, but with Blockchain, the provenance of items can be traced back to a specific manufacturer. In addition, compliance with standards is ensured, or non-compliance is detected.
- 3) Improved Efficiency - To avoid administrative errors, excessive time consumption, or fraud during document exchange and payments, smart contracts can be introduced that bypass these issues and improve the workflow.
- 4) Accelerated payment process - Blockchain technology keeps a detailed record of communication logs between the involved parties and allows for the tracking of any activity during the process. This is reflected in increased security and a low risk of fraud or errors in money transfers.
- 5) Simpler and faster document processing – transporting goods globally involves a large number of different forms and types of documents. In addition to the usual "paperwork," which is often subject to alteration, theft, misuse, manipulation, etc., there are significant financial and time losses. Blockchain technology enables the electronic transfer and processing of documentation via computers, tablets, and mobile phones, which ensures complete security and protection of information, as well as its real-time updating^[45].

7. RISK MANAGEMENT IN LOGISTICS AND THE SUPPLY CHAIN

Paulsson (2007) states that the term risk is attributed different meanings in the literature: a threat or danger ("flood risk"); probability ("driving a car without wearing a seatbelt implies an increased risk of injury"); a general assessment of the probability and magnitude of an outcome; a measure of dispersion ("taking out insurance involves a reduction in risk"). Attempts to define the term "risk" have led to a large number of approaches that draw on various scientific and professional disciplines. To highlight the breadth and diversity of approaches to defining the concept of risk, some authors somewhat jokingly state that the number of definitions of risk is equal to the number of papers written on the subject.

Risk, as a concept with applications in all spheres of human activity, has been extensively studied from various standpoints and perspectives. Generally, risk is most often viewed as a product or function of several components. According to Waters (2007), risk represents the probability that an unexpected event will harm a business organization and is defined in terms of the level of uncertainty (given by probability or frequency) and the magnitude of the impact (consequence) of a given event. Similarly, the International Organization for Standardization⁷⁵ defines risk through two basic components: consequence and the uncertainty of its occurrence.⁷⁶, where the consequences can be positive or negative. A definition of risk similar to the previous ones is given by Culpa (2001), according to which risk is defined as any source of chance that can have a detrimental impact on people or business. It has already been noted that the understanding of the concept of risk, and consequently its definition, depends on the context of observation or study. In his doctoral thesis, Vilko (2012) provides several definitions of risk depending on the scientific discipline within which risk is studied. According to this author, and almost identical claims were made by Manuj and Mentzer (2008), from a financial perspective, risk is viewed as the probability of an expected outcome⁷⁷. This aspect of viewing risk is probably the oldest, as it was used a hundred years earlier in the business of insuring merchant ships. From a strategic perspective, risk is used to adjust the size of the return on capital when investing⁷⁸. From a marketing perspective, risk encompasses the nature and significance of purchase objectives and failures to satisfy psychological or qualitative goals⁷⁹.

Risk management in logistics and the supply chain involves mitigating the negative impact of the environment on the management of various types of risks within the logistics and supply chain sector, including identification, assessment, and control. This includes collaboration with all participants throughout the entire chain⁸⁰.

Supply chain risk management has gained importance in recent years due to globalization, expanding supplier networks, market demands, and competition. Supply chain risks can be defined as events that can have a negative effect – either quantitatively or qualitatively – on the flows of production, transportation, or delivery of an item. Therefore, risk management consists of anticipating, identifying, assessing, and mitigating these possible disruptions in each logistical process^[50].

Risk management is a systematic mechanism for managing the risks or threats an organization faces in its logistical activities, enabling it to identify events that could lead to accidents or adverse consequences and to establish the best course of action for identifying, assessing, understanding, handling given situations and risk reporting.

Below are the steps we must follow for supply chain risk management.
We will look at them in detail later.

The risk management process aims to identify and assess risks to enable a clear understanding of them and their effective management. A key step that links identification/assessment

⁷⁵ ISO 31000

⁷⁶ McCormack et al., 2008.

⁷⁷ Beaver, 1966.

⁷⁸ Christensen and Montgomery, 1981.

⁷⁹ Cox, 1967.

The key to managing risks is understanding them. Risk management also involves taking appropriate measures to minimize threats. For risk management to be effective, it must be a continuous process^[51].

Effective risk management plays a key role in every company's pursuit of financial stability and superior performance. Risk management seeks to protect the organization and its revenue without hindering its growth. Moreover, investors are more willing to invest in companies with good risk management practices. Adopting a risk management framework that embeds best practices into a company's risk culture can be the cornerstone of the organization's financial future^[55].

Risk management in logistics is an ongoing process. Regular monitoring and evaluation of risk mitigation strategies are necessary to ensure their effectiveness. Key Performance Indicators (KPIs) are desirable and recommended for monitoring the performance and progression of logistics processes and comparing the system's state before and after any corrective measures are applied, as will be shown later in this paper. By analyzing these metrics, logistics professionals can identify areas for improvement and refine their risk management strategies^[55].

In essence, risk management is an essential component of effective and resilient logistics operations. By identifying, assessing, and mitigating potential risks, logistics companies can minimize disruptions, ensure timely deliveries, and protect their reputation. Adopting strategies such as supply chain diversification, enhanced visibility, collaboration with partners, contingency planning, insurance coverage, and regulatory compliance can help mitigate risks effectively. Continuously monitoring and enhancing risk management strategies allows logistics professionals to stay ahead of challenges and maintain a competitive edge in an ever-evolving industry and world of innovation that brings daily new developments, such as the application of Blockchain and cryptocurrencies in payments, which is the focus of this work.

There are at least six key components or steps that must be considered when creating a risk management framework for logistics processes and supply chains regarding the aspect of cryptocurrencies and Blockchain, which will be presented below.

7.1 Risk Identification

Risk identification focuses on the activities of observing and recording internal and external events that have the potential to affect the company's business in the logistics sector. The result will be a list of identified risks and causes, the degree of impact of the risks the company faces, and the consequences of the identified risks on the company's logistics operations. When identifying risks that affect the achievement of the organization's goals, it is necessary to pay attention to and separate risks from opportunities that can be utilized to conclude new business, enter new markets, develop the company, etc.

The identification of potential risks in the logistics and supply chain sector from the perspective of blockchain and cryptocurrencies will be conducted by observing available data on the operations and business practices of three major companies listed in **chapter 6.1 - Ways of Applying Blockchain in the Logistics Sector**,

„Walmart", "Provenance", "Everledger". The aforementioned companies have applied Blockchain technology and adopted cryptocurrencies in various ways and represent an excellent basis for assessing the potential risks of implementing these innovations, which can then be assumed to also occur in the case of other companies. The list of all identified risks is presented below in **Table 3**.

LIST OF POTENTIAL RISKS OF IMPLEMENTING BLOCKCHAIN AND CRYPTOCURRENCIES IN LOGISTICS PROCESSES AND SUPPLY CHAINS		
Serial No.	Risk Label	Identified Risk
1.	R-01	Need for additional employee training
2.	R-02	Difficulties in securing expert personnel for the development and creation of a new system

3.	R-03	Disinterest of employees and users in the logistics sector
4.	R-04	Insufficient technological literacy
5.	R-05	Aversion and distrust of innovation
6.	R-06	Software errors
7.	R-07	Poor maintenance / lack of maintenance of infrastructure
8.	R-08	Need for technological integration
9.	R-09	Poor software maintenance/no maintenance
10.	R-10	System Downtime
11.	R-11	System Overloads
12.	R-12	A sensitive process of designing, developing, and implementing an application in an area where there is little experience
13.	R-13	Lack of procedures, regulations, and standards (new area)
14.	R-14	Vague or incomplete instructions
15.	R-15	Poorly set performance measurement parameters
16.	R-16	High process maintenance costs
17.	R-17	Long duration of the development and implementation process
18.	R-18	Relatively small amount of professional literature and scientific papers
19.	R-19	Concerns related to the application of AI – artificial intelligence
20.	R-20	Collapse of the crypto system
21.	R-21	The need to open a new sector within government institutions
22.	R-22	Losses due to potential failure of technology implementation
23.	R-23	Lack of financial resources
24.	R-24	Lack of legal regulation of the crypto market
25.	R-25	Risk of investing in new technology
26.	R-26	Neglecting new technology for "more important" issues
27.	R-27	Introducing a new way of working
28.	R-28	Opposition and disruption of cryptocurrency use by environmental groups
29.	R-29	Collapse of the crypto system
30.	R-30	Hacking attacks
31.	R-31	Consumption of a huge amount of electricity
32.	R-32	Media portrayal of a negative image of the crypto market and Blockchain tech.
33.	R-33	Decline in popularity and interest in cryptocurrency trading
34.	R-34	Differences in the legal status of cryptocurrencies as a means of payment in other countries
35.	R-35	A surge in the financing of drug and human trafficking, terrorist attacks, and other criminal activities through the crypto market
36.	R-36	The stance of other countries on recognizing cryptocurrencies as a legal means of payment and their impact

Table 3 – List of all identified potential risks in the logistics sector and supply chains

7.2 Risk Classification

When it comes to risk management, there is a need to classify all risks into specific categories. There are two important reasons for this:

- Focusing on one class/group of risk sources makes it easier to identify specific risks
- the risk class sometimes defines the possible actions that should be prepared (e.g., if it is an external risk, it is not possible to influence the occurrence of the risky event with internal organizational measures, but only to mitigate its consequences)

When examining the parts of an organization and a typical business process, the following classification is applied when establishing a good risk management system:

- 1) external / internal
- 2) strategic / operational or long-term / short-term goals
- 3) an expanded list of 4M and/or 7M (man, machine, method, material, money, marketing, management, measurement, mother nature, maintenance, market)
- 4) by organizational units

At the third level, specific risk sources are defined, while each source can also be classified according to the remaining levels of the breakdown. This actually means that for each specific risk, it must be determined whether it is an external or internal factor, whether it affects strategic or operational goals, and which organizational unit it pertains to. When it comes to the third level of the division, the basic idea for it was the classification of causes often used when creating an Ishikawa diagram, which will be applied in the following section, **Chapter 7.2.1**. In the case of the present work, the emphasis is placed on the organization's logistics sector and supply chain when it comes to risks associated with the application of cryptocurrencies and blockchain as an innovative component of their functioning.

OVERVIEW OF RISK CATEGORIES		
Ref. No.	Risk Category	Explanation
1.	People (man)	- These are shareholders, employees, customers, suppliers, partners, members of the supply chain, unions, the community, etc. - It includes characteristics of people (knowledge - expertise, psychological profile, propensity for certain vices, marital status, health and safety, malicious intent or accident, ethics, demographics, etc.)
2.	Machines, devices, and infrastructure (machine)	- Covers critical and other, new and old, markets, labor, technological level, maintenance, etc.
3.	Working procedures – for production or service provision (method)	- Include technological level, review, product/service defects, technological equipment adequacy, staff training, work procedures, control procedures, cost monitoring, time tracking, delivery delays, bottlenecks...
4.	Material (material)	- These are: raw materials, semi-finished products, finished products, consumables, packaging, external and internal transport, incoming and outgoing inspection, nonconformities, attestations and certificates, hazardous materials, recovery and recycling procedures, etc.
5.	Money (money)	- This includes revenues, expenses, accounts receivable, accounts payable, loans, interest rates, exchange rate differences, cash flow dynamics, etc.

6.	Marketing (marketing)	- Includes the organization's reputation, brand, pricing policy, target market, research and development, product promotions, etc.
7.	Leadership, management	- Includes: staffing solution, bad decisions, management style, mobbing, leadership, vision, discipline, respect for rules, etc.
8.	Measurement (measurement)	- Includes: personnel, training, equipment, procedures, legal requirements, use and interpretation of results, communication and reporting of measurement results, etc.
9.	Environment (Mother Nature)	- Includes: rain, wind, sun, lightning strikes, electrical discharges, floods, earthquakes, volcanic eruptions, tsunamis, environmental protection laws, endangered species, hazardous materials, SO2 and CO2 emissions, etc.
10.	Maintenance	- Includes: reliable machine operation, processing quality equal to the designed, machine productivity equal to the designed, machine safety equal to or higher than the designed, low costs...
11.	Market	- Includes: price fluctuations, recession, currency exchange rates, changing customer demands, invoice payments, new technologies, competition, product obsolescence, alternative raw materials, fashion/trends...
12.	Social community – politics	- Includes: events such as wars, terrorism, labor union activity, legislation, socially responsible behavior, etc.
13.	Communication, information transfer, information system	- Cases of information system failure, accidental data changes, integrations, viruses, software errors, hacking attacks, document loss, oral transmission of information, unclear or ambiguous communication, etc.
14.	(Non)advancement	- Risk can also involve a possible positive outcome of an uncertain event

Table 4 – List of Risk Categories^[40]

The listed risk categories are not exhaustive, nor do their given components represent everything that should be considered for that category. They can only serve as a guide, and any specific situation may generate/add its own specific categories or components. Each of the risks shown in **Table 3 – List of all identified potential risks in the logistics sector and supply chains** is classified according to its category from **Table 4 – List of Risk Categories^[40]** and analyzed in more detail in the following section of the paper.

1. People

- **Insufficient technological literacy**

The rate of technological advancement in the world has increased exponentially over the past century and will continue to do so in the future. Technology plays an ever-increasing role in people's lives, permeating their work as well as their leisure activities. For this reason, technological literacy has become an indispensable part of everyday life, and it is important to align with it so that it brings benefits rather than being a hindrance.

Technological literacy shows how to get the most out of technology while avoiding its pitfalls and scams.

This concept generally implies the ability to use, understand, manage, and analyze technology safely, effectively, and responsibly. It includes the use of technology—devices such as computers, smartphones, tablets, etc.—as well as the internet to evaluate, create, and integrate information, which is crucial for using cryptocurrencies and Blockchain in daily business. Today, digital literacy means possessing the skills to thrive in a society where communication and access to information are increasingly conducted through digital technologies, such as online platforms and mobile devices. According to statistical research:

- one in three people (between 5 and 69 years old) is computer literate (computer literacy rate 34.3%)
- every other person (between 5 and 69 years old) is digitally literate (digital literacy rate 57.2%)
- 22.9% of households owned a desktop or laptop computer⁸¹

What was once considered a bonus is now a critical component of almost every role. Today, employees must assume they will be forced to continue improving their digital skills. The expectation is that when an employee starts a new role, they either have the digital skills to do the job or will learn them quickly. Every time technology evolves, especially when it comes to completely new technology, such as Blockchain and the active use of cryptocurrencies, the workforce's skill requirements increase.

- ***Disinterest from employees and customers in the logistics sector***

As with any innovation, there is a risk that logistics service users and employees alike will show little interest in adopting the new technology and adapting to a new way of working. Cryptocurrency trading and blockchain technology are relatively new fields that a large percentage of potential users and companies are still unfamiliar with. A low response rate for the use of new technology can be expected due to fear of the unknown, preoccupation with other more important business issues—which would make learning about something new seem like a waste of time—and a sense of being threatened by potential misuse, and so on.

- ***Resistance and mistrust of innovation***

Any change can potentially cause resistance due to a lack of an objective assessment of its effects, as well as the acceptance of the current state as unchangeable or difficult to apply to new, required conditions. The reasons for this perspective on change are, first and foremost, rooted in subjective perceptions of what is fact, in reducing the assessment of that fact to a personal level, and in a lack of seeing the bigger picture that the change is intended to bring. On a personal level, the demand for change is perceived as coercion, and accordingly, resistance is offered to prevent such a subjective feeling⁸².

The adoption of any innovation takes time and involves the emotions of the users, in this case, employees of logistics companies as well as users of logistics services, who are guided not only by what the consumer thinks about the innovation but also by how they feel about it. Research from multiple social science domains shows that emotions are fundamental to decision-making, consumer behavior, and social change⁸³. Emotions contribute to shaping the perception of the risk of using new technology, which is often critical for its acceptance and can lead to aversion⁽⁸⁴⁾. The introduction of major changes into people's daily lives that come with innovations may also require users to acquire new knowledge and skills to be able to use them. On the other hand, this requires will and time, so it is possible for users who are not sufficiently interested or prepared to engage in such a process to reject the application of innovations.

⁸¹ <http://www.statistics.gov.lk/Resource/en/ComputerLiteracy/Bulletins/AnnualBuletinComputerLiteracy-2021.pdf>

⁸² Jovanović, M (2018) "Resistance to Change"; http://upravljanjepromenama.gov.rs/blog/отпорпроменама/?up_abc=la

⁸³ Turner and Stets, 2005

⁸⁴ Rogers, 1983

2. Machines, devices, and infrastructure

- **Software errors**

Software development is a complex process that depends on factors such as the team's competence and collaboration, or the type and complexity of the technology being used or sought to be developed. It is important not only to select the right experts and tools, but also to effectively manage the entire process, during which many challenges and obstacles will inevitably arise, especially in the case of developing new and under-implemented products. The result can be various errors that affect the user experience and the quality of the service in the supply chain that the software is intended to provide. These errors can relate to the application of an incorrect methodology in software development, a lack of competence among team members who cannot handle obstacles or lack the knowledge and experience to design a specific system component, failure to apply the latest technology, etc.

Errors are an integral part of any development process; however, the later they are discovered in the process, the higher the costs of fixing them and preventing their recurrence. For this reason, it is very important not to allow lapses in system control and testing, and in setting and monitoring the quality parameters for the software's work.

- **System Overload**

It is expected of every system to reach the highest possible level of capacity utilization; however, it is a mistake to think that there are no specific limits to this. The risk of system overload is very real during operation and requires considering the use of an overload protection system that analyzes certain parameters and regulates their values to prevent system failure.⁸⁵ Any overload for the user means poorer service provided by the logistics system, whether it involves a slowdown and a complete shutdown, and for the process owner, securing experts who would restore the system parameters to acceptable values, potential costs of replacing components, and citizen complaints.

This risk can also refer to the problem of cryptosystem scalability, which is related to the network's capacity, i.e., the number of transactions that can be processed in a given period of time. This represents a known drawback of applying Blockchain and cryptocurrencies in business, which can be influenced by changing parameters such as memory capacity, encryption speed, hardware and software performance, computer efficiency, and the like.

- **The need for technological integration**

One of the significant problems and risks facing blockchain technology is that its integration with existing systems can be a difficult and complicated process. The essence of this problem is that this technology is not an industry standard, and therefore, a large number of other necessary technologies for its operation cannot be integrated without various adjustments. Therefore, before implementation in the logistics sector, companies must seriously consider the issue of integration and assess their existing system¹⁷¹.

3. Work Procedures

- **Lack of procedures, work regulations, and standards (new area)**

If there are no procedures, instructions, and regulations on how something should be done, there is no way to know the correct way to operate. It is also impossible to track whether a specific process has been improved if there is no established standard to compare it against. The wide variety of frameworks means that there is a lack of standardization. This is potentially one of the biggest risks facing current Blockchain projects.

By standardizing processes and introducing procedures, guidelines, and instructions related to the application of blockchain and cryptocurrencies in logistics and supply chains, it is not only ensured that the process is always carried out in the same correct way, but it also allows for the monitoring of workflows and the creation of a basis for continuous

⁸⁵System overload and system utilization; https://link.springer.com/chapter/10.1007/978-1-4757-5920-4_2

improvements and the overall stability of the system. Also, in the event of a problem, it will provide guidance and assistance in resolving it.

- ***Difficulties in securing experts for the development and creation of a new system***

Cryptocurrencies have only existed for about a decade. Since this is a new field the world is encountering for the first time, the selection of literature and experienced professionals is understandably very limited, especially in the educational sphere, as there are few accredited educational institutions offering courses on this industry.

- ***A lengthy development and implementation process***

When it comes to developing a new system, application, or other digital product, as with any project, the process doesn't always go according to plan, whether an external company or an internal team is hired for the job. Downtime, obstacles, going back to a previous phase to fix bugs, and the like are risks that are especially present when implementing the brand-new blockchain and cryptocurrency payment technology, which has only existed for a few years. Working to resolve unforeseen errors and issues, implementing different project phases, or adding new functionalities can take much more time than originally planned. This risks missing deadlines, paying penalties, increased bug-fixing costs, and user dissatisfaction. (if a launch date is announced), etc.

- ***Vague and incomplete instructions***

Vague, inadequate, or incomplete instructions for the implementation of new technology can pose a problem for both logistics staff, who must ensure the system's functionality, and for users of logistics services. Detailed instructions are necessary for users of new technology so they know how to manage a digital wallet in the case of crypto payments, handle new models and methods for more efficient shipment delivery, and understand the concept of applying artificial intelligence in combination with blockchain and IoT devices.

IoT devices (Internet of Things) are non-standard computer hardware – such as sensors, actuators, or devices that connect wirelessly to a network and can transmit data to other devices. They extend the use of the internet beyond standard computing devices such as computers, laptops, and "smart" phones to traditional physical devices and everyday objects without an internet connection. In this way, with the presented embedded technology, these devices can also "communicate" with other devices as well as via the internet, with the possibility of remote monitoring and control (for example, various sensors for tracking the transport of goods, changes in process parameters, energy consumption, robots, etc.).^[67]

These devices are used for better monitoring of goods transport throughout the entire supply chain, environmental conditions, warehouse surveillance, and the like. Otherwise, frequent complaints, user backlash, abandonment, losses due to operational and management errors, and other problems are at risk. Instructions and guidelines must be available in writing in a clear and understandable form for troubleshooting, monitoring parameters, or simple daily use, with appropriate training provided.

- ***A sensitive process of designing, developing, and implementing a model in an area where there is little experience***

As previously stated, for cryptocurrency projects, the availability of expert personnel for the project team and of available literature and experience is limited. Over the past few years, various logistics companies have identified and tested a wide range of applications and use cases for Blockchain and cryptocurrencies to help solve operational problems, improve processes, and reduce business costs. Moreover, to date, no technological innovation has had greater potential to advance the supply chain industry than Blockchain technology.

However, a completely new and developing field carries with it serious risks, which means that development and implementation projects will very easily have a greater chance of failure, missed deadlines, unplanned costs, problems securing expert personnel, going back to

previous phase of development or application, and so on. An example is the "300cubits" cryptocurrency project, presented in 2017 as a pilot project and created for the world of logistics to ensure an element of trust that a certain quantity of products will be secured for delivery when scheduled⁸⁶. Although the potential benefits of the new technology are quite promising, there are still obstacles preventing blockchain technology from achieving widespread adoption.

- ***Concerns regarding the application of AI – artificial intelligence***

The integration of blockchain and artificial intelligence can revolutionize supply chain and logistics operations by improving efficiency, transparency, and decision-making. AI can provide intelligent insights and data-driven analysis, while blockchain secures and tracks data, ensuring its accuracy and integrity. Together, these technologies can enhance inventory management, route optimization, traceability, and end-to-end visibility throughout the supply chain. By leveraging artificial intelligence and Blockchain, organizations can enhance decision-making and optimize supply chain operations, leading to increased customer satisfaction and business success⁸⁷.

As artificial intelligence algorithms become more powerful day by day, it also raises concerns about their ability to make fair decisions for the betterment of humanity. As AI slowly achieves human-level cognitive abilities and makes up a significant portion of applications used in logistics, the issue of trust becomes increasingly important. A striking example that sparked controversy and raised significant questions about the application of AI in high-frequency trading occurred when Amazon's blockchain and artificial intelligence-based algorithm for

"...same-day delivery" unintentionally showed bias toward predominantly Black neighborhoods. Expert programmers could not determine on what basis the AI acted in this manner⁸⁷.

By analyzing historical data and real-time information, AI-based systems can predict demand patterns, inventory fluctuations, and potential disruptions, which allows for optimizing inventory levels, minimizing stockouts, and streamlining supply chain operations^[69]. In essence, the main factor on which all deep and machine learning models are based is the availability of data and resources to feed them. This means that these systems have access to a vast amount of data from various sources, which is generated by users worldwide, whether they are companies or legal entities. Various sensitive data are thus exposed to potential cyberattacks and can potentially fall into the hands of individuals who could misuse it. This risk is inevitable when it comes to the application of blockchain and crypto technology and must be taken seriously into account, given that in the event of a sensitive data leak, logistics companies risk losing business partnerships, potential lawsuits, loss of image and customer trust, and so on.

- ***Need for additional employee training***

Employee development should be a continuous effort to strengthen performance through approaches such as group training or leadership mentorship. With the introduction of innovative blockchain technology into business processes, a large number of employees, especially those from older generations, lack the knowledge and skills regarding blockchain and cryptocurrencies, as well as their application in the workplace. Learning about blockchain and its fundamentals, as well as how cryptocurrencies work and are applied to payments, should be included in employee training programs to help them advance their careers and stay current with trends in the blockchain technology industry.

4. Material

- ***A relatively small amount of professional literature and scientific papers***

As already explained, cryptocurrencies and blockchain technology in general have only been known for about a decade, which means that all research, statistical data, and other literature have been published recently

⁸⁶ [https://builtin.com/blockchain/blockchainsupply-chain-logistics-uses*](https://builtin.com/blockchain/blockchainsupply-chain-logistics-uses)

⁸⁷<https://www.educba.com/artificial-intelligence-problems/>

often by insufficiently qualified individuals. This can often mean the existence of unconfirmed or incorrect facts that are then applied in the design, development, and implementation of new blockchain and crypto payment technology in the logistics sector, thereby risking errors, various costs, and downtime in transportation, production, and the like.

5. Money

- ***Lack of financial resources for the development and implementation of new technology***

The implementation of new blockchain technology in the logistics and supply chain sector requires projects that are very often pilot projects which, to date, have not been applied in any company; therefore, there is no experience or evaluation of their success. This requires the assessment of significant parameters and the modeling of an innovative process, which in itself carries a high degree of risk of failure. For this reason, a situation where the project budget is exceeded due to the need to resolve unforeseen problems and errors, or where deadlines are missed, is not ruled out. It is expected that not all logistics companies have the financial means to turn to the application of new technology, which can potentially lead to significant losses.

- ***Risk of investing in new technology***

Investing in new technologies carries a significant financial risk, as it is unclear which processes will be economically viable in the long run. Innovation is full of uncertainty, and choosing the right timing and method of investment, to be distributed among different technologies and various parts of the supply chain, can be seen as a key success factor. There is a risk of misallocating investment to poor or often immature technologies, such as blockchain and crypto payments. On the other hand, there is the risk of delaying investment, where companies can miss trends and, consequently, the opportunity to enter the market.

- ***High process maintenance costs***

Given that there are not many examples of blockchain technology and cryptocurrency applications in the logistics sector, future process maintenance costs during their implementation are a matter of assumption, and this risk must be taken into account.

- ***Losses due to the potential failure of the new technology's implementation***

The risk of losses from failure is present during the development and implementation of any project. When it comes to newer technologies, especially cryptocurrencies and blockchain, which are innovations with divided opinions among critics, there are not many examples of successful implementation, and they are very often experimental projects with no guarantee that they will contribute to business improvement as intended. The chance of potential losses from a project failure is much higher than with proven, long-term technologies. The goal, of course, is to avoid a loss scenario at all costs. The foundation for a successful project is laid during the development phase, and its performance should never be stopped. To this end, performance and risk indicators are established and monitored to timely detect changes and events that could lead to undesirable outcomes.

6. Marketing

- ***Media portrayal of a negative image of the new technology & restrictions on public advertising***

This risk is very real and expected, especially when it comes to media outlets, journalists, and critics who, since the advent of cryptocurrency and blockchain technology, have largely emphasized the negative aspects and risks of their implementation, while ignoring the potential they have for improving logistical processes. Misleading advertising and a lack of verified, appropriate information pose a problem and risk for companies implementing new technology, as this can lead to doubt, a lack of trust, reservations, and prejudice among potential customers, as well as cause them to refuse to do business with a particular logistics company.

In the United Kingdom, the Crypto Asset Taskforce, composed of HM Treasury, the Bank of England, and the Financial Conduct Authority (FCA), published a report in 2022. In 2022, the UK's Crypto Asset Taskforce, composed of HM Treasury, the Bank of England, and the Financial Conduct Authority (FCA), published a report which found that mis-selling and a lack of appropriate information are key consumer protection issues for blockchain and crypto assets, and that their future media portrayal will be subject to FCA rules⁽⁸⁸⁾.

On the other hand, in 2022, the Monetary Authority of Singapore initiated an approach that could lead to the creation of aversion and uncertainty when using cryptocurrencies and Blockchain, ignoring their advantages. Guidelines were issued to restrict the marketing of cryptocurrency in business activities on electronic media, third-party websites, and public events, as well as the use of crypto ATMs, which are also considered a form of marketing. Essentially, the Monetary Authority of Singapore has limited promotions to companies' own websites and official social media accounts. This creates a negative image of the new technology and fosters prejudice, given that the restrictions come from an official government institution that citizens trust.

Spain had a better approach to the issue of cryptocurrency promotion. In January 2022, the National Securities Market Commission issued a regulation requiring that marketing of crypto payments and blockchain include a risk warning, as well as assurances that the promotion is "clear, balanced, and fair." in order to protect both individuals and companies from fraud^[70].

7. Leadership and Governance

- ***The need to open a new sector within government institutions***

Since crypto trading was an area that had not received attention until recently, there was no need to hire people in government institutions to deal with this field. However, this situation is changing with the development and application of cryptocurrencies and blockchain in regular business in all industry sectors, including logistics and supply chains. In this situation, a problem can arise where states are delayed in implementing changes to the existing system and restructuring it so that the new technology can be used in regular business operations, creating a gap between those states that have enacted legal regulation and those that have not. This includes the need to hire personnel and establish a special sector within government institutions, which requires time, restructuring the existing system, rebalancing the budget, allocating financial resources, and more.

8. Measurement

- ***Poorly set performance measurement parameters***

Performance measurement is a significant part of the process of introducing blockchain technology and using cryptocurrencies in the logistics and supply chain sector. Since this is an innovative technology that has not been applied until now, we are dealing with the complex task of establishing appropriate parameters to show a realistic picture of its success, which carries a risk of error. In complex areas, it is very difficult to determine what should even be considered an indicator of success, and in some cases, it is hard to establish the threshold that indicates success.

9. Environment

- ***Requirements for the consumption of a huge amount of electricity***

The use of cryptocurrencies in daily life as well as in business requires the consumption of a huge amount of electricity. To understand the impact of cryptocurrencies on the environment in this way, one must first understand the relationship between the virtual currency and the physical infrastructure and operational inputs it requires. As a digital representation of value, cryptocurrencies use distributed systems to securely store and transfer ownership through a cryptographic process based on complex calculations. The more blockchain and cryptocurrency payment technologies are applied in business, the greater the environmental impact if serious steps are not taken to protect it.

⁸⁸ Financial Conduct Authority: the body that regulates the financial services industry

The PoW model consumes ever-increasing amounts of energy, especially if the price of energy is lower than the profit made from mining. Bitcoin's energy consumption, for example, is estimated based on a further estimate of the computing power the network uses.^[5]

- ***Opposition and disruption of the widespread adoption of new technology by environmental groups***

Although many believe that crypto trading and blockchain technology are the future, there are certain downsides to this system related to its environmental impact, which are highlighted by environmental groups. Specifically, the cryptocurrency mining process is known for its negative environmental impact, and its increase in the current era of a major climate crisis would pose a significant danger. A single Bitcoin transaction uses the same amount of energy that the average American household consumes in a month – which is roughly a million times more carbon emissions than a single credit card transaction.

Because cryptocurrency mining requires so much energy, it is often located near the cheapest, least-regulated sources of power. Miners most often rely on harmful fossil fuels like coal, which produce toxic emissions that pollute the environment and are linked to asthma, cancer, acid rain, and climate change. There has also been a noticeable increase in electronic waste.

Taking these facts into account, with the growing popularity of cryptocurrency payments and their expansion into all sectors of industry, including logistics, numerous environmental groups have expressed their concern about its long-term impact on the planet. In some countries, campaigns have been launched against the Proof-of-Work system, for example. In the US where Congress has been asked to commit to serious legislative regulation, defining "green" mining procedures and monitoring miners to reduce carbon emissions, energy consumption, and electronic waste, and to prevent climate chaos^[20].

10. Maintenance

- ***Poor software maintenance/no maintenance***

Software maintenance is the process of changing, modifying, and updating software to meet user needs and changes in the field for which it was created, in order to remain competitive and relevant^[48]. Software maintenance is performed after the product has been launched for several reasons, including improving the software as a whole, fixing problems or bugs, increasing performance, and so on. Using the right software maintenance techniques and strategies is a critical part of keeping the software operational over the long term and ensuring user satisfaction.

Neglecting this issue risks that bugs will not be discovered until they lead to a system failure, which requires significant costs and manpower to restore, and can lead to system obsolescence, thereby slowing down the speed and quality of service. The software risks failing to keep up with market changes and demands, thereby losing users and more.

11. Market

- ***Decline in popularity and interest in cryptocurrency trading***

Due to the nature of the internet, it is impossible for cryptocurrencies and Blockchain to simply disappear. The only possible scenario is a drop in their value, which could lead to users abandoning their use as a means of payment. Also, there is a possibility of their popularity declining due to bans on trading and declarations of illegality in a larger number of countries, or a potential more serious hacking attack. However, the chances are slim that all of the aforementioned situations will lead to a significant drop in the number of users.

- ***Collapse of the crypto system***

Bitcoin has succeeded in its mission to become a currency that is beyond the direct control of major world governments, but this has brought uncertainty and unpredictability when it comes to its value fluctuation. In the world of cryptocurrencies, there are no conventional safeguards that governments use to ensure market prices remain stable before they get out of control. For this reason, crypto trading can bring users

huge "rewards" in a short period of time but also experience catastrophic drops, leading to losses. In the short history of crypto trading, there have been several serious system crashes that were overcome, and the value of the coin grew again⁸⁹. However, there is a risk that these fluctuations could at some point experience a prolonged decline after which the situation would never recover, which would lead a large percentage of logistics companies to abandon their use in business, and thus result in losses in the time and financial resources used for their implementation in application.

12. Society – Politics

- ***Implementation of a new way of working***

Implementing a new way of working requires time, financial resources, and the engagement of human resources from the perspective of logistics companies that want to apply new technology, while service users need time to accept, approve, and get used to the changes and innovations. There is a possibility of errors and the new way of working may not fit into the business environment it is introduced into (the society has not reached the necessary level of development for the application of a certain innovation, users do not understand the new way of working, the introduction of cryptocurrency payments in regions where a large percentage of the population still does not have even a stable internet connection, and so on).

- ***A surge in the financing of drug and human trafficking, terrorist attacks, money laundering, and other criminal activities through the crypto market***

One of the advantages of cryptocurrency is that it facilitates the anonymity of the person making a payment with it. Thus, it is possible to track and determine the path of a coin, but not the identity of the person making the payment. The party receiving the digital currency cannot learn the identity of the parties who have previously used it, but can have insight into a series of ownership changes. The anonymity of currency and capital holders has led to great interest from those seeking to launder money from criminal activities such as drug, arms, and human trafficking, etc. Undoubtedly, this is one of the illicit opportunities provided by the use of cryptocurrencies⁹⁰. In essence, the advantages of digital currencies in terms of security and the speed of changing ownership and maintaining anonymity can easily be turned into a potential danger or risk, and thus lead to their exclusion from business use. The extent of this risk largely depends on the legal regulation of cryptocurrency use in business, an area where government authorities must make an effort.

- ***The stance of other countries on recognizing cryptocurrencies as a legal means of payment and their impact***

Today, the status of cryptocurrencies varies from country to country. In some countries, cryptocurrencies are recognized as a legal means of payment, while in most countries, their status is somewhere between legal and illegal; they are considered neither legal nor illegal and are simply partially or completely undefined but not banned. This means that if a certain criminal act or other undesirable violations and legal disputes occur during business operations where a company's headquarters are located in one country, and the other company in the dispute in a second country with a regulated system cannot protect the legal entity under its jurisdiction, i.e., its citizens, because the transaction was carried out with a user from a country that either bans crypto trading or has no regulation whatsoever. This can also affect companies' international business. This issue can create distrust in the use of new technology, cause misunderstandings, lead to logistical disruptions, a sense of insecurity, and a reluctance to use it.

- ***Lack of legal regulation of the crypto market***

Crypto assets have existed for more than a decade, but it is only now that efforts to regulate them have moved to the top of the policy agenda. This is partly because only in the last few years has crypto trading developed from an interesting innovation in search of a purpose to a potential legally recognized payment instrument. The enormous growth of the market capitalization of crypto assets and their entry into the regulated financial sector

⁸⁹ A history of cryptocrashes; <https://milkroad.com/crashes>

⁹⁰ Dogo and Grujić, 2014.

The enormous growth of the market capitalization of crypto assets and their entry into the regulated financial(89) A history of cryptocrashes; <https://milkroad.com/crashes>(90) Đogo and Grujić, 2014. Monitoring the crypto market is difficult because data is inconsistent, and government agencies find it hard to track thousands of users who may not be subject to typical data disclosure or reporting requirements. Finally, in addition to developing a framework that can regulate both the actors and activities in the crypto ecosystem, national authorities must take a position on how the underlying technology used to create crypto assets fits with other public policy objectives – such as the case with the immense energy intensity of "mining." of certain types of crypto assets. All of the above affects the operations of companies in the logistics sector because the legal regulation of cryptocurrency and Blockchain use is essential for protecting both their business and their customers in the event of unwanted situations, legal disputes, hacking attacks, and the like.

13. Communication, transmission of information, information system

- ***Hacking attacks***

The use of various innovative information and communication technologies and tools also brings new risks, which include hacking attacks that can lead to the theft and misuse of data online, theft of funds, and other inconveniences. The consequences of these risks can vary depending on what data was accessed and whether it was only retrieved or also modified.

The functioning of cryptocurrency trading and Blockchain is explained at the beginning of the work. Crypto transactions are recorded in "blocks" that form a digital public ledger protected by data encryption. Due to the way this system works, the data can be very difficult for hackers or any third party to alter or access. Transactions require a two-factor authentication process, which involves entering a username and password to initiate the transaction, followed by an authentication code received via SMS to a personal mobile phone.

However, although it is considered a very secure payment method, cryptocurrencies attract hackers because some cryptocurrency activities cannot be traced. These individuals typically try to gain access to individual accounts or crypto platforms by stealing private keys and using them to access a user's or company's digital cryptocurrency wallet. Additionally, there is the possibility of stealing funds directly from a wallet on a specific device, computer, or mobile phone by hacking its system. The target of hacking attacks can also be the "bridge" or the moment of transferring cryptocurrencies from one blockchain to another, which would be exploited to illegally seize the coins of the affected user/company.

- ***System crashes***

Software and human errors are the most common causes of malfunctions and crashes in computer systems. Computers and devices used in tandem are becoming more reliable, but the level of failures caused by errors hidden in programs remains the same. For this reason, the type and quality of the software used are a key factor in a system's reliability. Errors can be the result of improper handling and human error or unexpected environmental changes. If system crashes are frequent and are not properly addressed, this can lead to user dissatisfaction and a rejection of new Blockchain technology.

14. (Non)Progression

- ***Neglecting new technology due to "more important" issues***

With the implementation of any new technology, there is a risk of misinvesting in poor or often immature technologies, which can include Blockchain and crypto payments. On the other hand, companies must be open to monitoring trends and innovations because there is a possibility that by focusing on some, at that moment "more important" business issues and problems, too much time and resources are spent stabilizing the situation in one area, and trends are missed. This can lead to missing out on

an opportunity to improve a specific part of the logistics process before the competition by applying new technology, gain a competitive advantage by providing faster delivery service, or take advantage of some other previously mentioned benefits.

The following section presents an Ishikawa diagram outlining all the identified risks that could potentially lead to problems in the logistics and supply chain sectors regarding the application of cryptocurrencies and Blockchain. These risks were identified through the analysis of 14 risk groups, which were detailed in the previous section of this paper.

All risks are grouped into 5 categories based on their similarity:

1. People / Users
2. Devices and Infrastructure
3. Process
4. Legal and Economic Issues
5. External factors

7.2.1 Ishikawa diagram

A **cause-and-effect diagram** is the result of a general analysis of the influences (causes) that determine a specific outcome of an observed phenomenon (work process). In efforts aimed at improving the quality of products and work processes of companies and service organizations, this method has a wide range of applications in work process quality assurance procedures, with regard to:

- identification of all actual causes of a specific state (outcome) - the result of the work process of a company or service organization,
- The identification and analysis of cause-and-effect relationships in the flows of material, energy, and information provide the basis for effectively solving problems when the state—the outcome of the work process—is considered outside the limits of permissible deviation from the established objective function.

The main characteristics of this method are: a clear visual representation of the possible causes of the phenomenon under observation and the consequences of their effects^[40].

The fishbone diagram, as it is also called in the literature, is a method for the detailed analysis of the relationship between a specific state of the observed system (the consequence) and the influencing factors that cause the occurrence of that state (the causes)⁹¹. In analyses related to improving the quality of products and work processes in enterprises and service organizations, the term:

- **Consequence** – refers to a specific outcome of the work process at the observed cross-section of the system, at a specific time and under specific conditions; as the outcomes are the result of influences related to the action of the environment and disturbances in the work processes, these are random variables and can be classified into two main categories: *within* and *outside* the limits of permissible deviations of the established target function.
- **Causes** – refers to the set of environmental conditions and disturbances in the system's work processes that result in a specific state, the outcome of the work process. From the standpoint of achieving the designed states/effects, these are disturbance-related quantities

Based on the given definitions, it is possible to identify a range of outcomes—effects of the system's work processes—that differ in character and unit of measurement and which, in this context, occur as *consequences*. In research on the possibilities of improving the quality of products/services and work processes, these are the quantities: quality elements and the quality of the product/service as a whole, or the elements and the overall quality of the work process, which indicates a large number of quantities that can be considered consequences in a specific, concrete case. Each of the given quantities is the result (**Figure 23 – Basic Cause-and-Effect Relationship**) of the action of a series of influences—*causes*, which differ in character, magnitude, and mode of action depending on the nature of the observed effect.

⁹¹Ishikawa, K. (1943)

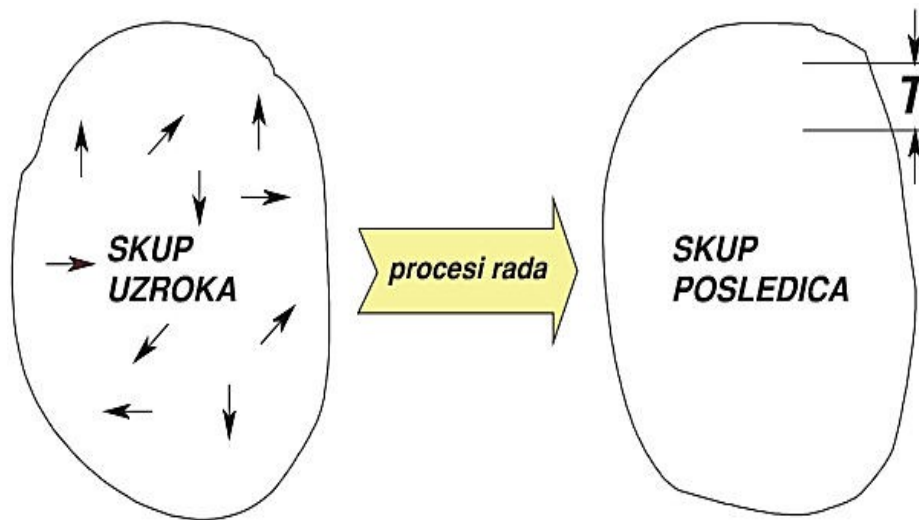


Figure 23 – Basic cause-and-effect relationship

In a cause-and-effect diagram, a set of causes on one side and effects on the other are organized according to the following principles:

- *selection* - identifying the actual causes of a specific outcome of a work process - a specific consequence
- *sorting* - grouping the selected causes according to their character, significance, and mode of action
- *logical linking* of the observed consequence and the isolated causes^[40]

The basic characteristics of the subject method are:

- a clear visual representation of the possible causes of the phenomenon under observation and the consequences of their effects,
- it provides the possibility of analyzing the interrelationship of individual influences - causes, their significance for the observed consequence, and their place in the overall structure of possible causes - a single cause is can appear in multiple places in the diagram (in multiple different groups of causes),
- Cause-and-effect relationships and the interrelationships between causes are, in general, qualitative and hypothetical in nature and serve as a basis for more effective problem-solving by another, appropriate method.

7.2.2 Application Procedure

As the basis for creating the Ishikawa diagram, the division of risk sources into 14 categories will be used. Based on these categories, all identified risks were identified and further explained in **the previous chapter**, Error! Reference source not found. Below, these risks will be displayed graphically and further grouped by similarity for the diagram's clarity.

1) Defining the Problem

In the majority of cases, a cause-and-effect diagram is used when a specific problem is defined as the effect—such as poor quality of work, an assembly, or a product/service, the occurrence of a failure mode, or other issues for which it is necessary to visually represent and analyze the causes. In this case, it is necessary to identify the causes of the occurrence of a specific problem as the causes. It is also possible for a specific desired effect to be defined as the outcome, in which case the goal is to identify the conditions that lead to achieving that effect.

The effect is depicted graphically, typically denoted by a rectangle symbol on the right side of the diagram, as shown in **Figure 24**. This leaves space on the left side of the future diagram for entering the causes.



Figure 24 – Graphical representation of defining the problem-consequence when creating an Ishikawa diagram

In the case of the analysis of the selected topic, the consequence will be the problems of implementing the logistics processes of organizations through the application of cryptocurrencies in business.

2) Cause Identification

After the step of defining the problem under observation, the process of identifying the causes that lead to the predefined problem follows. This step consists of:

- Formulating a review of all possible causes of the problem being analyzed

It is important that the review of causes be complete, meaning that no cause should be preemptively excluded that could, in a later analysis, result in the primary cause-and-effect relationship (Figure 25).

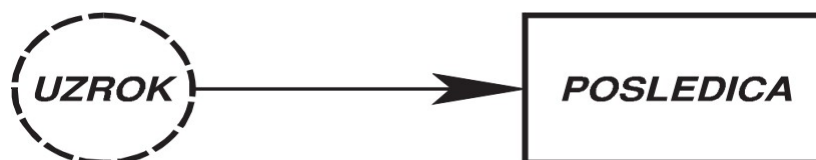


Figure 25 – Basic form of a cause-and-effect relationship ^[40]

- Classifications of causes by character, mode of action, and similar characteristics.

Within this sub-step, the set of all causes is sorted into basic groups, and each individual cause is assigned to a specific group. The number of groups is not limited, but consideration must be given to the diagram's readability and, consequently, the efficiency of the method itself. Sorting is most effective when applying the simplest possible form of a cause classification system. In the case of the problem under analysis, the causes will be divided into 5 groups based on similarity:

1. People / Users
2. Devices and Infrastructure
3. Process
4. Economic and Legal Issues
5. External factors

3) Choice of the basic structure

The subject phase of the analysis involves the selection of a specific cause-and-effect diagram structure. The basic groups of causes are entered by dragging the corresponding lines onto the cause action line in the diagram from step 1, resulting in the basic diagram structure as shown in Figure 26.

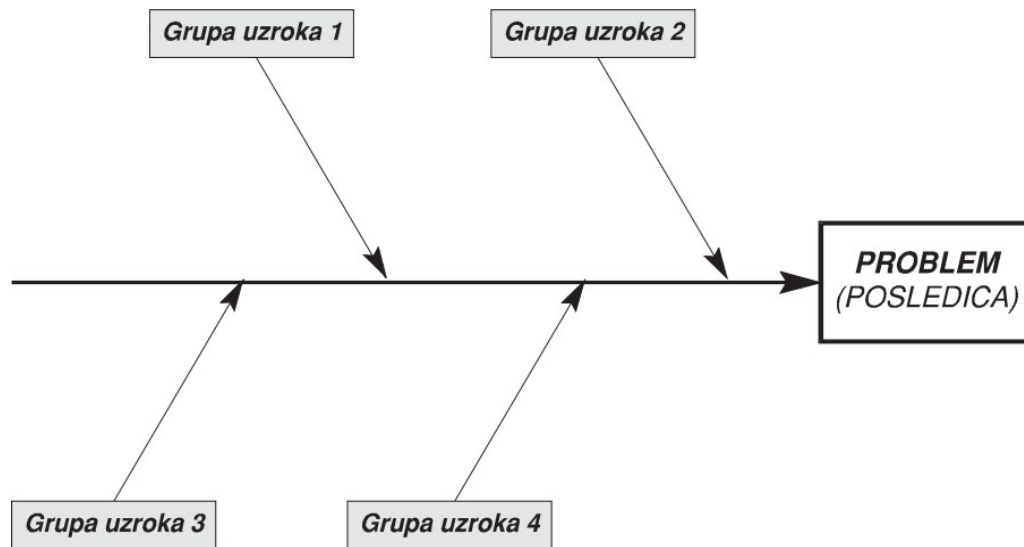


Figure 26 – Basic cause-and-effect diagram structure ^[40]

4) Diagram Elaboration

For the selected basic diagram structure, the causes previously located in a given group must be added to the lines of the basic cause groups. This is done by dragging the links of each cause onto the line of the basic cause group, as shown in Figure 27.

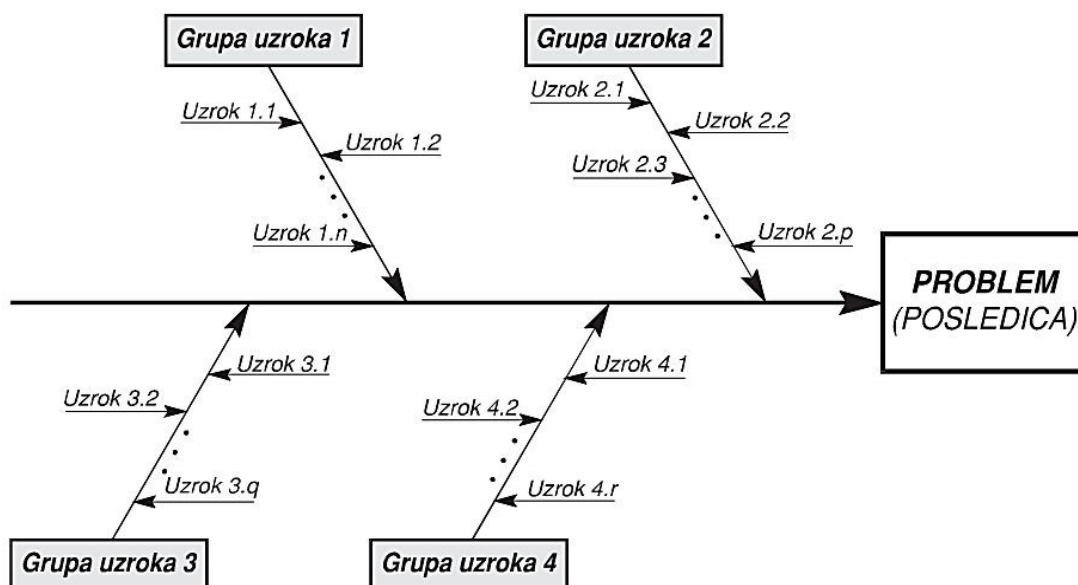


Figure 27 – Causal Diagram Elaboration Phase^[40]

5) The Expansion (branching) procedure

Causes identified and sorted into groups are not necessarily independent, and in most cases, they are not. It is common for two or more causes to also be causally related. Given this, the subject relationships can be sequential and parallel in nature.

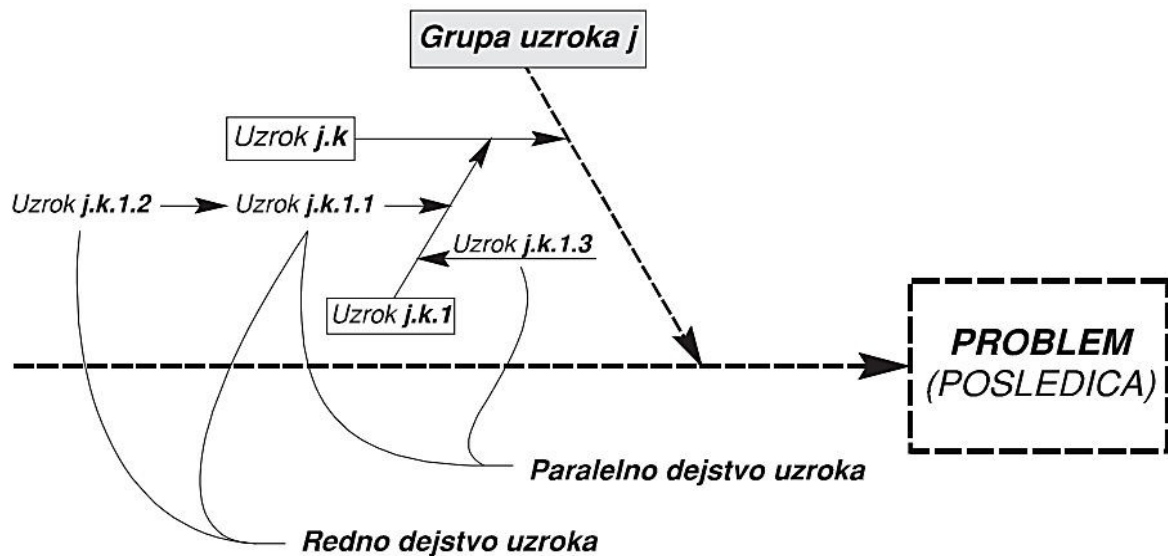


Figure 28 – The Expansion (Branching) Procedure ^[40]

6) Diagram Analysis

Once all identified causes have been entered into the diagram at a certain level and the logic of each branch has been verified, the analysis is conducted in two directions:

- Identification of the most likely risks or causes of the problem being analyzed and their labeling in the diagram.
- This procedure, in addition to focusing on the root causes of the problem, allows, in certain cases, for the identification of the line of critical causes, which is certainly one of the significant results of the method in question.
- A cause-and-effect diagram by itself is not sufficient to solve the problem; it only points to its root causes and causal relationships ^[40].

In the present work, the Ishikawa diagram shown below will be used to graphically represent all risks that potentially affect the implementation of logistics processes when it comes to the aspect of cryptocurrencies and Blockchain. Furthermore, later in the paper, the Ishikawa diagram will be used as a basis for the application of the FMEA method.

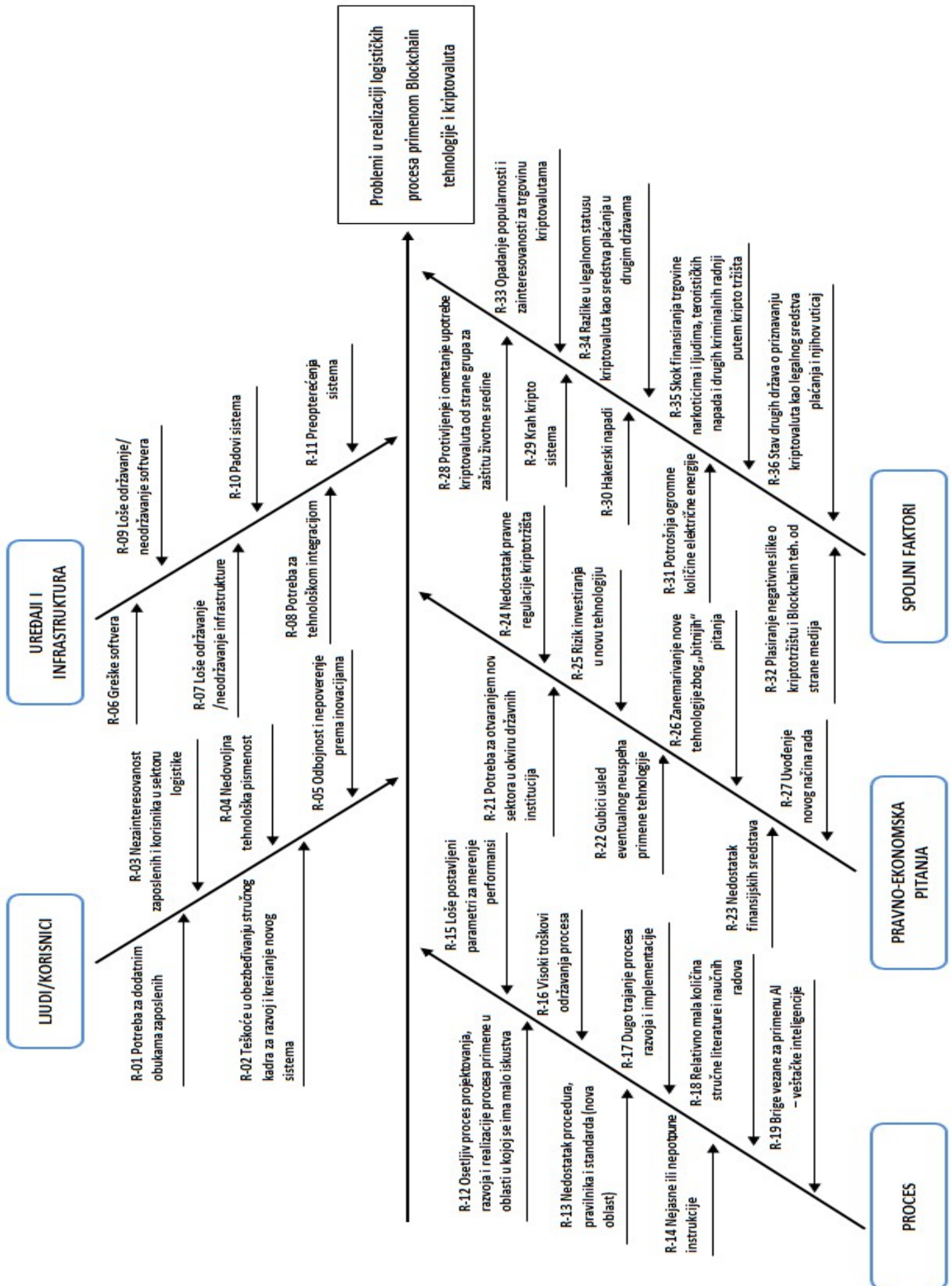


Figure 29 – Graphic representation of all identified risks via the Ishikawa diagram

7.3 Risk Analysis of the Impact of Cryptocurrencies on Logistics Processes – Identifying the Highest-Impact Risks

After the previous identification of all risks that may affect the success of implementing cryptocurrencies and blockchain in logistics processes, it is necessary to evaluate all risks and determine which of the identified risks is the most dangerous and harmful to their implementation. The goal is to focus on eliminating/mitigating the aforementioned risks. For this purpose, the FMEA risk assessment method will be applied. Given the nature of the analysis and the difficulties in quantifying risks, the assessment will be conducted using a qualitative method. As a basis for applying the method, the previously created graphical representation of all potential risks using the Ishikawa diagram method, as shown on Error!, will be used. Reference source not found..

7.3.1 FMEA method

FMEA (Failure Mode and Effect Analysis) is a systematic method for identifying problems, assessing and reducing risks, and preventing problems before they occur. It can be used for both products and processes. The FMEA method focuses on preventing deviations and increasing the degree of safety^[40]. The recognition that all well-managed companies are interested in preventing or at least minimizing risk in their business is the basis for the existence of the concept of risk management analysis. The FMEA methodology can improve systems, product design, and manufacturing processes by helping to identify potential problems and thus enabling corrective action to be taken.

The FMEA method reveals:

- ❖ all potential hazardous events (including those caused by the human factor)
- ❖ the consequences of these hazardous events
- ❖ the mechanism of the hazardous event
- ❖ a way to avoid or mitigate the consequences^[40]

The method was developed for system design purposes, but it is also successfully applied in:

- development of quality assurance programs
- system reliability, maintainability, and availability analyses
- system testing
- maintenance activity planning
- system safety analyses^[44]

The input data required to implement the FMEA method are:

- flowchart of the process under consideration
- parameters that can affect the outcome of the activity
- possible consequences of a hazardous event
- experiential data (if available)^[43]

7.3.2 Application Procedure

Step 1 - Reviewing the process, product, or problem

- The FMEA team should review the project or the product's technical drawing, or the detailed flowchart of the process being examined. This ensures that each individual on the FMEA team gains an appropriate understanding of the process or product the team will be working on.
- If the product's technical drawing or the process flow diagram is not available, the FMEA team should create it themselves before beginning the FMEA process.
- For a "product" FMEA, the team must have access to the actual product or its prototype in order to see and examine it. For a "process" FMEA, the team should, under real-world conditions, go through all phases of the process, in the manner shown by the flowchart^[44].

- In the present work, the significant elements for understanding the chosen topic, i.e., the problem of the impact of cryptocurrencies and Blockchain on logistics processes, have been covered and examined through the previous chapters.
which allows the problem's content to be clearly understood.

Step 2 - Identifying potential deviations/risks using the brainstorming method

Once all team members understand all the details of the process, product, or problem, they can begin to consider potential deviations, or in other words, impactful risks, as is the case in the present paper.

After the first brainstorming session, team members should write down all ideas, regardless of their logic. The collected ideas are then sorted into appropriate categories based on their similarity. Grouping can be done by types of failure modes (e.g., electrical, mechanical, human factor), by where in the process a failure can occur, or even by the potential severity of the failure's consequences (based on the team's assumptions at that time)^[43]. As the basis for the FMEA analysis of the subject problem, an Ishikawa diagram will be used, during the creation of which all potential risks that can lead to problems in the logistics processes when applying cryptocurrencies and Blockchain were listed. All risks are then, as already stated, grouped by similarity into 5 categories:

1. People / Users
2. Devices and Infrastructure
3. Process
4. Legal and Economic Issues
5. External Factors

Step 3 - Establishing potential effects for each failure mode/risk group

After entering all failure modes into the FMEA matrix, it is necessary to consider all effects that could occur with the occurrence of a failure/risk realization, i.e., the materialization of a specific risk. Individual failure modes may cause only one effect, while for others, multiple effects may occur. When performing this step, you should think in an "IF-THEN" pattern (IF a failure mode/risk realization occurs, WHAT WILL BE the consequences?). All the answers obtained are entered into the pre-prepared FMEA table (**Figure 30**).

Step 4 - Rating the severity of the consequences of each failure mode/risk realization

This step involves assessing the consequences that could result from the occurrence of a specific deviation/risk realization. Sometimes this is straightforward, especially if there is experience from a previous period, but sometimes the severity of the consequences is determined based on the knowledge and expertise of the team members. For the assessment, each effect is assigned a specific severity rating (**Table 5**).

RATING	EFFECT	IMPACT
5	Extremely large	Impact leads to failure of the model's application
4	High	Influence has a significant effect on the model's implementation and can lead to its failure
3	Medium	Influence has an effect that impacts the performance of the applied model
2	Negligible	The impact has a negligible effect on the model's application
1	No impact	The impact has no effect on the model application

Table 5 – Scale for assessing the consequences of an adverse event

Step 5 - Assessing the occurrence probability for each type of deviation

The best method for determining probability is to use actual data from the process. If such data are not available, a subjective estimate is made of how often each type of deviation can occur or a risk may materialize. Also, as in the previous assessment, a table is compiled where each descriptive probability of occurrence is assigned a score (Table 6).

RATING	OCCURRENCE PROBABILITY	DESCRIPTION OF CRITERIA
5	Extremely high	The occurrence happens multiple times a month
4	Large	Occurs once every 2 months
3	Medium	Occurs once every 6 months
2	Moderate	The phenomenon occurs once a year
1	Small	Occurs once every 5 years

Table 6 – Scale for assessing the probability of an adverse event

Step 6 - Assessing the detectability of a deviation for each type of deviation and/or its effect

This step begins by identifying existing control measures that can detect a deviation or its consequence. If no appropriate measures exist, the probability of detecting the deviation in a timely manner will be low, so the score for this factor would be very high. A table is then created assigning a score to each described criterion (Table 7).

SCORE	POSSIBILITY DETECTION	DESCRIPTION OF THE CRITERION
5	Complex	A more complex examination is required to determine the occurrence
3	Moderate	The occurrence can be detected by observing the records generated during the process, standard parameters
1	Simple	No testing is required, the occurrence is obvious

Table 7 – Scale for assessing the detectability of an adverse event

Step 7 – Determining the relative importance and ranking of all risks

The relative risk of an adverse event occurring is determined based on **three factors**:

- 1) **Consequence Severity** - the potential consequences of a deviation, should one occur
- 2) **Likelihood of occurrence** - the probability that a deviation will occur or its frequency of recurrence
- 3) **Detectability** - the probability that the deviation will be detected before its consequences manifest themselves^[43]

To determine the significance of each risk and the priority for observation in order to implement appropriate measures, a numerical risk priority number (RPN) is calculated. This is a number that is simply calculated by multiplying the ratings for all three risk factors—assessed levels of consequence, probability of consequence, and detection—as follows:

RPN = Consequence * Probability * Detection^[40]. The total RPN is obtained by summing all RPNs, and its reduction after implementing recommended actions is an indicator of the overall effectiveness of the FMEA analysis.

To determine which risks require the most attention, a threshold RPN value must be defined, above which appropriate measures must be defined for all types of risks (**Table 8**). The thresholds will be defined such that risks with a maximum RPN value of up to 12 are considered to have a low impact on the proposed system. This risk level was determined by multiplying the following ratings from each of the three elements of the method: $2*2*3 = 12$, then the resulting number was rounded to 15 for easier scale setting.

The risk threshold for an unacceptable level of risk is set at 20 RPN, calculated by multiplying the scores from the three elements: $3*3*3 = 27$. Again, the number was rounded to 30 for easier scale application, and any risk with an RPN value greater than 30 will be considered unacceptable.

The medium risk level will represent the range between the low and unacceptable risk thresholds, i.e., 15 - 30 RPN.

RPN Problem Significance Rating	Risk/Significance Level
From 30	Unacceptable
15 - 30	Medium
to 15	Low

Table 8 – Risk Significance Assessment Scale

FOBA ANALYSIS									
Red. Bl.	If category *vora rūika	Oznaka rizika	Identifitvated rhit	Patentable effect/creation of a new method	Øzb iljnosz p oslediceF ,/OE/E	Causes of the occurrence of the unwanted event	rerovatno pojave	Mogućno: otkrivanji	RPN F/Z/E
1.  LJUDI		R-03	8e2airIteres0Van0st<ap0slerlih i user of logistics	Ddbojnost, smzntena pfkzsnost work, missing opportunities for Unpreparedness..	3	Resistance to innovation, PBSigvrno3t, years zügDSlerih, artistic and theological writerliness..	5	1	9
		R-04	Insufficient technology for change	Difficult to adapt to modern 60blJ, Znatnd fn Sre&tyz i YreiTle necessary for running modern processes, a rejection of their own technologies..	4	It is also not suitable for the expansion of existing systems. Ob0ku and improvement, nepoyerenje toward innovations..	4	1	1s
		R-05	Banish distrust toward innovators	Long time needed for the adoption of new technology, lack of interest, fear, failure of trends...	3	Sturiji zup0Slefli, fle0üayestenast, predrasude, negatiY0ü Sliit8 J medilijima, tehnolosfa nepismerast, 0Stüjarje r1"0rrf0r 2D i ..	4	1	12
2.	MASKs, LJREDALJs AND mra+Runuw	R-11	System overload	System status indicator on the MCC or DVI robot, user dissatisfaction, system crashes	2	Errors in the underlying infrastructure, excessive user load, poor integration with other systems	3	1	6
		R-08	Problem of technological integration	Three phases of development and implementation, the need for stručnim licima, veća šansa za greškama..	3	Failure to invest in technological development, High implementation requirements arise t9En0i0šić(9..	5	1	9
		R-06	Software errors	Dtkaz iistemz, potential hacker attacks and misuse, lo5 kvalitet usluge..	3	Sestrzfnost, preyidi tokom razvoja i implementacije, neodrzavante..	4	1	12

3.	PDSTIJPCIRADA- PRDIZVoaur in rRu us SERVICE	R-13	procedures and regulations work (new area)	Učestalije greške i nesporazumi, izdvajanje i ljudi za definisanje nejasno situations..	3	iskustvo za definisanje procedura i concepts.	5	1	15
		R-02	The process of ensuring professional development and the creation of a new system	Npdostatak potrebn0g StrrčnDt kadra, grete osled neisfzha ili restrvrčnosti..	3	Therefore, to ensure the necessary professional development and creation of a new system, it is necessary to create appropriate educational programs and provide adequate literature.	4	1	12
		R-17	Dugoutlined §i0cesarazvaja and implementacije	Svakim The demand forJ, the process of three-year development, and the ITLDTCP changes in trends and circumstances. Zbunjenost korisnika, povećanje vremena potrebnog za rešavanje nedoumica i žalbi, greške zaposlenih..	7	New and old, with a point-by-point presentation of the real case, a wide selection of professional staff, and sources from the literature.	2	1	4
		R-14	clear or incomplete instructions jektovanj		3	The task is to create extracts for supported flowjoast, zz SYP teLrt0l0jLe niy0e piSiTlpn0Sti i 2nūflj)ü..	4	1	12
		R-12	A sensitive process pro a, p. 7110cledU Subject to which the experienced one is dear	Errors in the model, I raised the cream trabivanje dodatnih finansijskih.	3	No need for a practical example, there is no existing one, but there is a wide Nova tehnologija, nedovoljna selection of professional jargon and sources of literature.	4	1	1z
		R-01	Need for additional tools zaposlenih	srečstava and ljoÉStYa, the need for creating guidelines and procedures..	2	needs to be clarified, old employees...	3	1	6
		R-19	Bf e yezane for the application of AI— artificial intelligence	Greste, respDrazumi, interruptions in the process, fnarISi(ski losses, gLbitak looks ipoyererja, nesigcrnast in the process..	4	No way to log in, SYP Yis9 Possibilities of AI on how to study and analyze faces, my mistakes and errors istraživanje objavljivanje naučnih	2 4	1 1	8 12
4.	tATfRIJAL	R-18	Relatively small professional caliber and scientific research	Errors in development, difficulties in rtyróiyürt"lJl tūcnih cir"erliCü..	3	Note Dblast— necessary time for literature..			

5.	MONEY	R-23	8ledastatakfinansijskihsredstava for development and implementation	For reasons of delay, project rejection, unplanned higher expenditure, lower quality...	4	Potencijalna situacija koja uvek For projects, the budget is estimated during budget planning, phases, reviews, changes...	2	1	8
		R-16	High-level process management	High-throughput, state-of-the-art process control and rare Strucra working group		and rare structural work 00vi dDdütüni proce5..	4	1	12
		R-22	Losses resulting from the potential failure to implement new technology	ZdStojl, nEdDStütak 3redstava 7ü drvge aktivnasti, potencijalna pDpa5t k0mpanije..	3	Investing in the revision of assets in the company tEb0DID In time, the countermeasure...			
		R-25	Risk of investing in new technology	ÊDt90Cí/al failure of predecessors, glJbiCi, zaSt0/i, grelhp, fall in confidence Ikorisriika and vgleda..	4	Pilot-preliminary, IDSa proposed, insufficient 3-D radar, corresponding ID-literature..	4	1	16
6.	MARKETING	R-32	Pushing a negative image as new gy agranicenja vezanaza javno recbmiranje	User prejudice, rejection of the proposal, claiming to have technical advantages, a misconception.	3	Personally, it's a godsend...			
7.	MANAGEMENT AND ADMINISTRATION	R-21	Need for a new arrangement 5eLtora M okVirM drzüVnih instructor	The main financial and material resources, rpbalaz budzets i restruduirarjp..	3	B0tk0t iriDVüCi(ü, lüzn0 izve5tüVünje, lola i3kctva 5a Cripa trzistü, Other government and institutional resources	4	1	12
8.	MEREKJE	R-15	Laser-based parameters for performance measurement	An unrealistic picture of the state/system, otkazi, zastoji..	5	Errors in design, oversight, negligence, technical flaws..	3	1	15
9.	SURROUNDING, LIVING ENVIRONMENT	R-28	MMV0dpnIM model of environmental protection	For example, in the implementation, the image of the technology, the risks, and the benefits, 3-pronged and potentially irreversible implementation, negative media portrayal of technology, stakeholders, and user feedback.	5	The process of "mining" requires the consumption of enormous amounts of electrical energy, a lack of proper regulation..	4	3	60
		R-31	Requires the consumption of a huge amount of electrical energy	Increase in resource consumption, pollution of the living environment, strikes and counter-protests by groups for protection of the environment..	5	Način funkcionisanja PoW ITlehanizma, nedDStatak pravnp regulation..	5	3	75

10.	MAINTAINING	R-09	Poor maintenance/no maintenance 5. Conclusion	Downtime, high operating costs overhauls, poor quality of service Complaints, constant errors, poor service quality, service delays, service interruptions, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality, poor service quality Service Specifications..	3	Human error and remark, or lamp procedure and instruction, lack of control..	3	3	18
		R-07	Poor maintenance/lack of maintenance of the infrastructure	Züst0ji U raálz, yisi tr0?k yi reit0rlü, riŽi I /ülitet lJslIzgpg, žalüp, Učē5tülü rrarovi, prekidi u and other factors...	3	Human error and oversight, failure to follow procedures and instructions, lack of control support..	3	3	18
11.	MARKET	R-33	Decline in popularity interest in trading cryptoyalmtdlTld and pritenM of BioChain	Change in trend0Ya in the industry, extraordinary situations such as wars, LukerSvi attacks and greskp/früf1e blockchaira..	4	Social and political developments, hacker attacks, reports on inflation, changes in trends...	2	1	8
		R-20	Crash of the Cryptosystem	Fall in interest and demand for tehriDI0gijU and p0ter\Cijüln0 rejection e, govici..	5	Dr. ÉtYs:rü P litlckÄ 0esüyürljÄ, hacker attacks, r9gDverenjp u iffi0Vücije, prDmera trendova	2	1	10
	DRUŠTVENA ZAJEDNICA - POLITIKA	R-36	The stance of other countries on recognizing cryptocurrencies as a legal means of payment and their impact	International problems during transactions, legal disputes, errors, non-compliance...	4	Social and political impact, hacker attacks, regulation and legal issues, market trends..	4	1	16
		R-34	Differences in the legal status of cryptocurrencies as payment methods in other countries	International problems during transactions, problems with third-party services...	4	Poor countries, the level of technological development, and the state of digital governance.	S	1	20
		R-27	Introducing a new way of working	In the race of time, the rowers and the swimmer on the prowl..	3	Inovacije i novine zahtevaju Adjustment of the law regvlatiyp..	4	1	12
		R-35	Boom in the financing of drug and human trafficking, terrorist attacks, and other criminal acts through cryptocurrencies	Doubts about technology, hacking, planting negative media? Is it a setup? A conspiracy?	4	Because of the potential for higher numbers and the fact that they would be encouraged to get involved in crypto trading	2	3	24

	R-24	Nedostatak pravne regulacije kriptotrišta	Nedostatak zaštite od strane države, gubici, krađe podataka, nesigurnost korisnika..	5	Siromašne države, nedovoljan nivo tehnološkog napretka država, bojkot novih tehnologija od strane vlada..	4	3	60
13.	R-30	Hakerski napadi	Krađa podataka i finansijskih sredstava, nemir i panika među korisnicima, pad ugleda..	5	Greške i mane sistema, kriminalne akcije, loša zaštita podataka, zakonski propusti..	2	1	10
	R-10	Padovi sistema	Nizak kvalitet usluge, žalbe, dodatni troškovi remonta i održavanja, manje korisnika usluga..	3	Vremenske nepogode, hakerski napadi, neodržavanje softvera/infrastrukture..	3	1	9
14.	R-26	Zanemarivanje nove tehnologije zbog „bitnijih“ pitanja	Propuštanje prilika, nadmoć konkurencije, zastarelost..	4	Problemi u drugim sektorima, loša organizacija i upravljanje, nepoverenje i neshvatanje značaja inovacija i unapređenja..	4	1	16

Table 9 – FMEA table used to determine the most significant risks

7.4 Defining and implementing measures to eliminate or reduce deviations/risks with a high risk level

By applying the FMEA method, the ideal case is the complete elimination of a specific type of nonconformity, but this is not always possible. In such a case, the goal is to lower the rated parameters as much as possible. Often, to improve the process, it is easiest to increase the detectability of nonconformities, which would lower the score for this criterion. Reducing the severity of consequences is important, especially in situations that can lead to injuries. However, the greatest opportunities for risk reduction lie in reducing the probability of occurrence, and the implementation of appropriate measures has the greatest impact on improving the process itself^[44].

As seen in the preceding content of this paper, cryptocurrency trading and the application of blockchain technology bring numerous benefits and advantages to its users, in this case, logistics companies and users of logistics services. However, it is very important to consider the greatest risks of implementing this technology that must be taken into account.

The result of applying the FMEA method to rank all risks of implementing cryptocurrencies and Blockchain technology in logistics processes is shown below in **Table 10**, where the risks with the highest RPN are highlighted:

RISKS WITH THE HIGHEST RPN ACCORDING TO THE FMEA METHOD			
Serial No.	Risk Code	Risk	RPN
1.	R-24	Lack of legal regulation of the crypto market	60
2.	R-28	Opposition and obstruction to the introduction of the model by environmental protection groups	60
3.	R-31	Requirements for the consumption of a huge amount of electricity	75

Table 10 – Highlighted risks with the highest RPN according to the FMEA method

It is noticeable that all the listed risks are in the environmental and legal domains, which is why all measures will be focused on these areas. The goal is to lower the values of these risks to an acceptable level and to ensure that the implementation of the new technology is safe and secure for all participants in the logistics processes, while also mitigating the negative impact of the technology on the environment.

Environmentalists have been drawing attention for years to this negative side of the crypto market and Blockchain technology, and with the current climate crisis, protests have become increasingly frequent worldwide. Demands from environmental activist groups and lobbies such as Greenpeace and Extinction Rebellion are constantly pushing for change and have clear demands regarding the future of crypto payments and Blockchain, and the way the energy resources needed for them are secured.⁹²

The effects of the emergence and popularization of crypto technology on the environment are significant. One of the impacts responsible for worsening climate change is the carbon dioxide emissions into the atmosphere caused by the enormous electricity demands of the crypto system. Unfortunately, the production of cryptocurrency still relies heavily on gas and coal power plants, especially in less developed parts of the world. When burned, coal and natural gas emit greenhouse gases, which warm the Earth and change the climate. As of 2022, it is estimated that such cryptocurrency mining is responsible for 0.1% of global greenhouse gas emissions⁹³. Another environmental impact is e-waste pollution due to the short lifespan of the equipment used for mining, especially

⁹²How crypto-owning climate activists balance saving the planet with supporting energy-hungry Bitcoin mines; <https://fortune.com/2021/11/15/cop26-cryptocurrency-owning-climate-activists-bitcoin-blockchain-energy-use/>

⁹³A deep dive into Bitcoin's environmental impact; <https://www.jbs.cam.ac.uk/insight/2022/a-deep-dive-into-bitcoins-environmental-impact/>

value-wise very strong cryptocurrencies such as Bitcoin and Ethereum, as well as the equipment used to support the Blockchain⁹⁴.

One of the measures to make crypto trading more sustainable and greener is to introduce a legal requirement that, to begin with, at least 55% of the energy used for the crypto system must come from renewable energy sources. Of course, the state's role in providing the infrastructure and opportunities for the use of renewable energy sources cannot be overlooked. It is necessary to provide subsidies for miners who, in regions where the use of solar panels and wind turbines is effective, express an interest in installing and using these tools.

Monitoring energy consumption and controlling its use is very important, so individuals or external companies will be hired whose task will be to collect the necessary information from users. Each user, a participant in crypto trading, will have a monthly reporting obligation and must disclose data related to energy consumption, which will be accessible to authorized personnel, third parties, or government institutions. Additionally, similar to the asset inventory requirement for every business that includes fixed and current assets, each legal participant in crypto trading will be required to report the type and quantity of electronic equipment they use. Building on this, users/companies will also be required to compile an annual report on the amount of electronic waste generated and how it was disposed of, certified with the seal and signature of a legal entity authorized to handle such waste. Users will be required to keep this documentation for a minimum of 5 years.

In addition to reducing the use of energy from non-renewable sources, users also have the option when it comes to miners to replace the mining process using computers and graphics cards with "Cloud Mining."-ing. Cloud Mining is a mechanism for mining cryptocurrencies using rented computing power in the cloud, without the need to install and directly run hardware and associated software. Cloud mining companies allow users to open an account and participate in the cryptocurrency mining process remotely on an online platform for a base price, making mining accessible to a wider number of people worldwide with significantly lower overall electricity consumption. Since this form of mining is conducted online via the cloud, problems such as the need for equipment maintenance or direct electricity costs are reduced. Each participant earns a proportional share of the profit in proportion to the amount of cloud processing power (Hash rate) that has been rented^[39].

Cloud Mining represents a more environmentally friendly concept within the PoW crypto system. However, despite this, PoW still represents a system that has a significantly worse impact on the environment compared to the PoS system. A PoS crypto system uses 100 TWh less than PoW because coin holders only verify the amount of crypto assets they hold (stake). For this reason, a more advanced solution would be to replace the PoW system with a PoS system, which would bring benefits not only in reducing environmental pollution through lower carbon dioxide emissions and the general consumption of less electricity, but also through lower hardware requirements. Something similar was implemented with the launch of the Ethereum 2.0 cryptocurrency which, unlike the previous version, is entirely based on the Proof-of-Stake system. Since the crypto system is independent of the state apparatus, it is impossible to compel the owners of cryptocurrencies themselves (who are often unknown) to convert from a PoW to a PoS system of operation, but this can be done with the users, or coin holders. As already stated, the introduction of a new state digital wallet model and legal regulation of the crypto market already implies that the state apparatus will have insight into the balance of crypto assets held by users in order to enable taxation and control. In addition to the information mentioned, the competent state authorities will also have insight into the type of system that crypto currencies use in a user's balance. A measure is proposed to limit the possession of crypto assets, such that 80% of the acquired assets must be obtained through a PoS system, while the remainder can be from a PoW system. In the case of acquisition methods involving Cloud Mining, 50% of the assets must be acquired through a PoS system and a minimum of 30% through a PoW system, but via Cloud Mining.

⁹⁴The Economic and Environmental Impact of Bitcoin; <https://ieeexplore.ieee.org/document/9385063>

Based on all the proposed measures for risk treatment, **Table 9** provides an overview of the resources required to implement each of the proposed additional measures and an estimate of the time needed to carry out each of the listed measures.

Risk indicator	Identified unacceptable risk	Recommended action to mitigate/eliminate the risk	Resources required to implement the action	Action Implementation Time Estimate
R-28	Opposition and interference from environmental groups to the implementation of the model	Enact a legal requirement that a minimum of 55% of the electricity for mining must come from renewable sources	Human Resources and Time: Staffing to monitor energy consumption or hiring an external professional organization to perform this service	A long-term 7-year plan is proposed. Two years for the introduction of the basic model and five years for the implementation of the corrective measures provided to make Blockchain technology greener and more sustainable.
		Users are required to create an annual report on the type and quantity of equipment used for crypto mining and trading, the amount of electronic waste, and its disposal method. Allowing access to reports for competent government institutions	Human resources and time required to define/amend laws and monitor their implementation	
		Switch to Cloud Mining or PoS	Human resources and time required to define/amend laws	
R-31	Requirements for the consumption of a huge amount of electricity	Legal regulation of the obligation to report energy consumption. Allowing access to reports for competent government institutions	Human resources and time required to define/amend laws and to monitor their implementation	
		Legislative limitation of the share of crypto assets acquired through a PoW system	Human resources and time required to define/amend laws	
R-24	Lack of legal regulation of the crypto market and Blockchain in business	Taxing all participants in crypto payments – introduction of a state registry of crypto technology users and a state crypto wallet	Human resources and time required to define/amend laws, financial resources	

		Amendment of laws/creation of new laws/regulations and procedures	Human resources and time required for defining/amending laws, financial resources	
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Table 11 – Resources and time required for the implementation of risk mitigation/elimination actions

7.4.1 Measures to mitigate/eliminate the impact of risks

Crypto assets have existed for more than a decade, but it is only now that efforts to regulate them have moved to the top of the policy agenda. This is partly because only in the last few years have crypto trading and Blockchain technology evolved from an interesting innovation in search of a purpose to a potential legally recognized payment instrument and a technology that enhances everyday business. The enormous growth in the market capitalization of crypto assets and their entry into the regulated financial system have led to increased efforts to regulate them. Applying existing regulatory frameworks to crypto assets or developing new ones is a challenge for several reasons. For starters, the crypto world is rapidly evolving. Governments are struggling to acquire the skills and knowledge to keep up with innovations given their overstretched resources and many other priorities, while companies in all sectors are seeking to capitalize on the benefits that new technology brings. Monitoring the crypto market is difficult because data is inconsistent, and government agencies find it challenging to track the thousands of users/companies that may not be subject to typical data disclosure or reporting requirements. Finally, in addition to developing a framework that can regulate both the actors and activities in the crypto ecosystem, national authorities must take a stance on how the underlying technology used in Blockchain fits with other public policy goals, such as environmental protection. The full, unhindered application of new technology in logistics processes across all countries remains, for now, only an aspiration.

In order to make the best use of Blockchain technology in the logistics sector, Focusing on the territory of the Republic of Serbia and the current status of crypto technologies, it is necessary to first define a legal framework that will protect participants and benefit companies, users, and the state through the taxation and regulation of crypto assets and transactions. Legislative regulation of this area would also require the introduction of a state digital wallet model, which would aim to establish legal rules that could provide a safe space for innovation and positive economic effects from the use of crypto technology for the state, and ensure a regulated, a monitored and controlled system for both users of logistics services and companies, which could take advantage of the wide range of new opportunities offered by crypto technology.

Taxing the use of cryptocurrencies is beneficial for users because it will ensure that investments in cryptocurrencies, Blockchain technology, and financial investments in this form are under the state's secure guarantee. This would significantly reduce the security risks that currently prevent high levels of trust in cryptocurrency technology. In order to define a tax system for crypto trading and ensure transaction security, the problem of complete anonymity for participants in transactions must first be resolved. In banking transactions, every participant, whether a legal or physical person, is a registered user of banking services. Therefore, in the event of any suspicion of potential money laundering, tax fraud, false identity, or other criminal acts, the burden of proof lies with the known individual(s) against whom the state is taking legal action. Consequently, the aggrieved party is backed by the state, which will attempt to compensate for or mitigate the damage and deliver justice. In the case of crypto trading, there is no obligation to report or register participants, which, combined with the way Blockchain functions, means the entire process is anonymous. Therefore, it is proposed to introduce a state registry of crypto traders, i.e., individuals and companies that offer the possibility of crypto payments, similar to how a registry of legal entities exists in the country, where the identity of each user/company and their physical address will be recorded. Users will, just as with any other form of property, be required to provide proof of origin for crypto assets and pay taxes on them.

purchased with cryptocurrencies, and cryptocurrency transactions will be required to have proof of origin and to pay taxes.

The project for the development and implementation of the state digital wallet will encompass the development of a complete infrastructure to enable its application—mobile applications and wallet software, mobile transaction processing software, as well as software to support the call center.

The state digital wallet will be the official cryptocurrency wallet of the Government of the Republic of Serbia, which will allow users to send and receive cryptocurrencies and/or dinars commission-free if they are Serbian citizens. The application will be compatible with all mobile phone models, and users will be able to download and install it on their mobile phones from the Apple or Play Store. Additionally, a state-owned digital exchange will be created alongside the application, where citizens will be able to exchange cryptocurrencies for dinars and vice versa, commission-free. The wallet will be connected to the Republic of Serbia's banking system to allow for the withdrawal of cryptocurrency in dinar equivalent, which can be done using the Bank Poštanska štedionica's ATMs.

In addition to standard ATMs, state-owned crypto-ATMs will be introduced and installed. Their number is planned to be equal to the number of traditional ATMs to ensure maximum accessibility to this service. On these machines, citizens will be able to buy and sell a wide range of cryptocurrencies by depositing physical cash or using a card. Unlike the three most well-known cryptocurrencies currently available—Bitcoin, Ethereum, and Litecoin—it will be possible to trade Solana, Avalanche, Polkadot, Cardano, and others for a 0.5% fee. Transactions between citizens of the Republic of Serbia and companies based in that territory will be commission-free, making it an attractive option for business.

Payment security will be ensured through two-factor authentication for approving each transaction. This can include facial recognition or entering a PIN and sending a confirmation code via SMS or email, which the user provides in their application. Quick payments in the wallet can also be made using the user's phone number, contact phone number, and passport number.

Additionally, even the most skeptical users can use the digital wallet, as a hardware wallet option is provided. In essence, this type of wallet is an electronic device that stores the private key offline and therefore out of reach of hackers. For each transaction, the device must be physically connected to a computer or mobile phone, and a pre-set PIN must be entered on the device to confirm the identity of the wallet's owner.

Citizens and legal entities will be able to use the state-provided hardware wallet for a monthly service fee of 500 dinars, which can be paid with their phone bill. The service fee includes insurance against potential hacking attacks, device servicing (excluding physical damage), and customer support, which can be accessed via a chat option on the application interface, by calling a landline number, or by sending an email.

Given the introduction of an innovation that is completely different from anything that has been implemented so far, it is very important to provide the simplest and most detailed instructions for use. Additionally, presentations will be made available on the YouTube platform, published by official government channels, along with brochures in digital and physical formats to familiarize users with how the system works.

After the legalization of cryptocurrency use, any legal entity providing logistics services will be required to accept cryptocurrencies as a form of payment if the customer wishes to pay that way. Additionally, if the service provider prefers payment in dinars, the user of the state digital wallet will have the option to convert the cryptocurrency to dinars on the spot when paying through the application. The owner of the state wallet will be able to use it to pay for all services, buy and sell products, as well as make transfers to bank accounts without additional commission fees.

On the other hand, if we look at the ecological aspect, it is important to consider that the industry recognizes it has an environmental problem. The negative impact on the environment is one of the important factors for the successful implementation of the new technology and the general widespread adoption of cryptocurrencies during an ecological crisis that permeates all economic and social spheres. This negative aspect of crypto technology largely negates all the potential positive effects of its application. However, if sufficient time, financial, and human resources were dedicated to implementing proposed measures that would benefit in reducing the carbon footprint and the amount of electronic waste,

Cryptocurrency technology could easily become a part of everyday life in logistics and other industries alongside existing currencies.

Companies can verify identities using facial recognition features, make faster payments, and track transactions along supply chains using blockchain technology. It then becomes easier for logistics organizations to provide a more consistent and seamless service.

7.5 Calculation of the residual risk after implementing the planned measures

After implementing measures to reduce the consequence, probability, and detectability of a nonconformity, the RPN number must be recalculated. If the RPN number has not decreased, it means the measures defined were not adequate. After successfully implementing the FMEA method, a risk reduction of at least 50% should be expected^[43].

After defining the preventive actions, the FMEA analysis, or rather the evaluation, should be re-conducted only for the identified unacceptable risks to determine if the objective of the corrective actions has been achieved—reducing the risk to an acceptable level below the value defined as the acceptability limit. For this purpose, the evaluation is conducted according to the same criteria as in the initial FMEA table. If all RPNs are below the established threshold, the issues have been satisfactorily resolved, meaning the corrective actions have achieved their intended effect. This process is repeated until the RPN is reduced below the acceptability threshold.

Risk design ation	Identified unacceptable risk	RPN befo re meas ures	Recommended Action	Consequence Severity F7/E	Likelihood of Occurrence	Detectability	RPN after meas ures
R-28	Opposition and interference from environmental groups to the implementation of the model	60	Introducing a legal requirement that a minimum of 55% of the electricity for mining must come from renewable sources	4	2	3	24
			Requirement for users to create an annual report on the type and quantity of equipment used for crypto technology and trading, the amount of electronic waste, and its disposal method. Allowing access to reports for competent state institutions				
			Transition to Cloud Mining or PoS				
R-31	Requirements for the consumption of a huge amount of electricity	75	Legal regulation of the obligation to report energy consumption. Allowing access to reports for competent state institutions	5	4	1	20

			Legislative limitation of the share of crypto assets acquired through a PoW system				
R-24	Lack of legal regulation of the crypto market and Blockchain in business	60	Amendment of laws/definition of new laws/regulations and procedures	4	2	3	24
			Taxation of all participants in crypto payments – introduction of a state registry of crypto technology users and a state crypto wallet				

Table 12 – FMEA table after the implementation of risk mitigation/elimination actions

From the repeated FMEA table, it can be seen that the corrective measures applied have contributed to reducing unacceptable risks when it comes to the application of Blockchain technology and crypto payments, which in the long term means a greater possibility of its successful implementation and the achievement of planned goals.

7.6 Defining a way to monitor and verify risks

After the repeated FMEA assessment confirmed that the defined corrective actions will mitigate the impact of the most significant risks, it is necessary to define a way to determine that the implemented measures have yielded the expected/desired results. To this end, it is important to define KPIs - Key Performance Indicators and KRIs – Key Risk Indicators.

Given that the proposed model for implementing the state digital wallet is intended for use by its citizens, a key indicator of its success is undoubtedly *user satisfaction over a specific period of time*. This indicator can be tracked through a survey that would contain a certain number of questions related to user satisfaction with the service and the ability to rate it based on multiple criteria (e.g., security, ease of use, interface appearance and accessibility, etc.). This survey would be sent to users via the email address they provide in the application, or they could access it through the application's interface. In that case, a pop-up notification would appear while using the app with a link to the survey. This type of survey would be conducted once every 3 months.

Additionally, an indicator of the model's success is *the number of new users on a monthly, quarterly, and annual basis*, as well as *the number of transactions completed via the digital wallet*. This data can then be used to track user churn and the wallet's usage rate. If a downward trend in their number is observed, the focus will be on determining the reason for it and, by extension, the model's shortcomings. Consequently, *a decrease in the number of users and transactions over time* represents a KPI for the model. Related to this, a significant measure of user satisfaction is also *the number of service complaints within a specific time period* (monthly, quarterly, annually), which can be considered a supporting KRI for the model. In that case, the leading KRI would represent *the trend of declining user satisfaction over a longer period*. Each user would be given the opportunity to express their dissatisfaction and report any potential problems by calling the customer service number, via the chat option within the wallet application itself, or by email. Within the aforementioned new government sector that will be responsible for the development, implementation, monitoring, and control of the digital wallet model and general crypto trading matters, it will be necessary to hire personnel to handle user complaints and problem resolution.

Another key accompanying risk indicator could be *the number of system outages on a monthly, quarterly, and annual basis*. Its implementation and data monitoring will indicate the quality of the design and implementation process, and the effectiveness of the system's deployment and maintenance. If more than two occur

system outages per month, the digital wallet's operational system will need to be analyzed in detail, the problems identified, and it will have to go back through the design process if the deficiencies are significant.

For environmental risks that have proven to be very significant for the overall success of the model's implementation, the following risk indicators will be determined, the values and changes of which will be recorded and monitored by the competent authorities and external legal entities.

These are the following monitored KRI:

- Number of protests by environmental activists during the year
- Annual electricity consumption in crypto trading

And leading KRI:

- Decrease in the number of users of solar panels and wind turbines
- Increase in the use of electricity generated from the combustion of fossil fuels
- Increase in the annual amount of electronic waste generated
- Increase in the use of PoW systems
- Uncontrolled electricity consumption in the crypto trade

The Statistical Institute will, in addition to monitoring existing economic and social indicators, also monitor the aforementioned KPIs on a monthly, quarterly, and annual basis, as well as KRI, to enable the comparison and monitoring of the model's performance over different time periods and to identify trends. All data will be published and made available to the public through the institution's online presentation and website. For this purpose, the creation of new additional jobs may be necessary, depending on the level of engagement required for the analysis of this area in subsequent years of implementation.

In addition to the statistical institute, the Ministry of Environmental Protection, which will deal with waste management and disposal issues, and the Ministry of Mining and Energy, which will be responsible for collecting data on electricity consumption and monitoring its use, will also be involved in monitoring and controlling the aforementioned indicators. Within this ministry, it is advisable to establish a sector dedicated to renewable energy sources, the development and implementation of projects in this area. This sector would also address enabling citizens, and crypto participants in particular, to switch to using electricity from clean sources. If necessary, new jobs can be created and new personnel hired in both ministries.

Regarding the documentation important for tracking the aforementioned KRI, it is proposed that every user/logistics company implementing crypto technology be legally required to record the amount of electronic waste and other equipment generated, as specified in the presentation of risk reduction measures in the previous chapter. This report, along with the seal and signature of the competent representative of the legal entity contracted for electronic waste disposal, will be submitted electronically by users to the Ministry of Environmental Protection at the end of each year.

In addition to all of the above, each risk identified for the application of the aforementioned model in the FMEA analysis can and should be verified after a certain period of time (e.g., once a year). using the same criteria, thereby ensuring the monitoring and early control of any potential problems in the years following the implementation of the model and risk reduction measures.

CONCLUSION

Cryptocurrencies are a phenomenon that economists, lawyers, IT experts, hackers, central banks, and security services are all concerned with. One of the reasons that initially attracted technology and internet enthusiasts, as well as investors, is the fact that cryptocurrencies are not subject to control by central banks or government institutions. They are protected from inflation by a mathematical function that prevents their available supply from increasing beyond a predetermined limit. Even if they suddenly lose value, which is a real possibility, crypto trading and technology will not disappear. The moment a problem arises that a particular currency cannot overcome, another currency will take its place, incorporating elements that address that shortcoming. The next significant feature of cryptocurrencies, recognized as an advantage, is the elimination of intermediaries in transactions because the crypto system is based on a decentralized system without a regulatory authority. This makes transactions cheaper, especially when it comes to international payments.

Cryptocurrencies can emerge as a preferred method for transferring money because cryptocurrency-based transfers do not rely on banks or other intermediaries used by modern systems. Consequently, there are no high fees or high costs. In this regard, if securities commissions and/or agencies responsible for supervising financial services were to support the establishment of funds tied to cryptocurrencies or approve similar innovations, the value of cryptocurrencies would increase even further.

With the adoption of blockchain technology in business and international trade, the potential for widespread cryptocurrency use is gaining importance, as companies would use them more among themselves, eliminating intermediaries. Blockchain is a technology that possesses key characteristics that enable the resolution of various current problems in the financial and non-financial spheres. This technology faces various ups and downs in the current world. People are divided into two groups: those who are genuinely interested in blockchain-based applications and those who are skeptical about this type of innovation. Therefore, the adoption of blockchain faces many challenges. The most significant of these is probably blockchain's ability to replace people in certain jobs, causing them to lose their jobs. On the other hand, it should be considered that Blockchain is an innovative technology that improves efficiency in various sectors and promotes globalization in general. Therefore, a large number of financial and non-financial companies are investing money in creating blockchain-based applications and are beginning to implement them in their business operations to improve their performance.

The adoption of blockchain in supply chain and logistics is currently slow due to the associated operational risks, which is why companies' skepticism toward the new technology is justified. However, it is very likely that it will soon gain their trust, as the use of cryptocurrencies for everyday payments, as well as blockchain technology in general, is in a phase of constant expansion across all industries. In a survey conducted on the LinkedIn platform, 75% of respondents believe that the use of cryptocurrencies is becoming increasingly common in the transportation and supply chain market and that this trend will continue to grow exponentially.

Logistics and supply chain management are considered domains where Blockchain technology fits well for a number of reasons. Throughout the product's lifecycle, as it moves along the value chain (from production to consumption), data generated at each step can be documented as transactions, thus creating a permanent history of the product. Additionally, blockchain technology can efficiently contribute to recording each individual asset (from the product to the container) as it passes through supply chain nodes, tracking orders, receipts, invoices, payments, and any other official document, as well as tracking digital assets (such as warranties, certificates, copyrights, licenses, serial numbers, barcodes) in a unique and parallel manner with physical assets. Furthermore, blockchain can effectively contribute, through its decentralized nature, to the exchange of information about the production process, delivery, maintenance, and consumption of products between suppliers and sellers, bringing new collaboration options when it comes to complex assembly lines. Challenges in logistical parameters, such as delivery delays, loss of documentation, unknown product origin, and errors, can be minimized or even avoided by applying blockchain technology.

However, the legal status of cryptocurrencies varies significantly from one country to another and is still undefined or changing in many of them. While in most countries the use of cryptocurrencies in itself

is not illegal, their status and usability as a means of payment in various industry sectors vary with different legal regulations. While some countries have explicitly allowed their use and trade, others have banned or restricted it. Despite its increasingly widespread daily use, this new form of currency and its innovative technology still remains in the shadows due to various risks associated with its use, which have been analyzed from a logistical perspective in this paper. However, even these risks will not prevent this new technology from becoming a part of our daily lives.

The chance of implementing the proposed measures that would allow for the free application and use throughout the territory of the Republic of Serbia, considering the current status of the use and legal regulation of cryptocurrency trading and possession, Given the high average age of the population and the high percentage of computer-illiterate or poorly literate citizens, the implementation of the proposed measures that would allow for the free application and use throughout the territory of the Republic of Serbia is very unlikely in the next 10 years. On the other hand, the part of the project concerning the possibility of legal regulation, i.e., declaring cryptocurrencies legal currencies with the accompanying taxation of crypto assets and enabling the recognition of asset purchases via crypto transactions, is very likely possible and will be realized in the near future. The field of environmental protection and the use of renewable energy sources are, as in most countries today, pressing issues that do not receive enough attention. Only about 25% of the electricity in Serbia comes from clean sources. The introduction and promotion of the same or a similar model, and of crypto trading in general, as presented in this paper, would inevitably mean higher consumption demands than before, as well as a greater annual waste generation. For this reason, any form of legal regulation and the introduction of cryptocurrencies into legal circulation "side by side" with other currencies can bring more benefits than harm, but only if all accompanying environmental protection measures are implemented.

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LIST OF FIGURES

Figure 1 – <i>Different participants in the logistics chain</i>	3
Figure 2 – <i>Functional decomposition of logistics</i>	4
Figure 3 – <i>Schematic representation of the company's logistics system</i>	6
Figure 4 – <i>The processes of movement (transportation) and holding (storage) in logistics systems of different structures</i>	7
Figure 5 – <i>Subsystems of the logistics system of an industrial enterprise</i>	9
Figure 6 – <i>Logistics within the business system</i>	10
Figure 7 - <i>The place of logistics in process manufacturing</i>	11
Figure 8 – <i>Comparison of characteristics of different types of production systems</i>	12
Figure 9 – <i>Structure of procurement logistics activities</i>	13
Figure 10 – <i>The place of distribution logistics in the supply chain</i>	14
Figure 11 – <i>Classification of distribution channels in terms of hierarchical structure and degree of distribution</i> ..	16
Figure 12 – <i>Connection between warehouse logistics and procurement, production, and distribution logistics</i> ...	17
Figure 13 – <i>Warehouse Receiving Operation</i>	18
Figure 14 – <i>Functional connection of the logistics system</i>	19
Figure 15 - <i>Classification of relevant factors of storage and inventory policy</i>	20
Figure 16 – <i>Order fulfillment scheme – conclusion of the sales contract</i>	30
Figure 17 – <i>Order fulfillment tasks</i>	31
Figure 18 – <i>A simple representation of the supply chain elements</i>	32
Figure 19 – <i>Supply chain</i> ³¹	32
Figure 20 – <i>Supply Chain Management Model</i>	34
Figure 21 - <i>Supply Chain without Blockchain Technology</i>	54
Figure 22 - <i>Supply Chain with Blockchain Technology</i>	54
Figure 23 – <i>Basic cause-and-effect relationship</i>	72
Figure 24 – <i>Graphic representation of problem definition – consequences when creating an Ishikawa diagram</i> ..	73
Figure 25 – <i>Basic form of a cause-and-effect relationship</i>	73
Figure 26 – <i>Basic structure of the cause-and-effect diagram</i>	74
Figure 27 – <i>Causal Diagram Development Phase</i>	74
Figure 28 – <i>The expansion (branching) procedure</i>	75
Figure 29 – <i>Graphic representation of all identified risks via the Ishikawa diagram</i>	77

LIST OF TABLES

Table 1 – <i>Characteristics of the logistics system</i>	8
Table 2 – <i>Internal and external transportation costs</i>	27
Table 3 – <i>List of all identified potential risks in the logistics sector and supply chains</i> 59	
Table 4 – <i>List of risk categories</i> ^[40]	61
Table 5 – <i>Scale for assessing the consequences of an adverse event</i>	79
Table 6 – <i>Scale for Rating the Probability of an Adverse Event</i>	80
Table 7 – <i>Scale for assessing the detectability of an adverse event</i>	80
Table 8 – <i>Scale for assessing risk significance</i>	81
Table 9 – <i>FMEA table used to determine the most significant risks</i>	90
Table 10 – <i>Highlighted risks with the highest RPN according to the FMEA method</i>	91
Table 11 – <i>Required Resources and Time for Implementing Risk Mitigation/Elimination Actions</i>	93
Table 12 – <i>FMEA table after implementing mitigation/risk elimination actions</i>	96